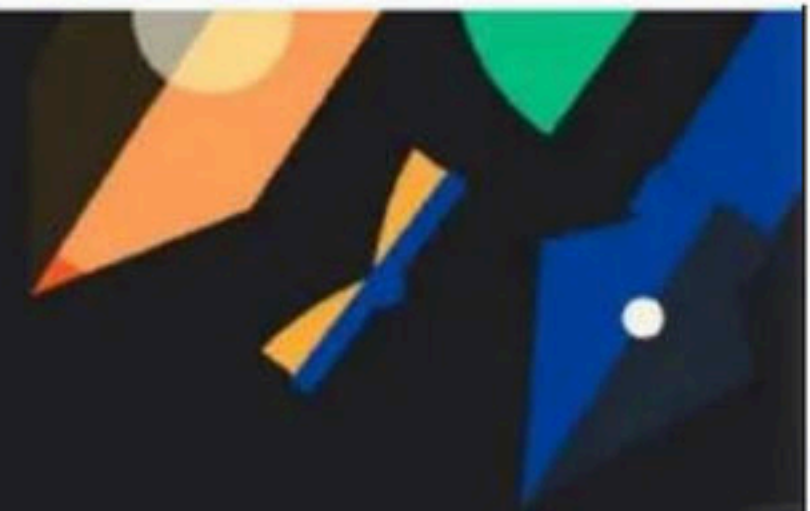




DC-Basics of Programming

Special class



L1-Basics of Programming

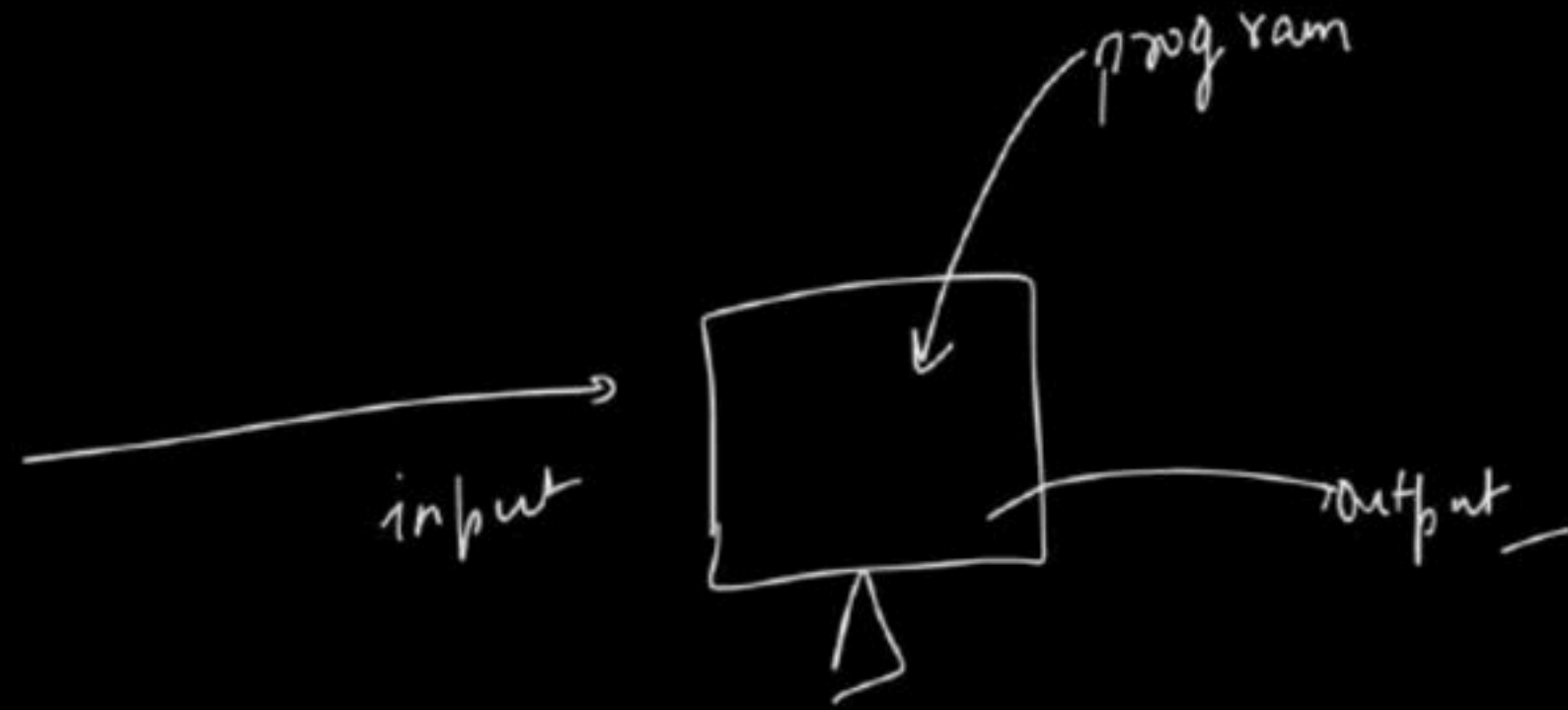
Special class

L1 - Basics of Programming

Flowcharts and PseudoCode

—> by Codehelp

What is Programming?



What is Algorithm?

Maggi →

Sherry Vohra

Step 1 ✓ → Water → Boil

2 ✓ → Maggi → Pan

3 ✓ → Masala → Pour

4 ✓ → 2 min wait

5 ✓ → plate —

6

⋮

⋮

⋮

Algorithm

Namkeen
Chawal



SI ~~→~~ oil → Jira

~~→~~ onion →

~~→~~ tomato

→ haldi / Lemon juice / Spice

→ Chawal

→ Water

→ Wait

Algorithm

Pasta $\rightarrow$?
=

~~✓~~ $> 90\%$

$< 90\%$
↓

How to approach a Problem ?

Thought process

- ① let's understand the Problem
- ② Analyse problem $\rightarrow$ given values / relate formula / constraints

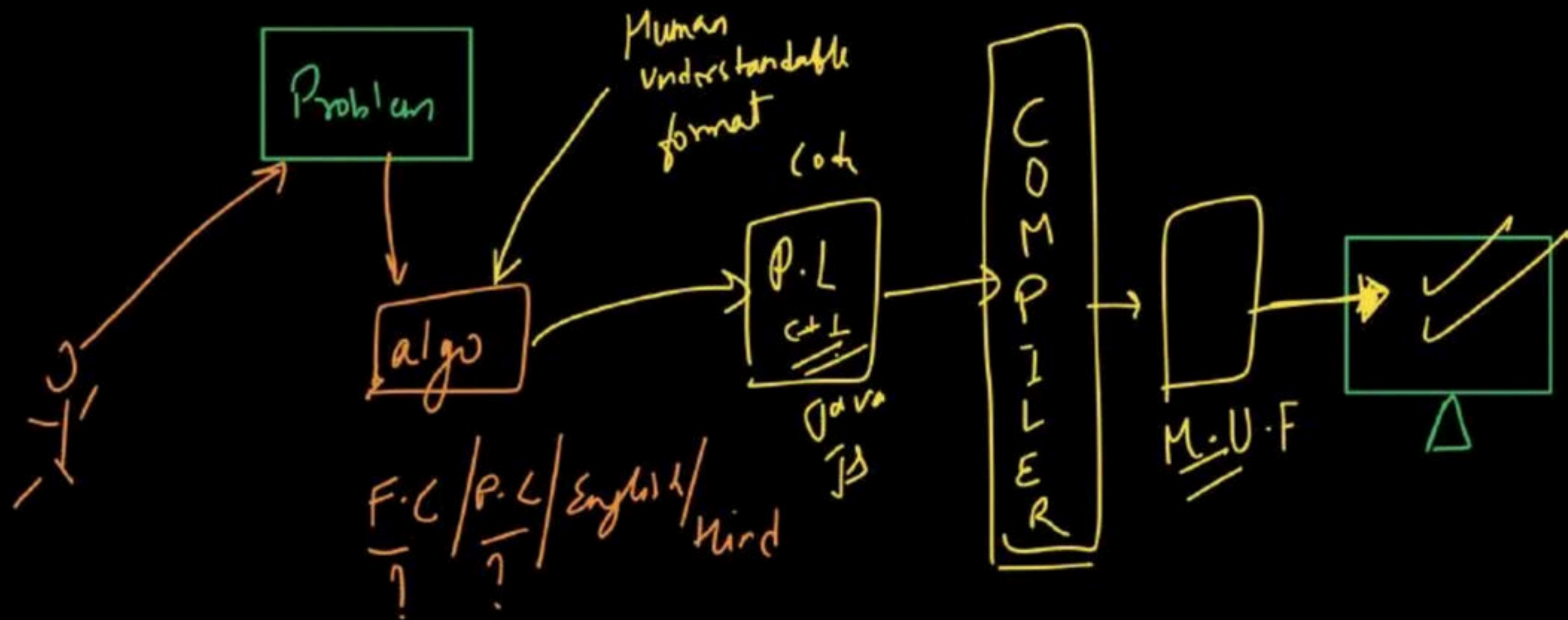
$$\text{side} = 4 \text{ cm}$$

- ③ Create a approach $\rightarrow$ Algorithm

Optimise

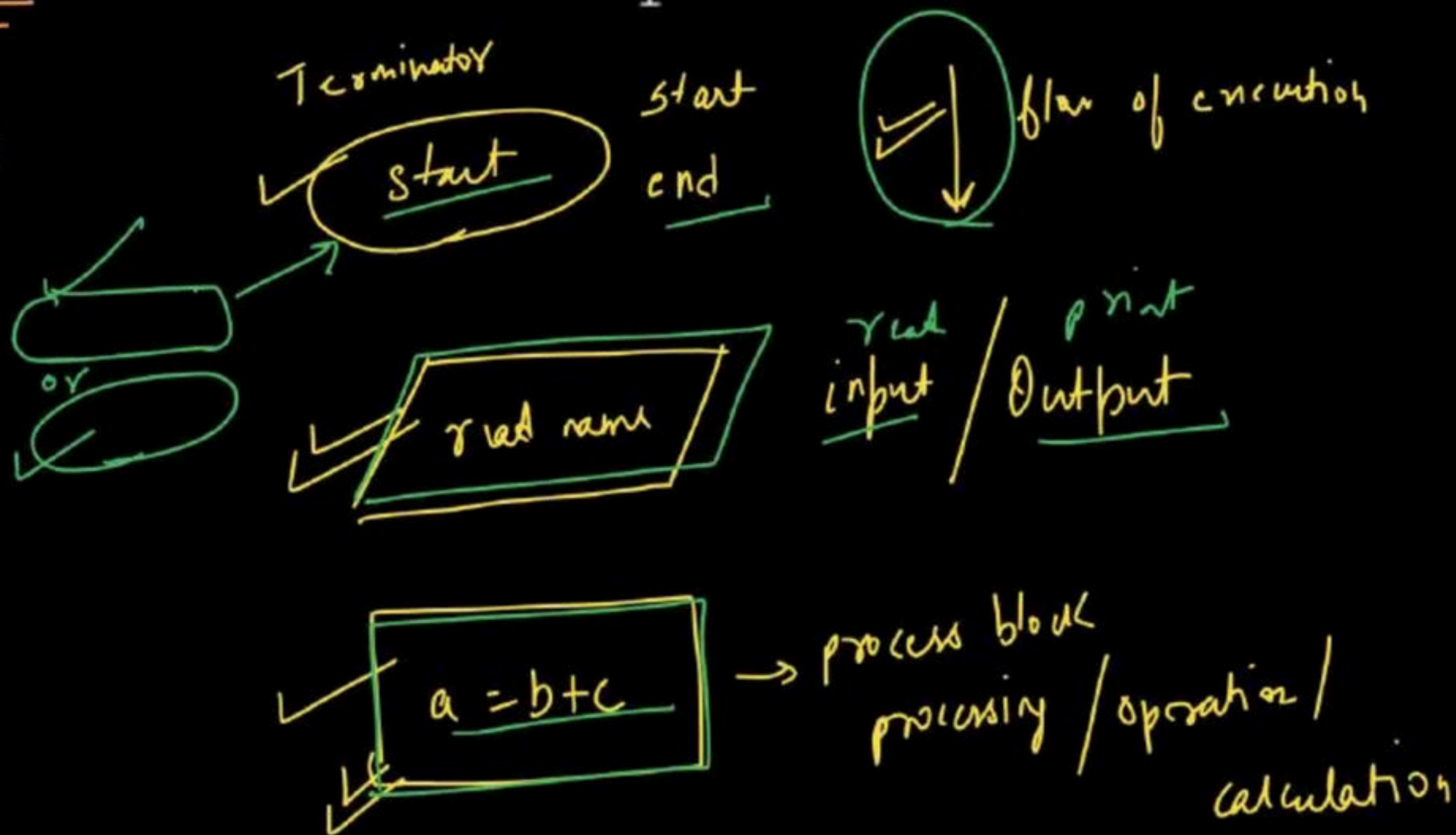
$$\begin{aligned} \text{area} &= 5 \times 5 \\ &= 4 \times 4 \\ &= \textcircled{16} \end{aligned}$$

Using a computer to solve a problem → ?

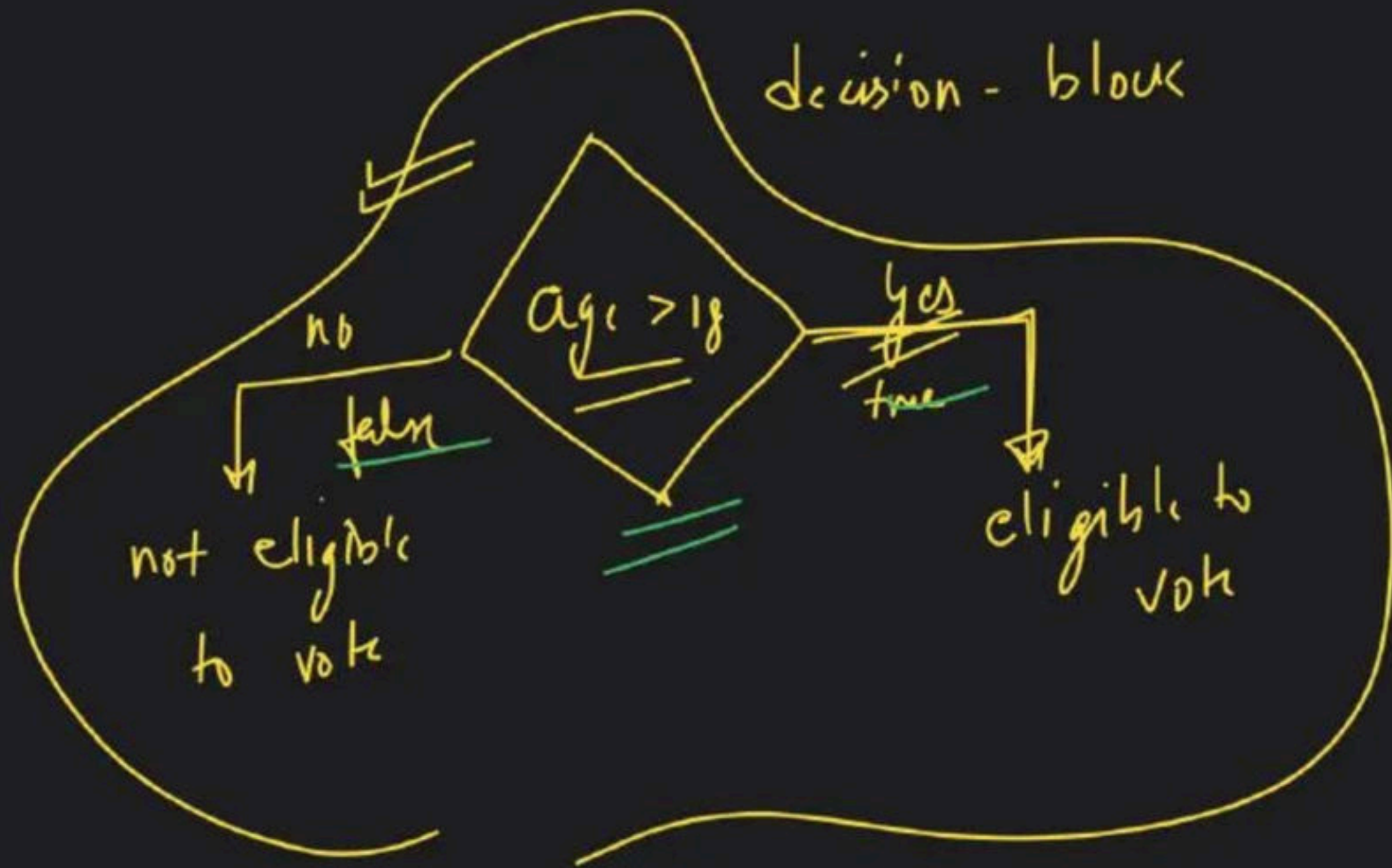


Flowchart and its components:

↓
diagrammatic
representation
of
your
Algorithm



decision - block



Pseudocode:

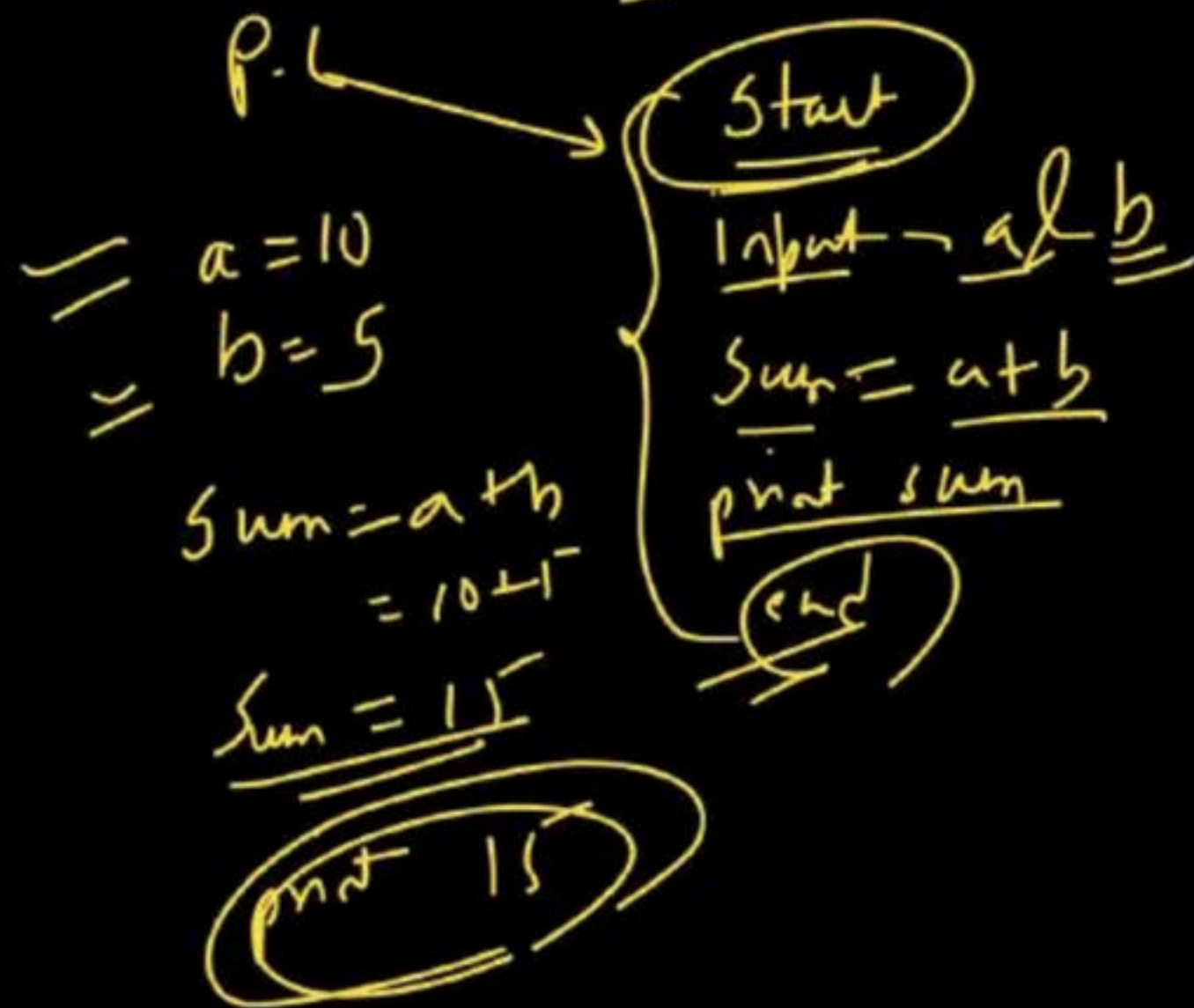
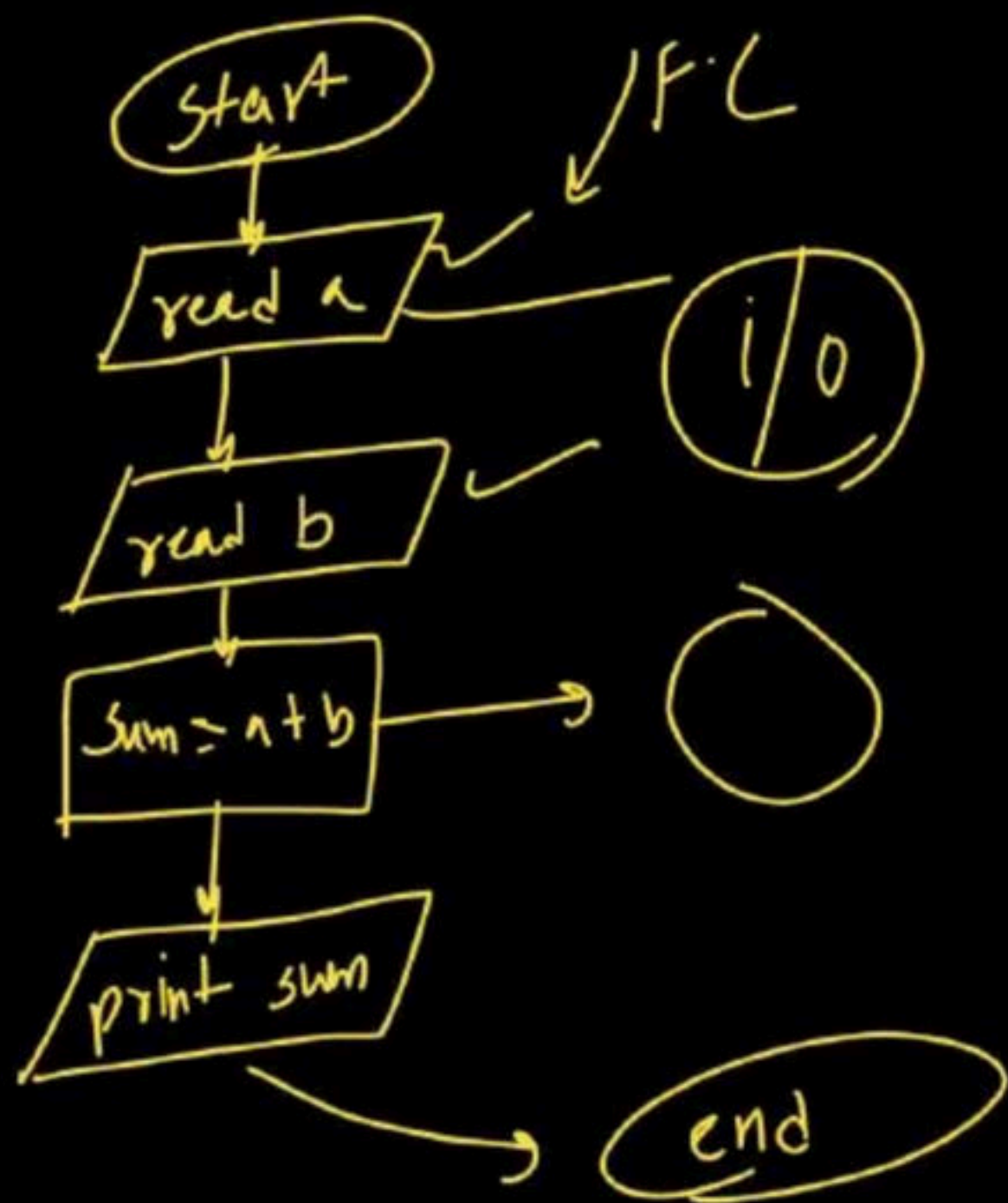
Nakli $\rightarrow$ code $\rightarrow$ denote $\rightarrow$ dui bhasa

Sum of 2 no.

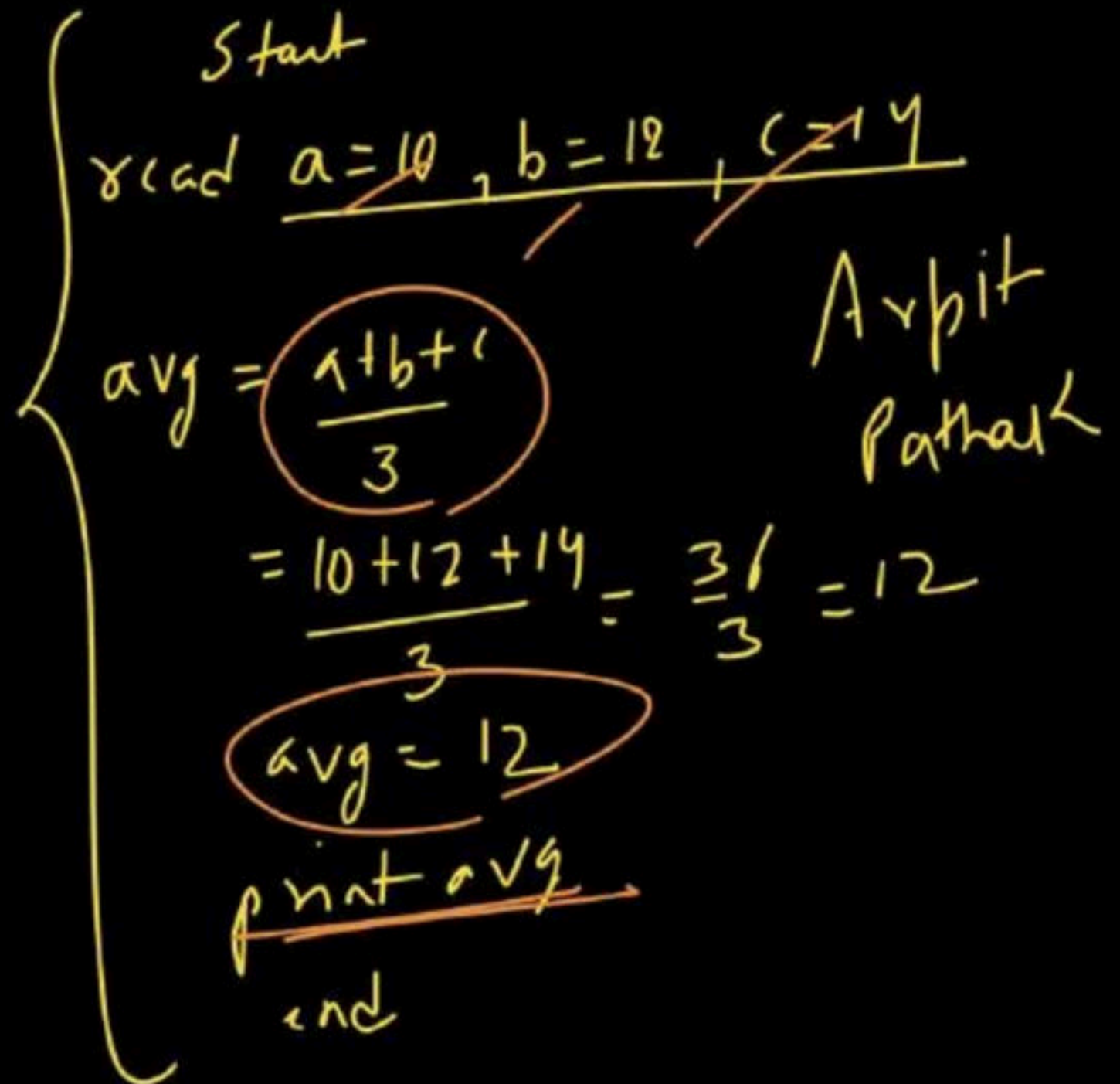
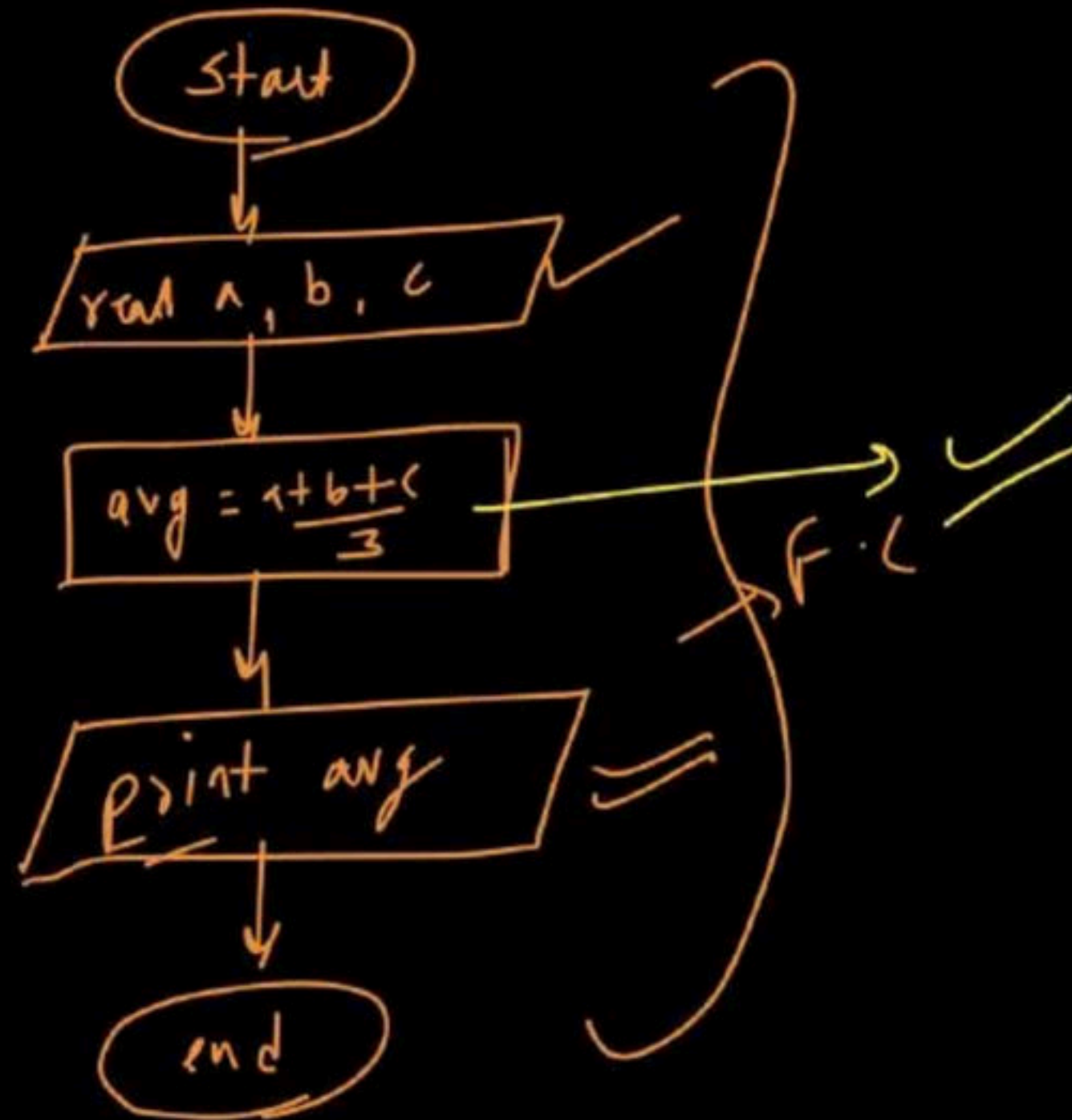
- $\rightarrow$ Start
- $\rightarrow$ read the value of a & b
- $\rightarrow$ $sum = a + b$
- $\rightarrow$ print sum
- $\rightarrow$ end

- $\rightarrow$ Start
- $\rightarrow$ read first name
- $\rightarrow$ read middle name
- $\rightarrow$ read last name
- $\rightarrow$ $f.n = f + m + l$
- $\rightarrow$ print full name
- $\rightarrow$ end

Design Flowchart - Print Sum of a and b

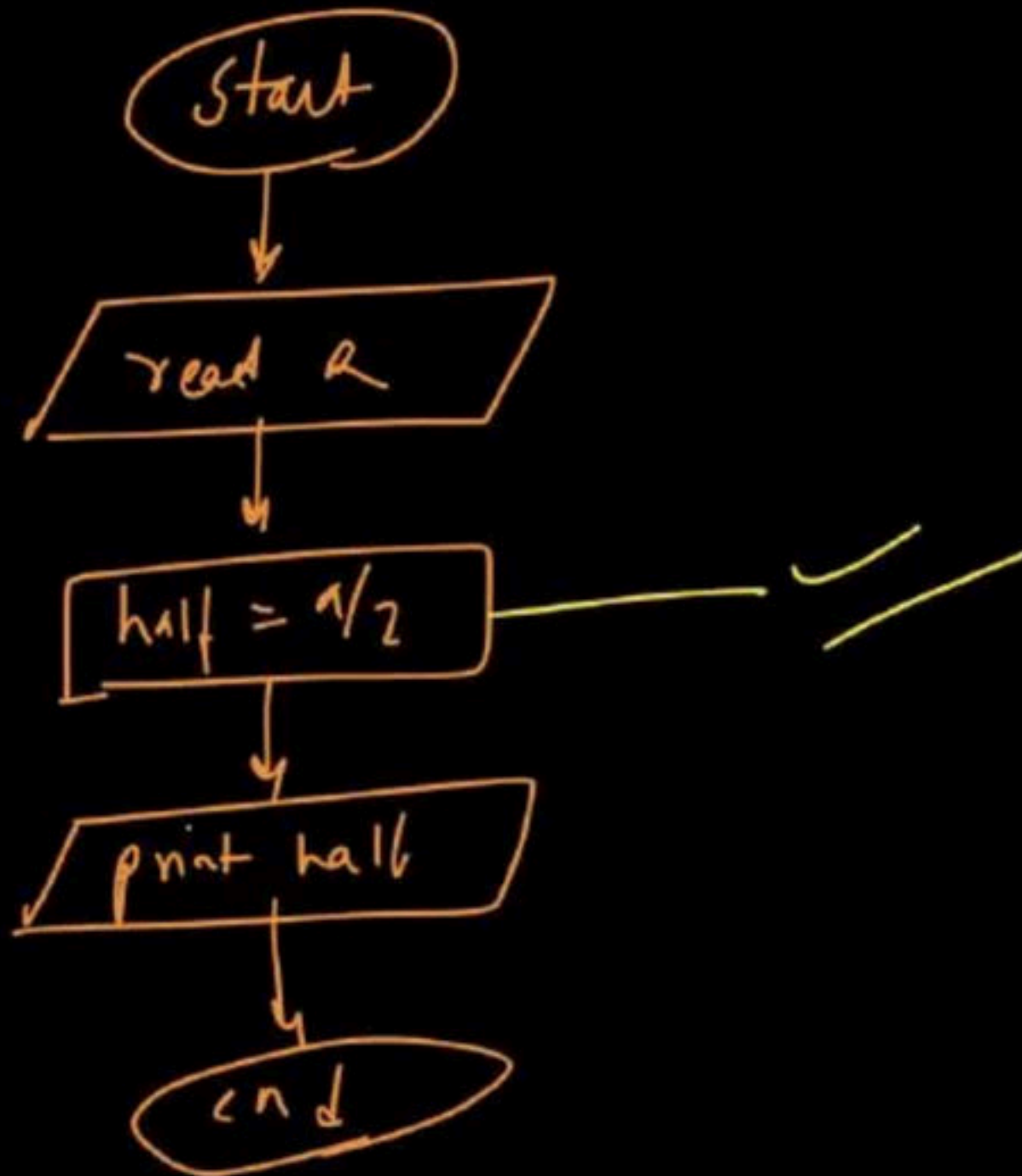


Design Flowchart - Print Average of a, b and c



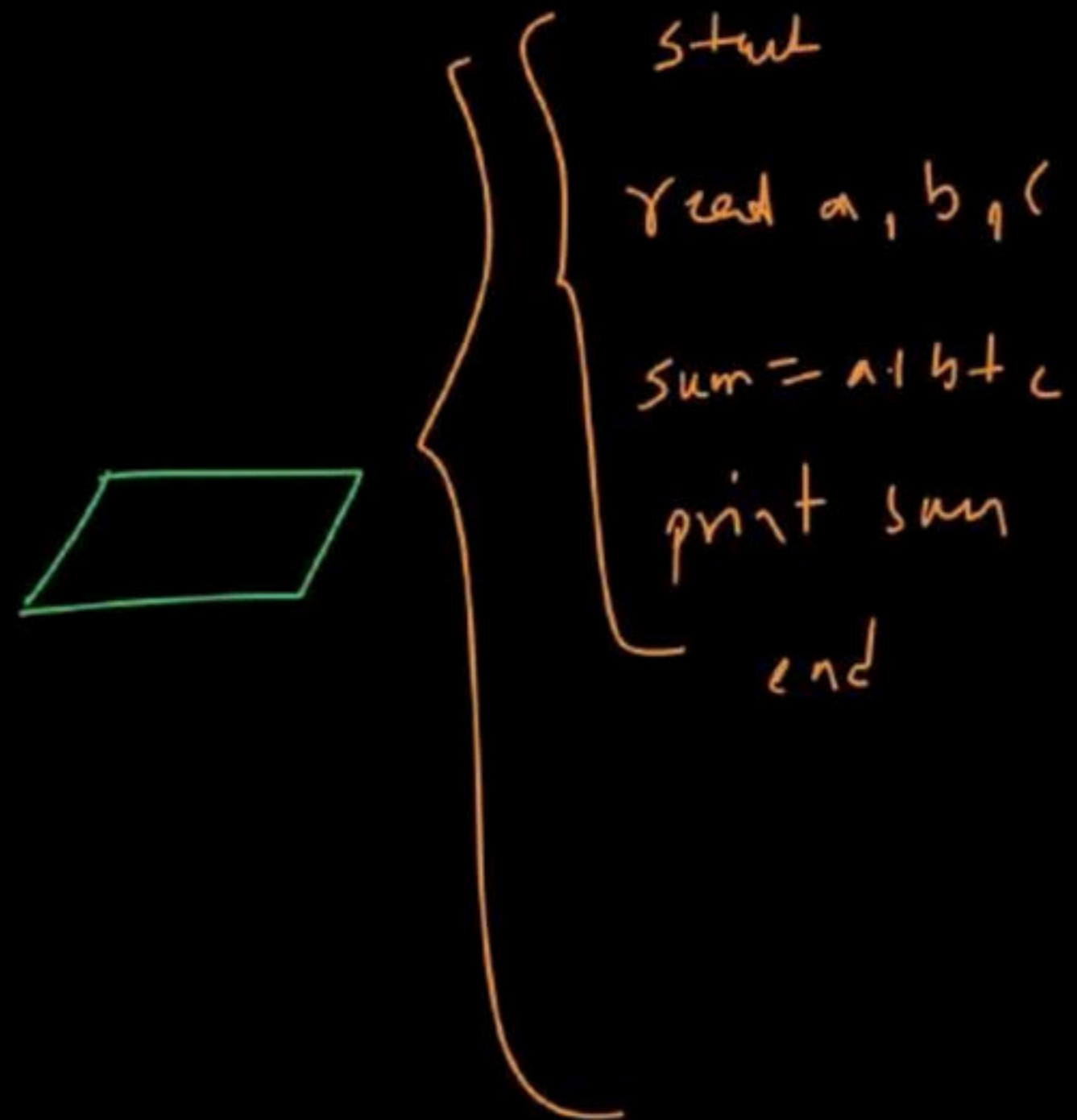
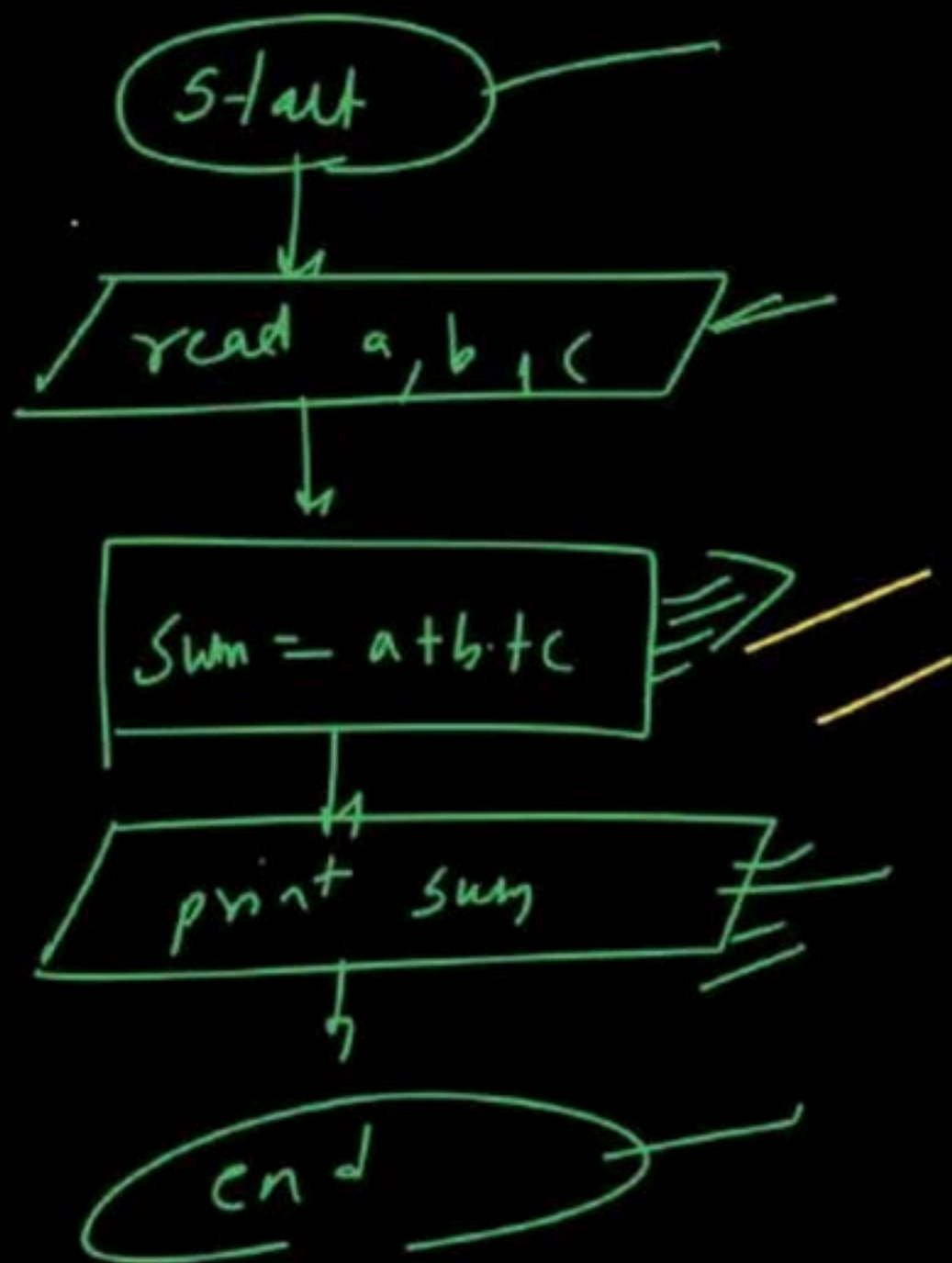
Design Flowchart - Print half of a

a/2



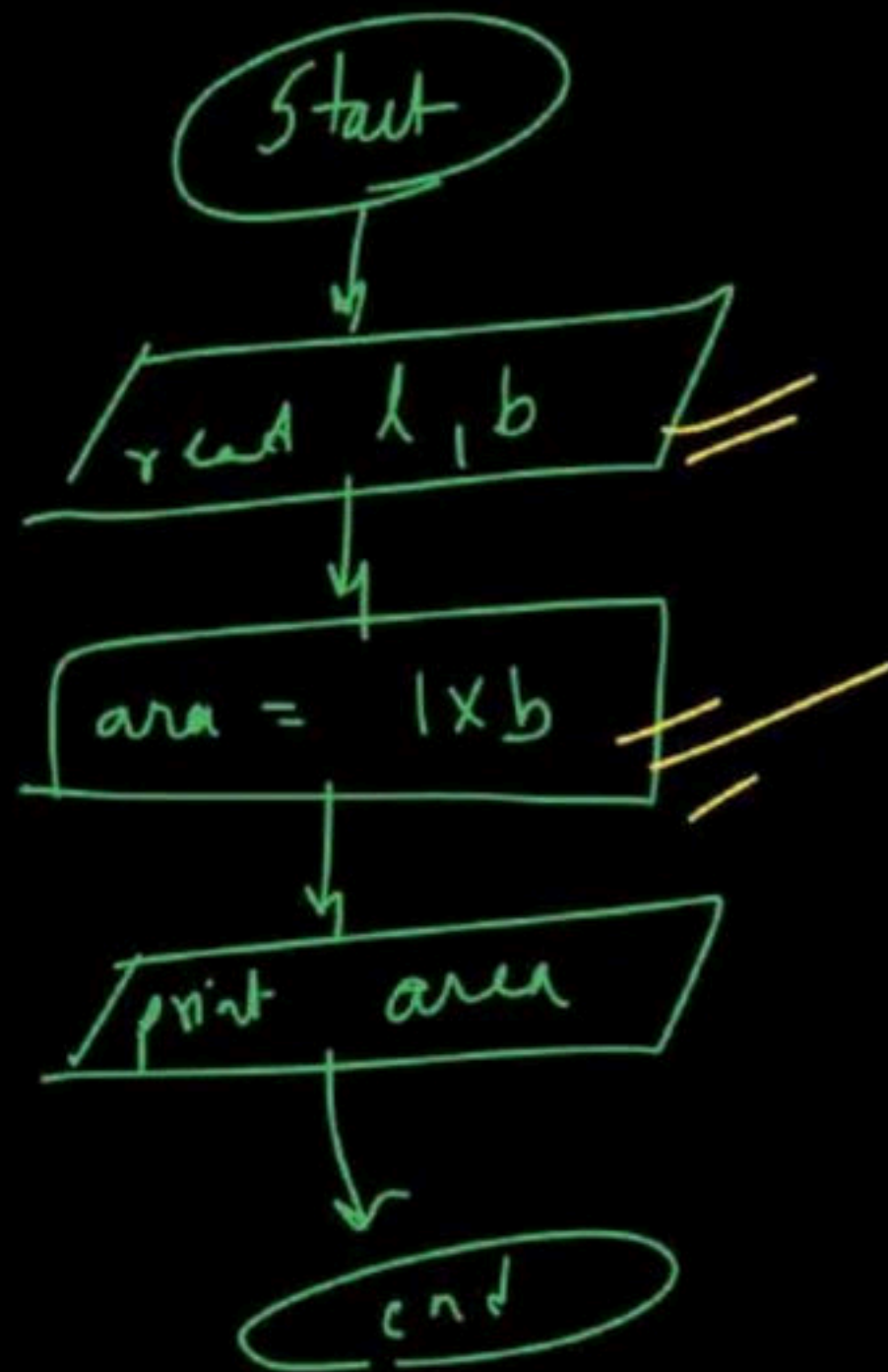
{ start
read a
half = a/2
print half
end

Design Flowchart - Take Input and add 3 numbers



Design Flowchart - Area of Rectangle

$$\text{area} = \underline{\text{len}} \times \underline{\text{br}}$$



bu
ad bnd

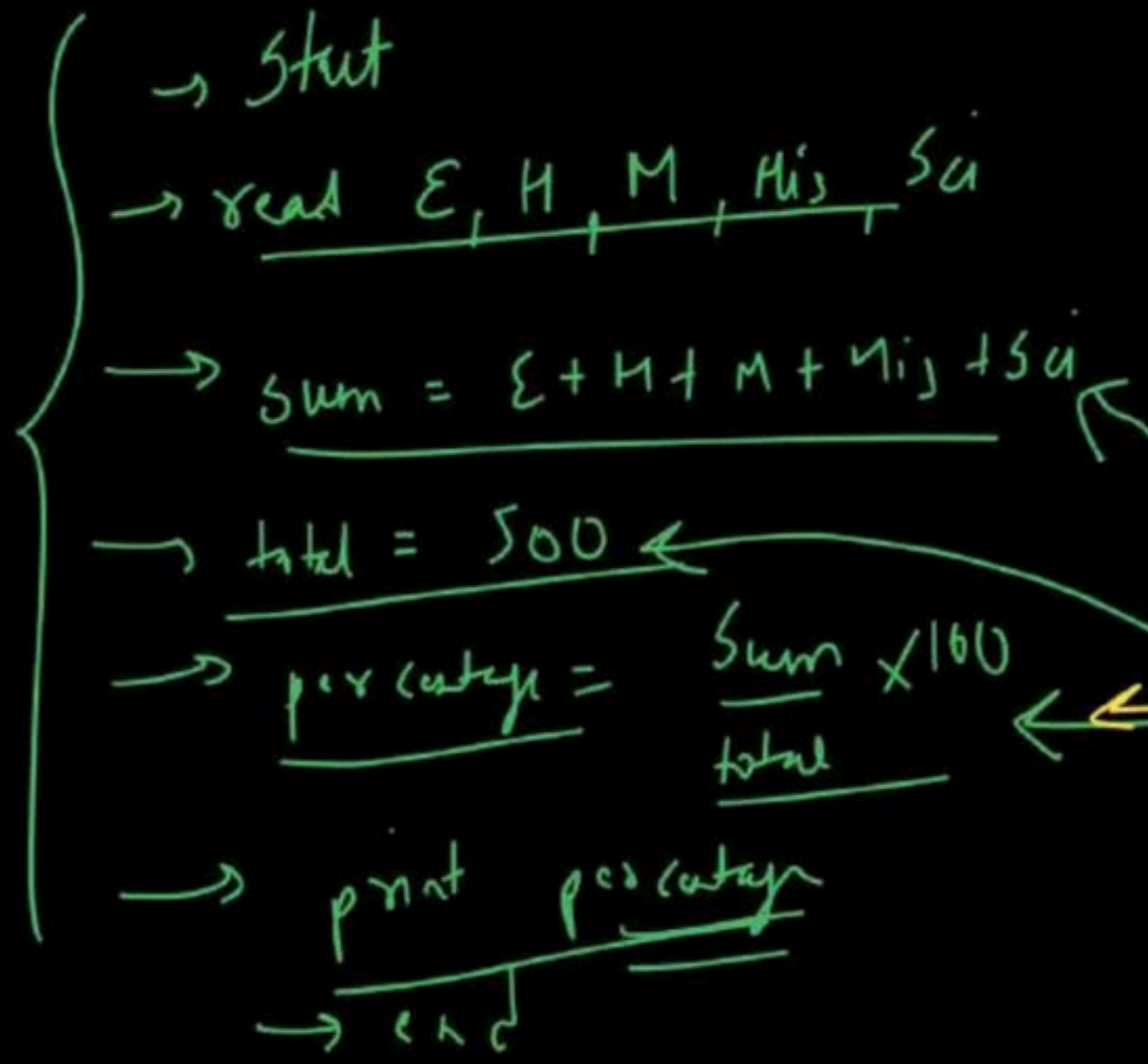
Start
read len, br
area = l x b
print area
end

2 min



Design Flowchart - Calculate Percentage

E	H	M	His	Sci
10	20	30	10	50



total
s/o :-

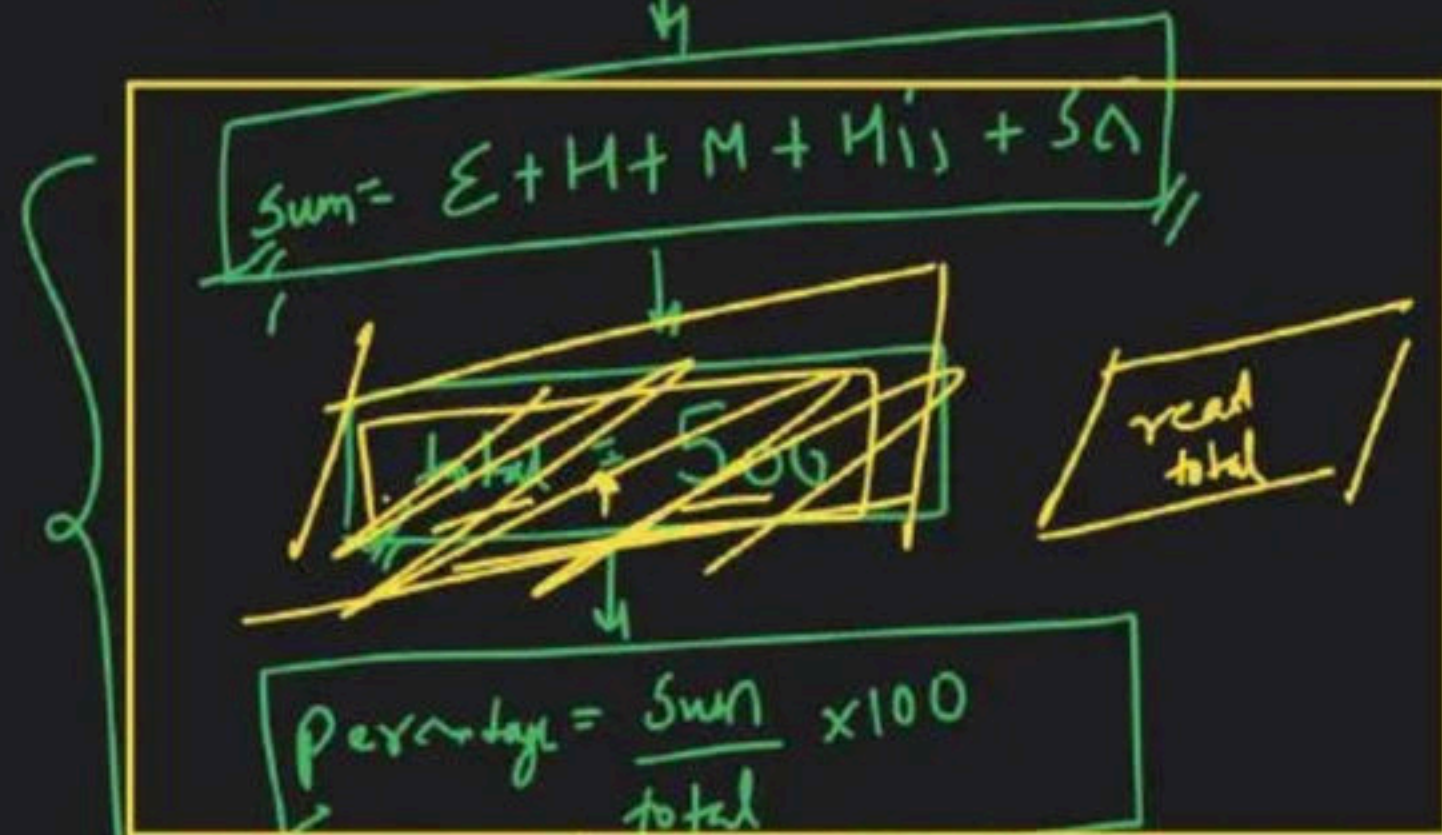
$$\frac{10 + 20 + 30 + 10 + 50}{500} \times 100$$

Sum of mark $\times 100$

total marks

start

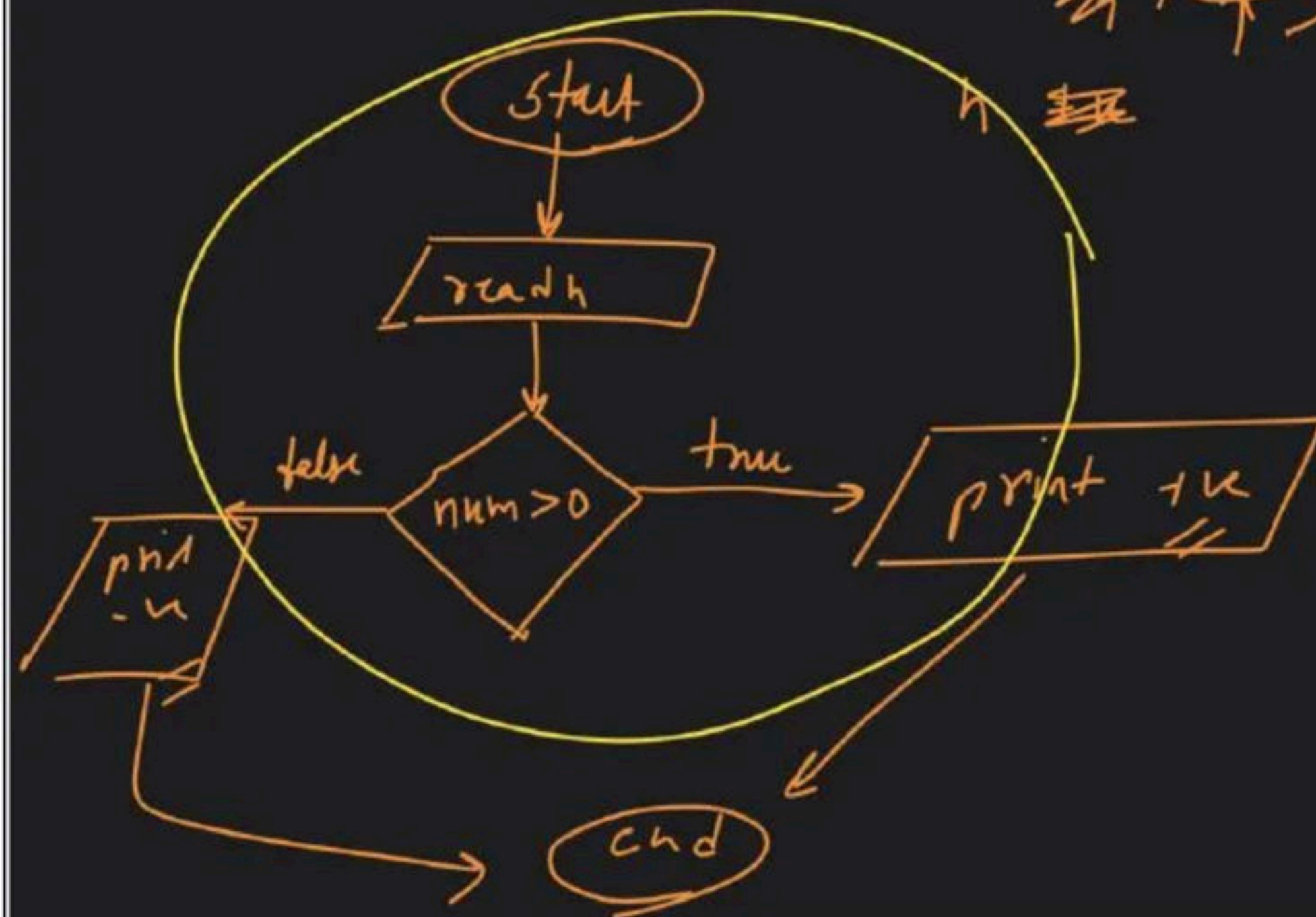
read E, H, M, Min, Su



print percentage

end

no $\rightarrow$ +ve / -ve

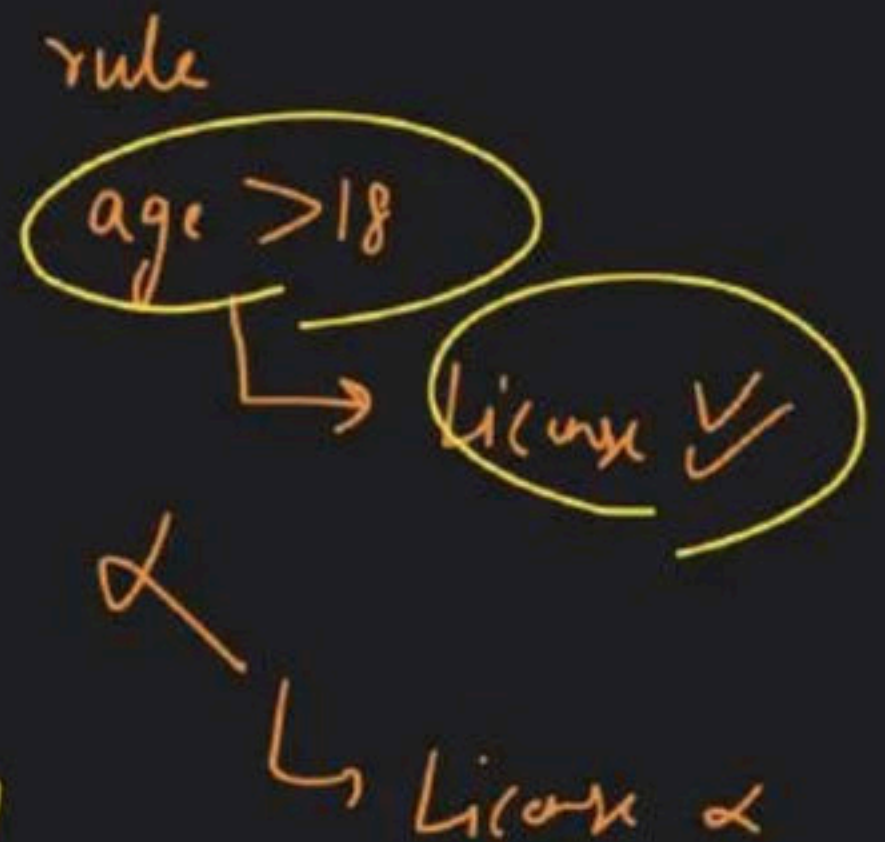
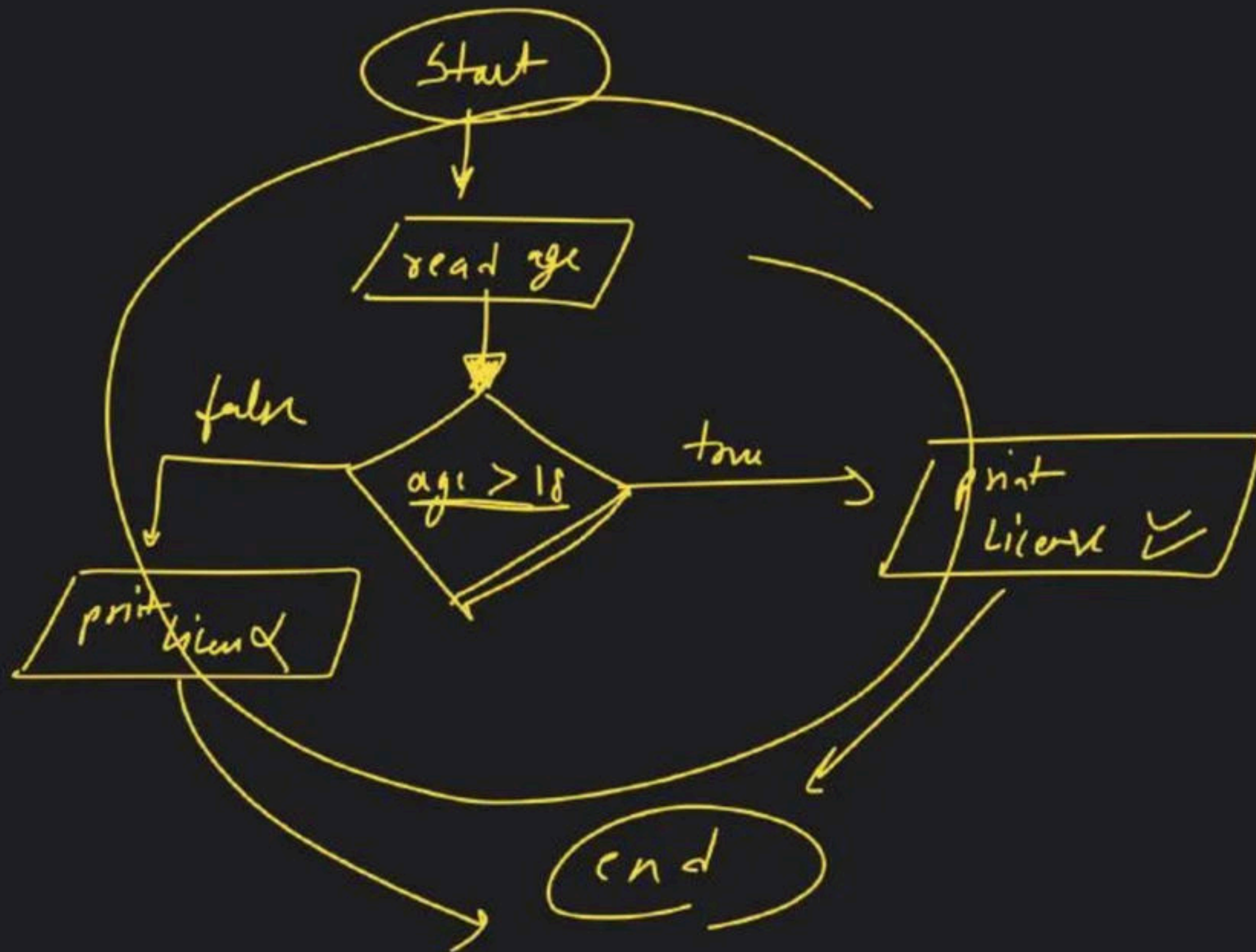


$$\begin{aligned} +ve &\rightarrow > 0 \\ -ve &\rightarrow < 0 \end{aligned}$$

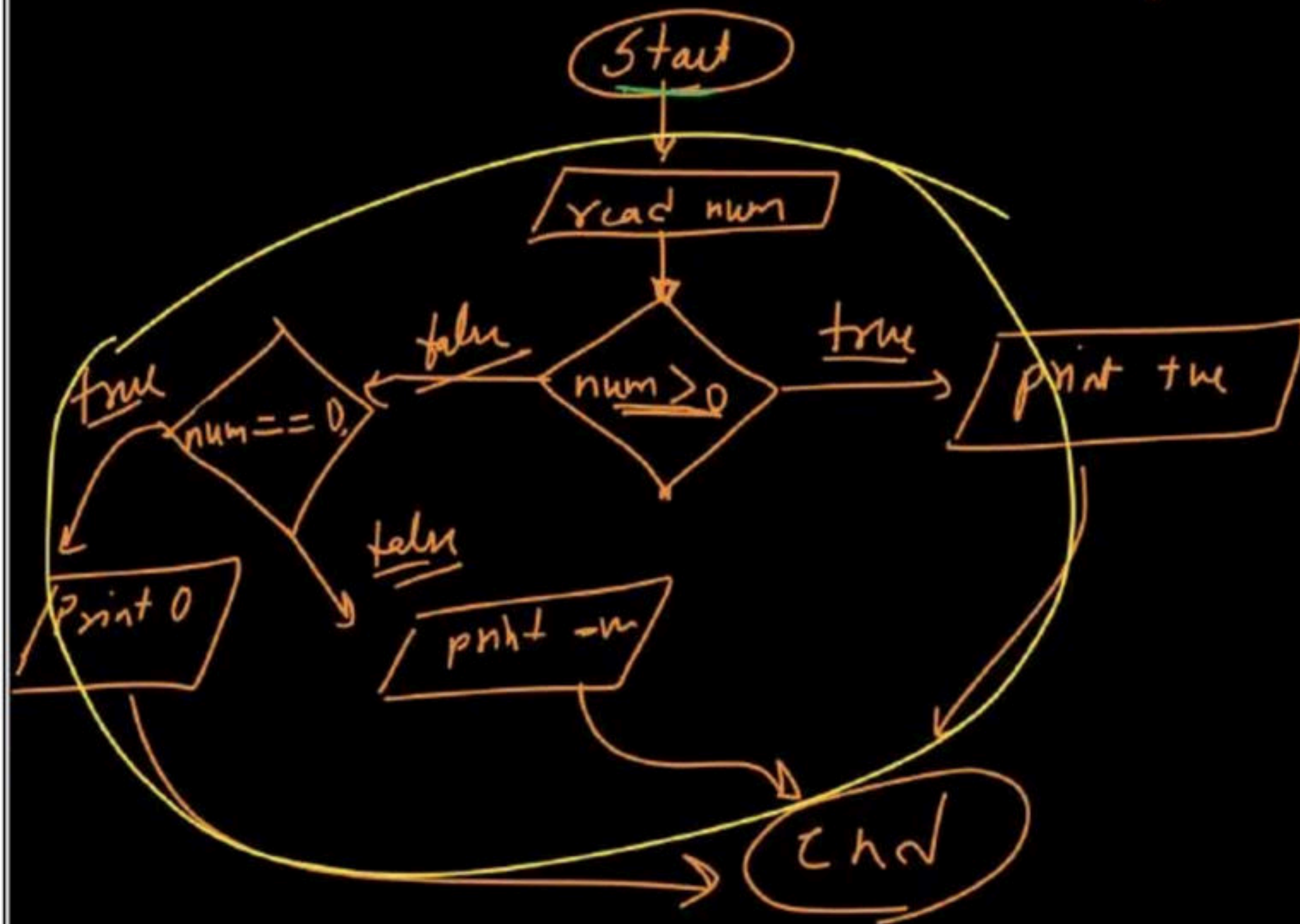
```

Start
Read num
if (num > 0) print +ve
if num < 0, print -ve
end

```



Design Flowchart - Check ~~positive~~, ~~negative~~ or ~~zero~~

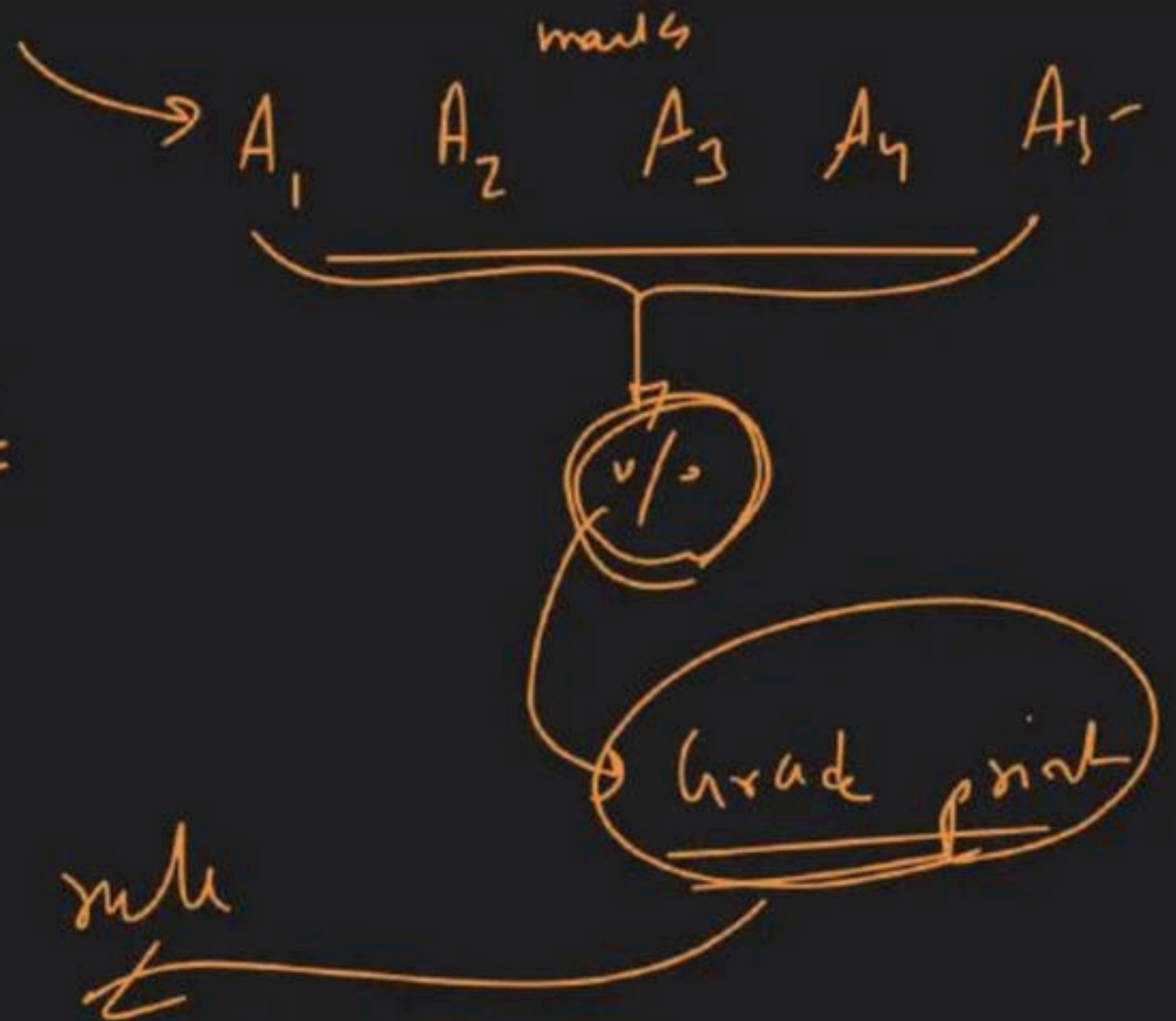


start
read num
if $num > 0$, print +ve
else if $num = 0$, print 0
else print -ve
end

H/w

20 min

$\left\{ \begin{array}{l} A \rightarrow > 90\% \\ B \rightarrow 80 - 90 \\ C \rightarrow 70 - 80 \\ \text{fail} \rightarrow < 70 \end{array} \right.$

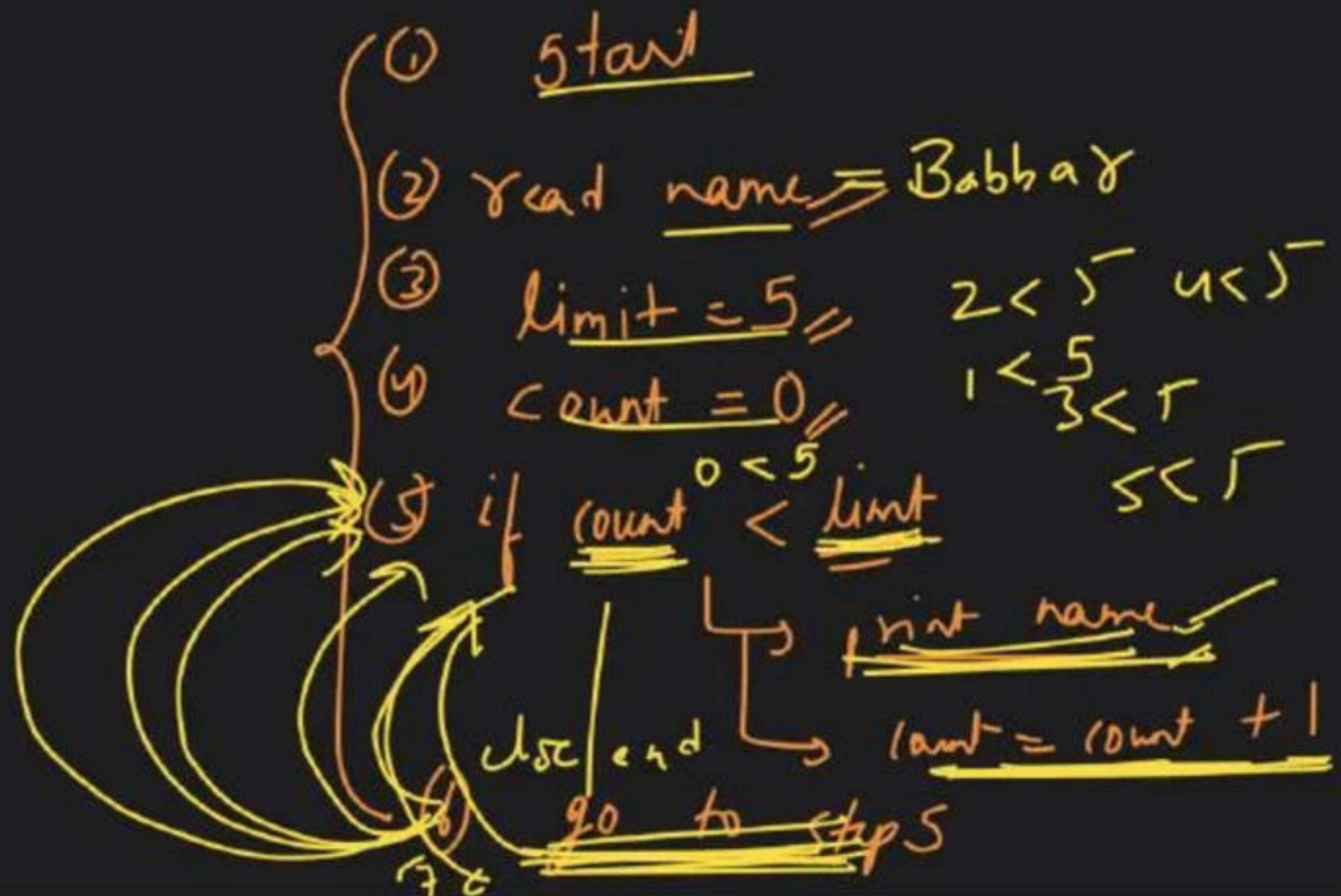


count = ~~0~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5

limit = 5

Name → 5 print

Babbar
Babbar
Babbar
Babbar
Babbar



① Start

② read name

③ $\text{limit} = 5$

④ ~~$\text{count} = 0$~~

⑤ if $\text{count} < \text{limit}$

→ print name

→ $\text{count} = \text{count} + 1$

else go to step 7

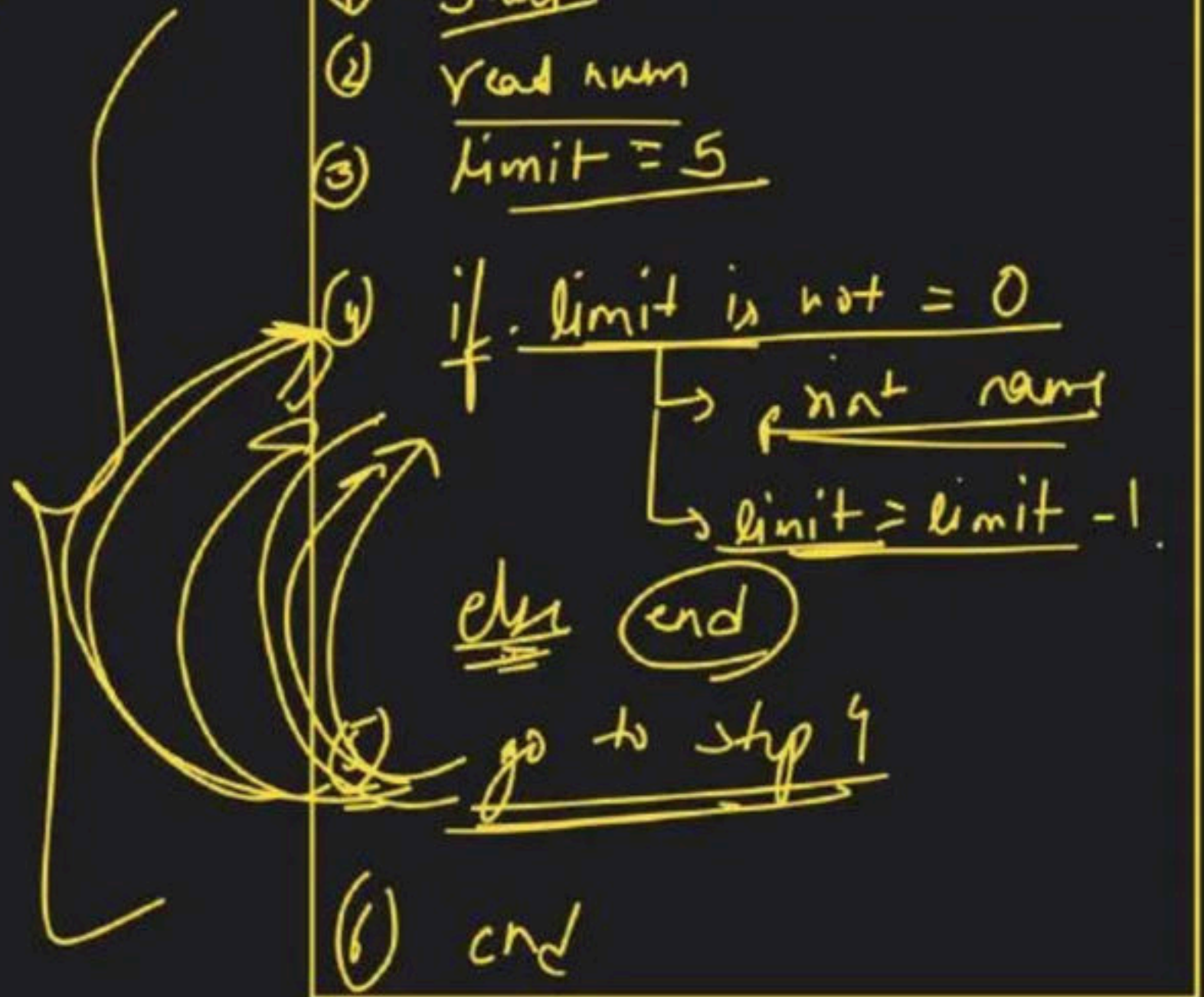
⑥ go to step 5

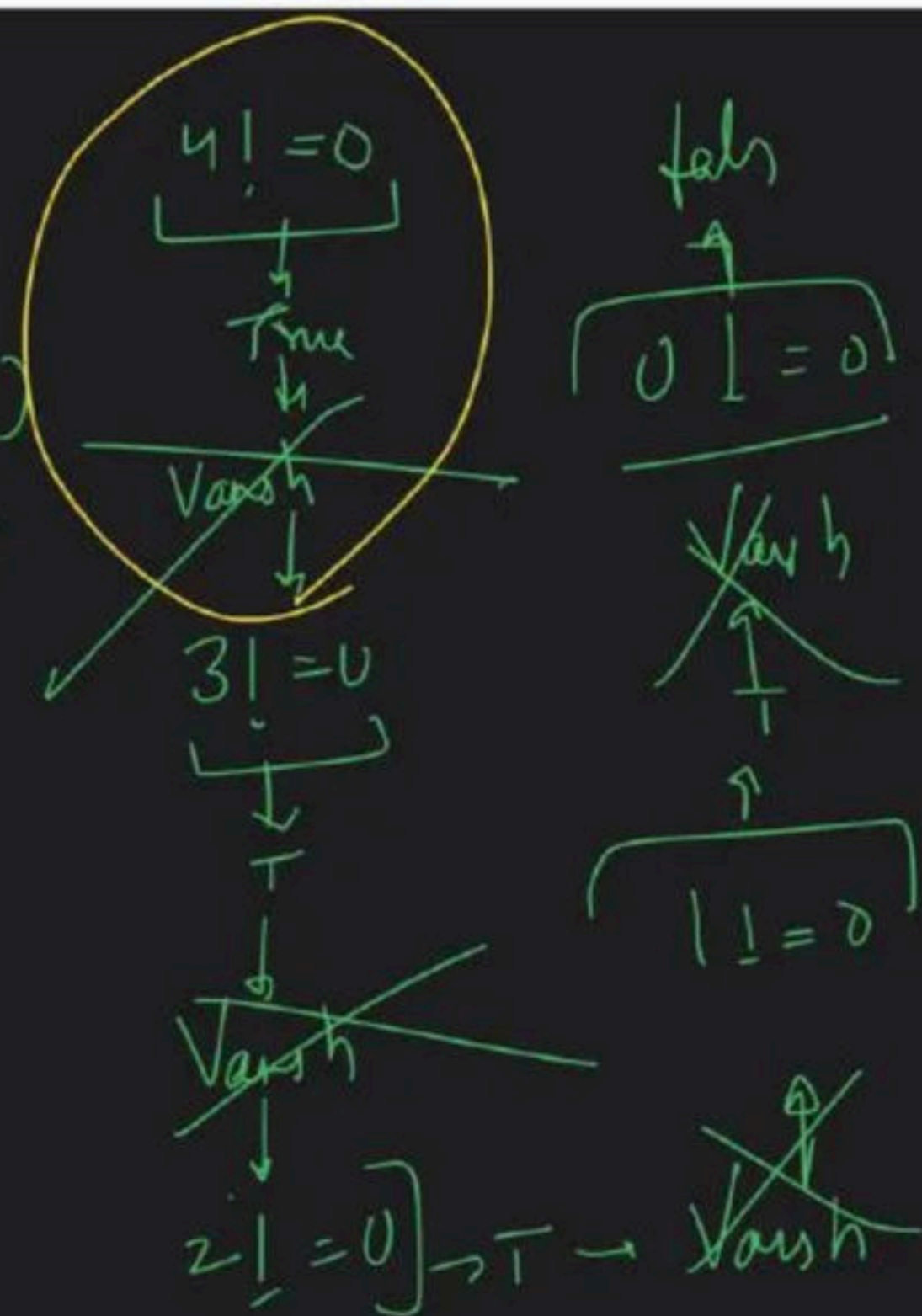
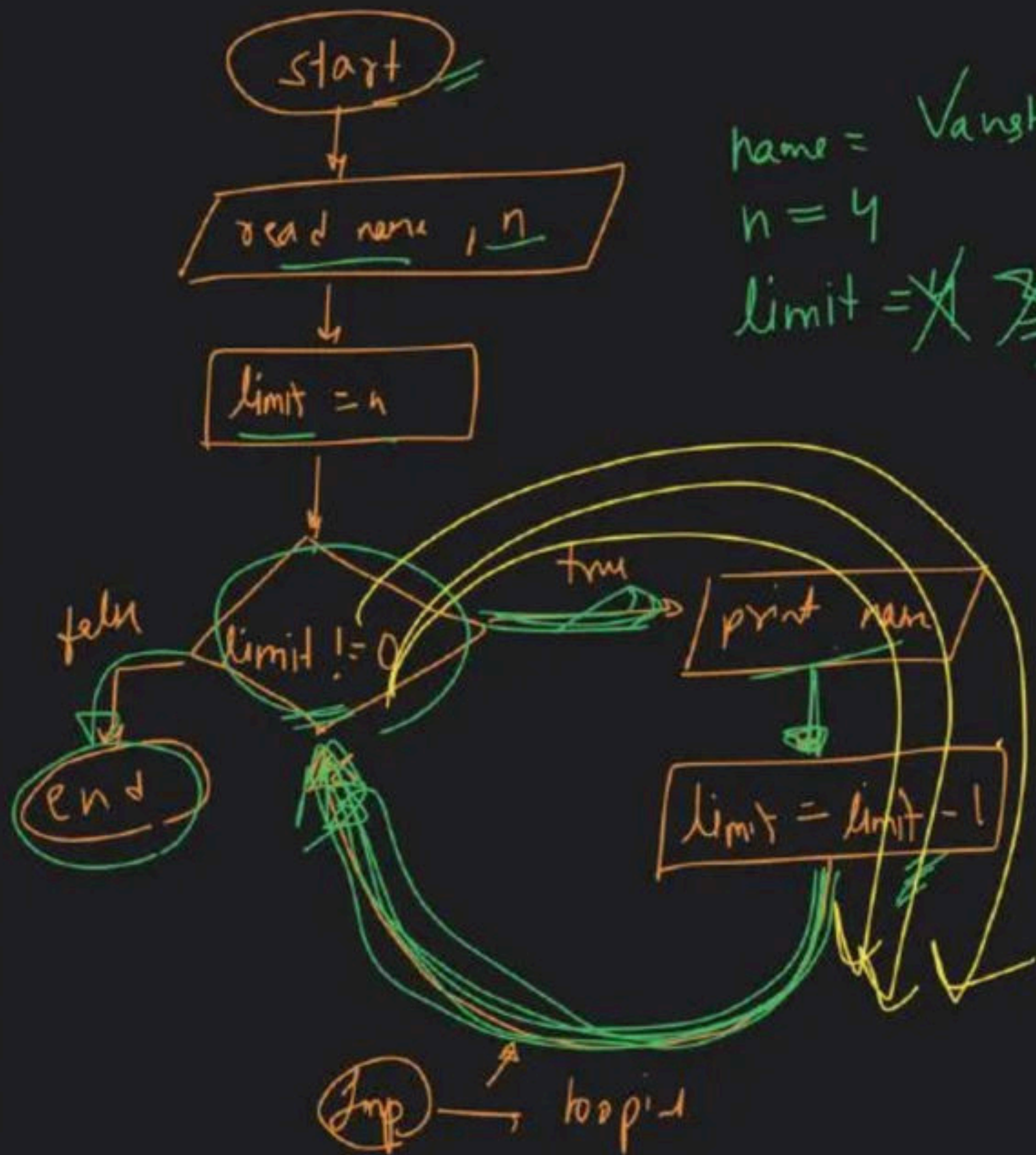
⑦ end



name = Babbar
limit = ~~4~~ ~~3~~ ~~2~~ ~~1~~ 0

Babbar
Babbar
Babbar
Babbar
Babbar

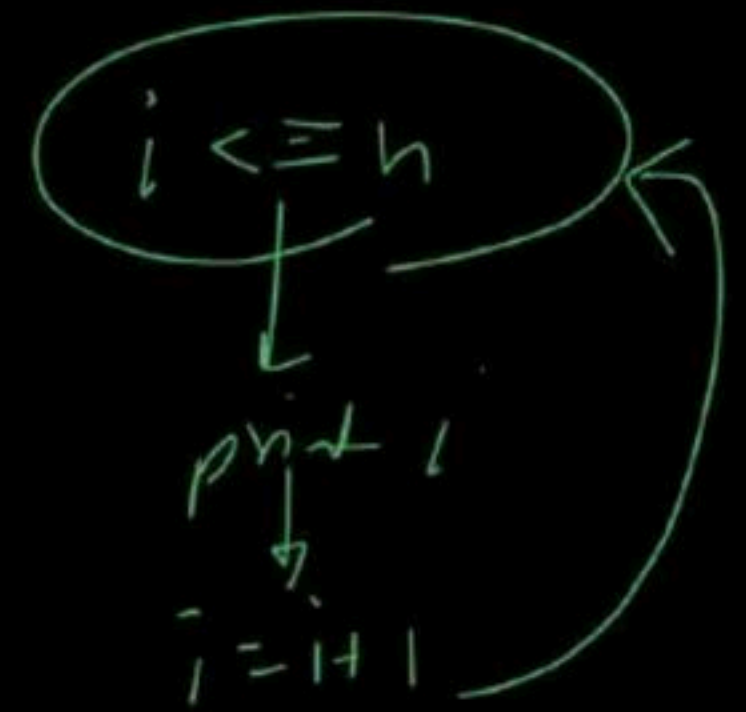
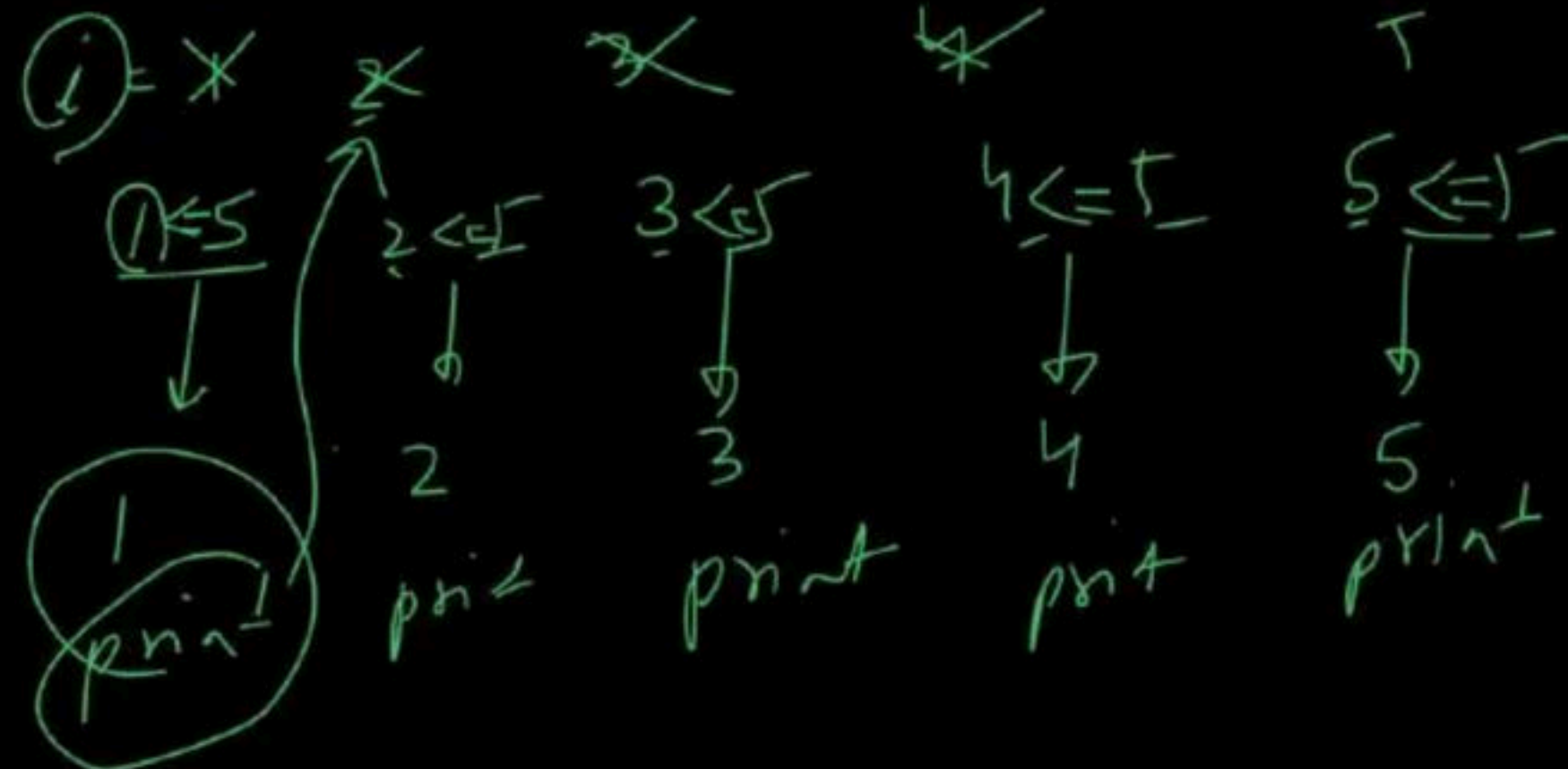




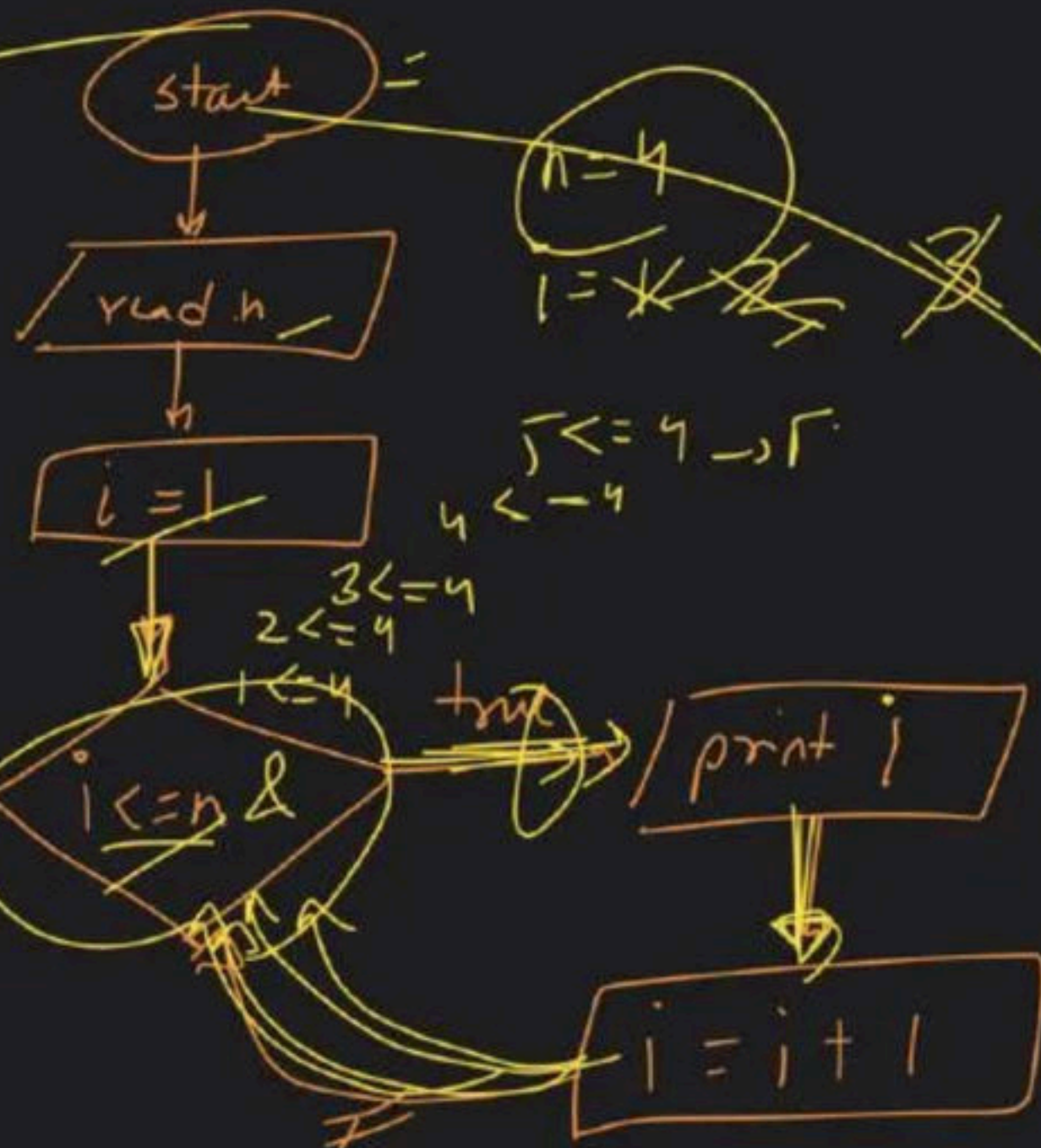
Design Flowchart - Print counting from 1 to N

Loops

$n = 5$



dry
run



$n = 4$

~~$i = 1$~~

~~$i = 2$~~

~~$i = 3$~~

~~$i = 4$~~

~~$i = 5$~~

~~$i = 6$~~

~~$i = 7$~~

~~$i = 8$~~

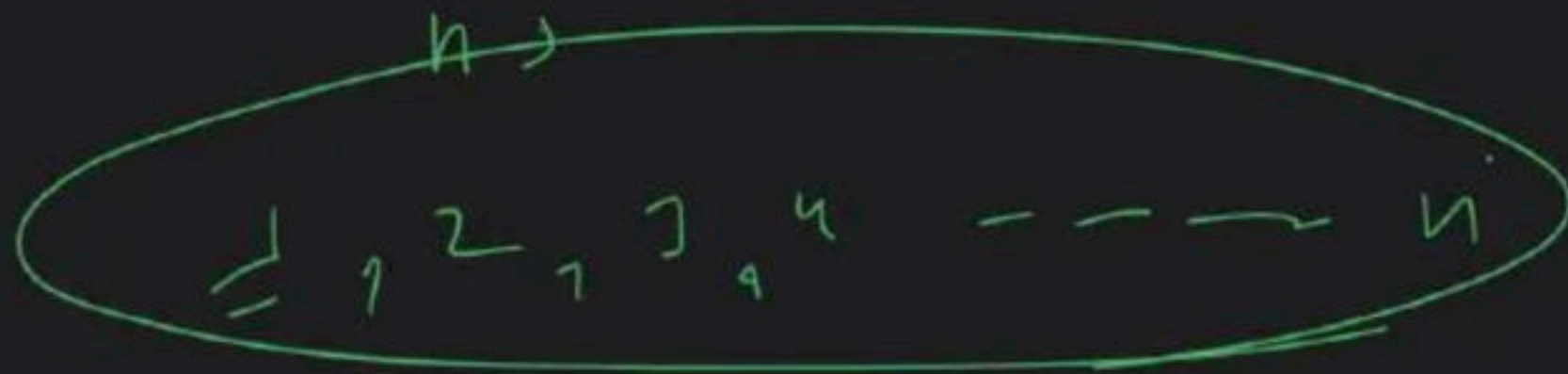
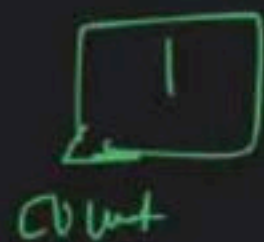
~~$i = 9$~~

~~$i = 10$~~

Learn (+)

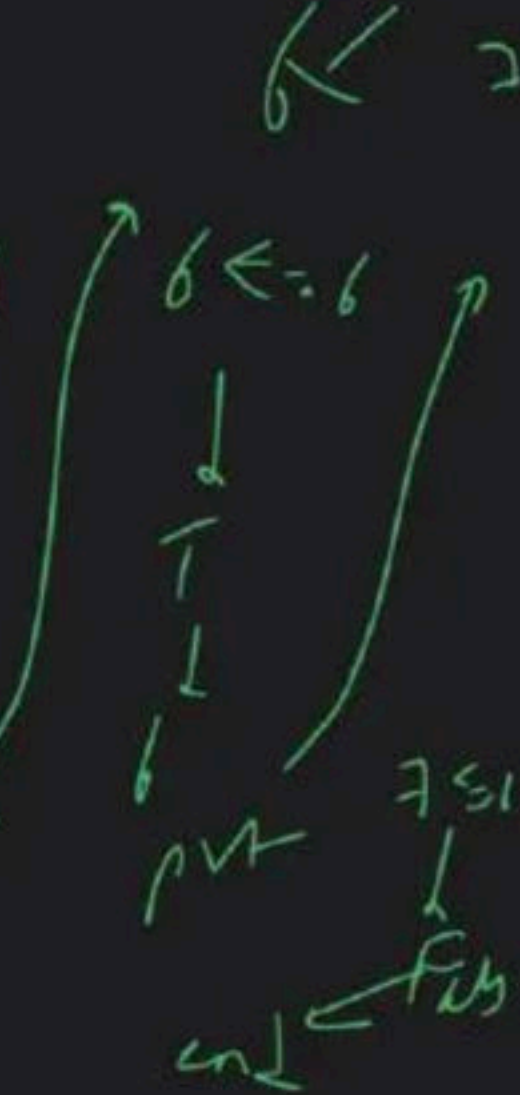
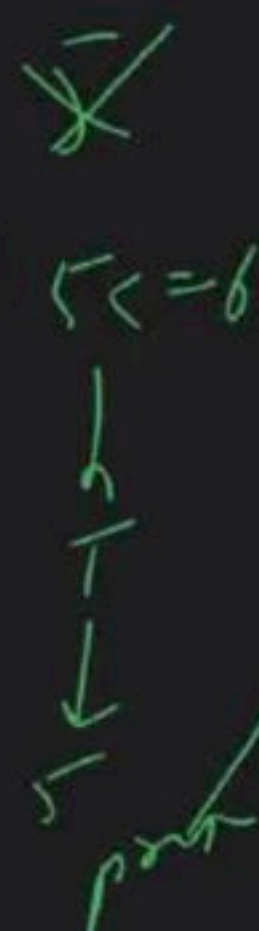
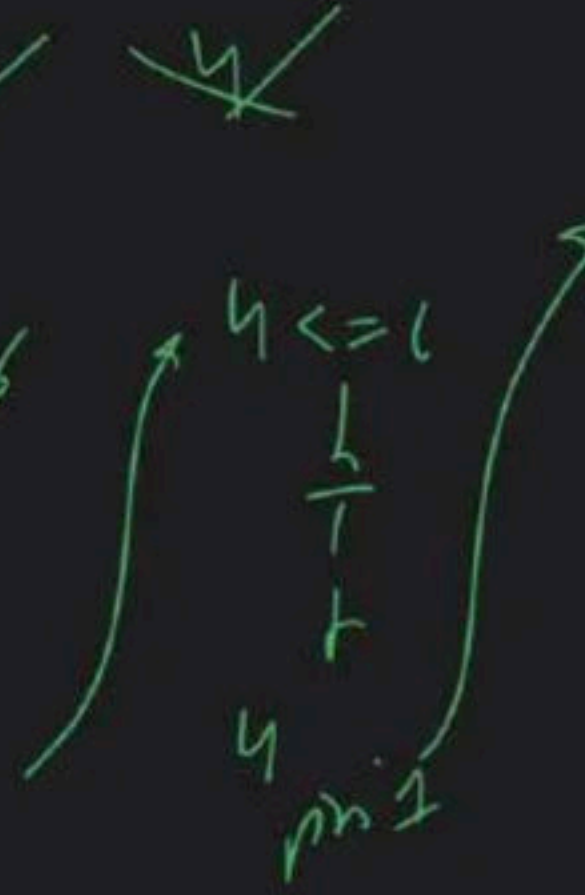
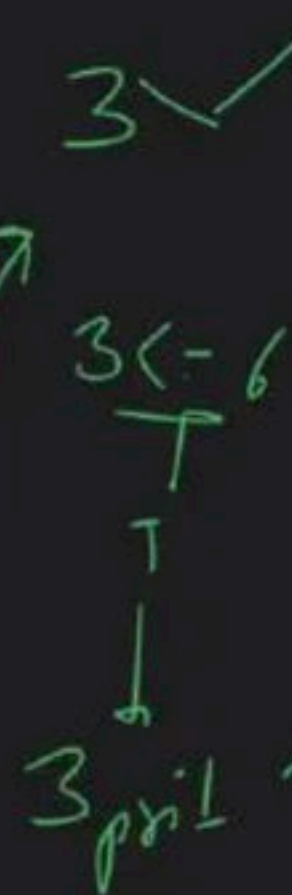
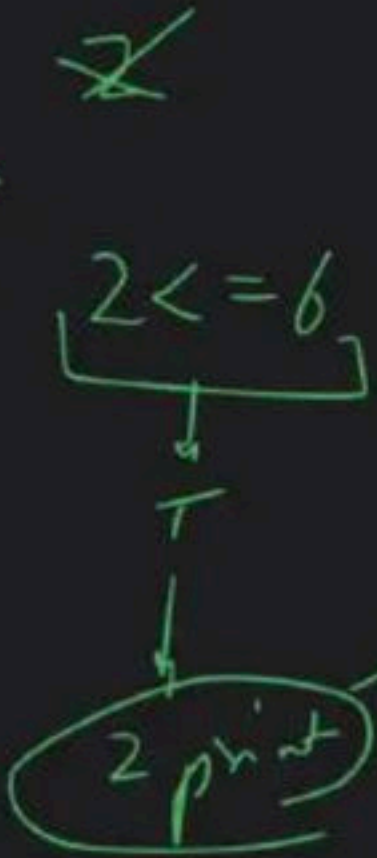
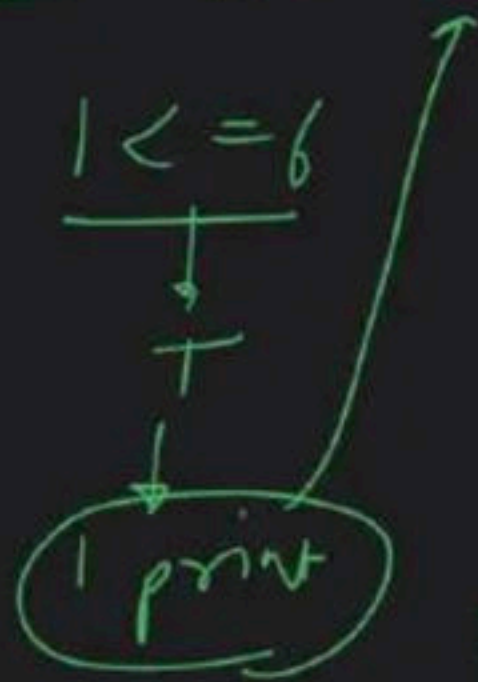
for loop

1
2
3
4

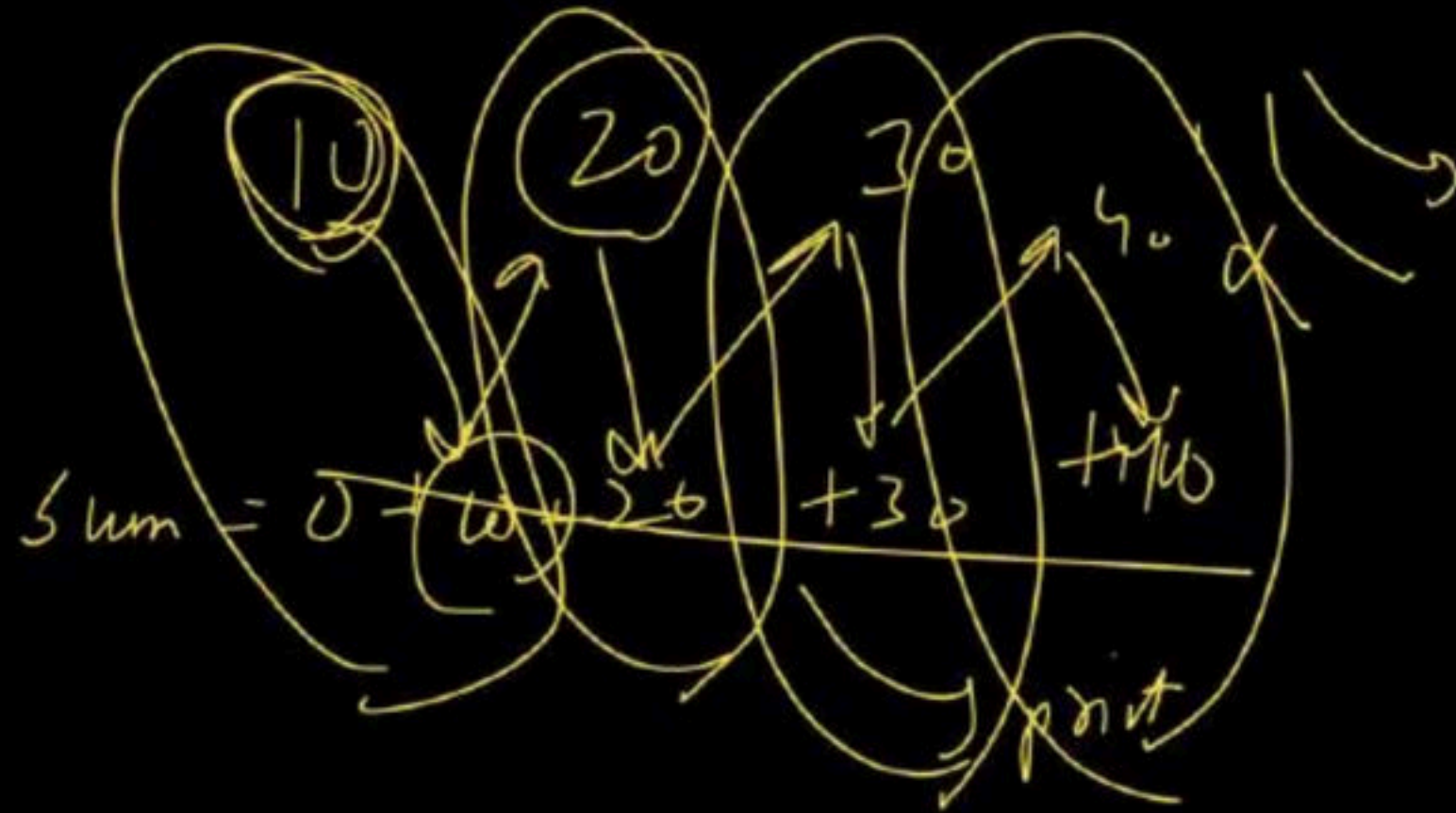


$n = 6$

count = ~~1~~



Design Flowchart - Add N numbers from user



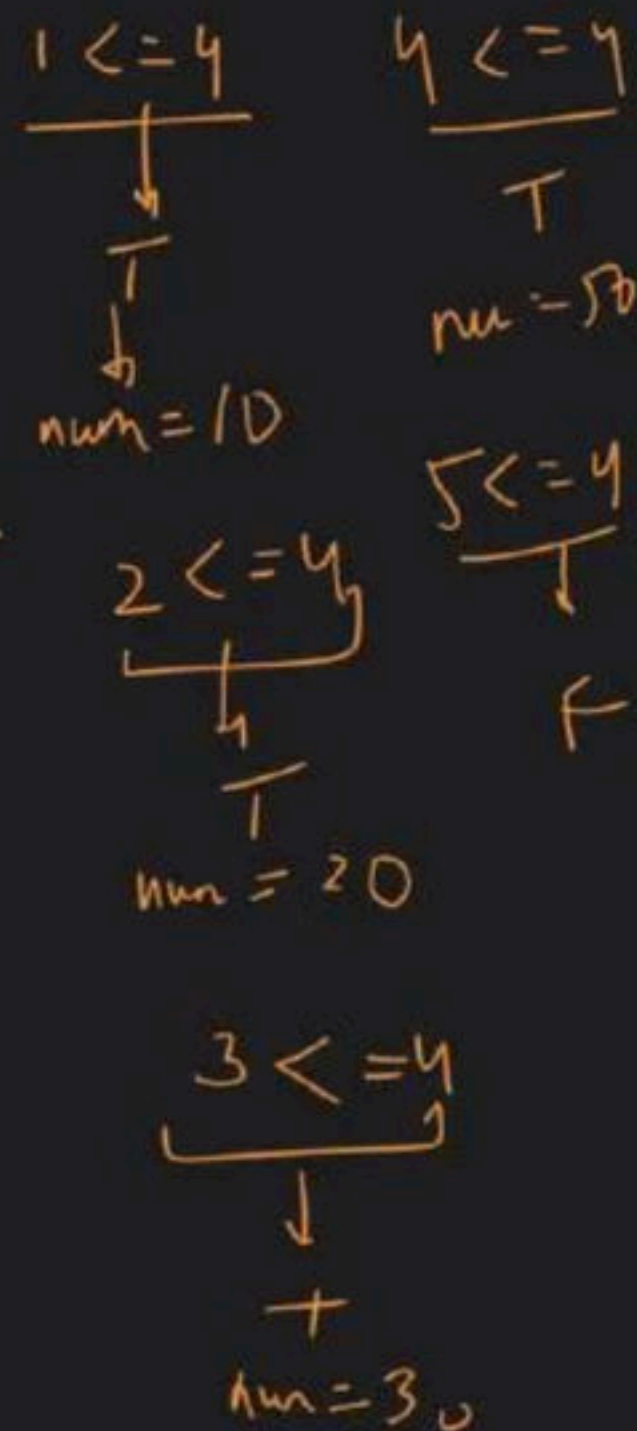
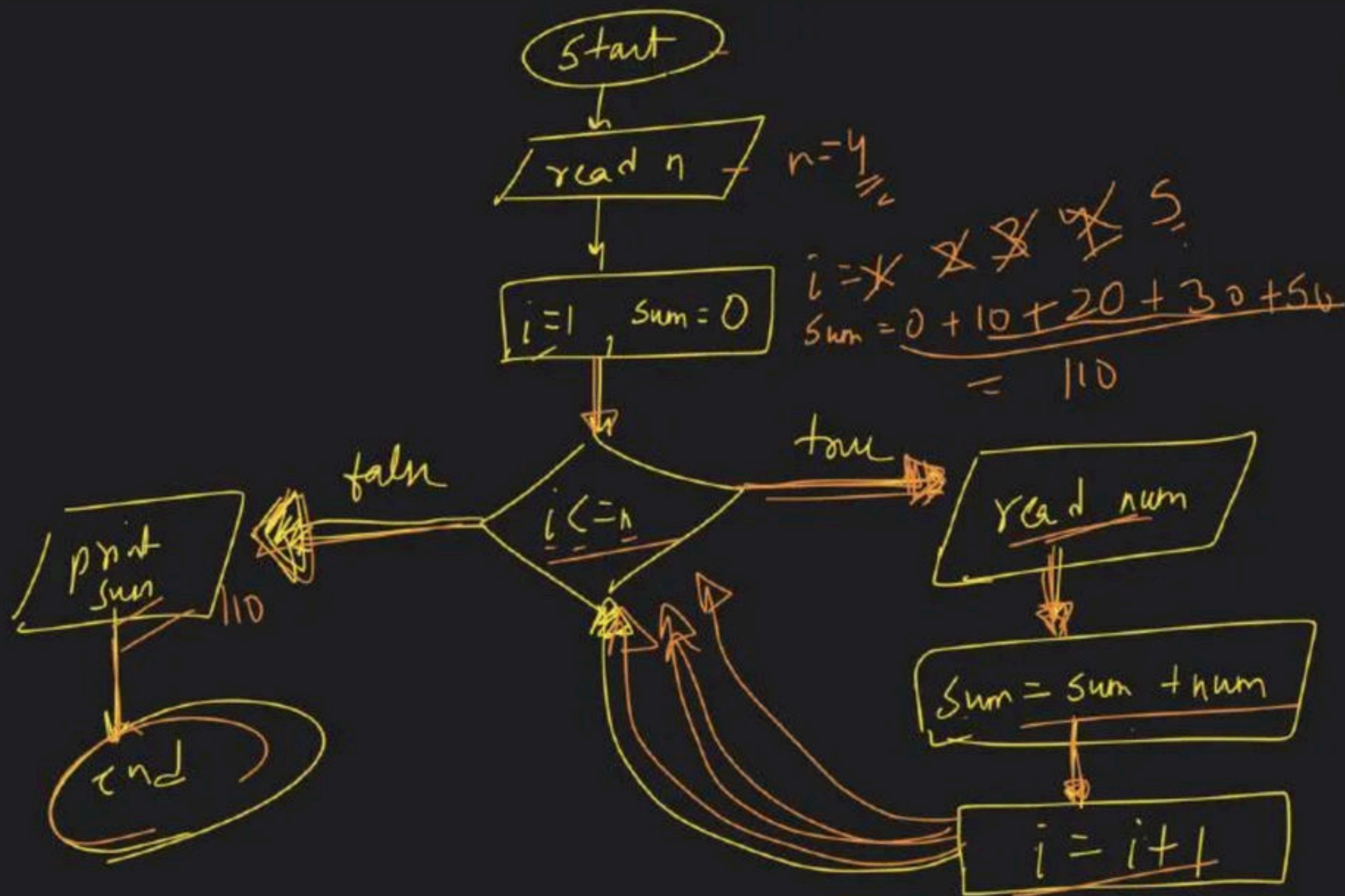
1 → Start

2 → Read n

3 → $i = 1$, sum = 0

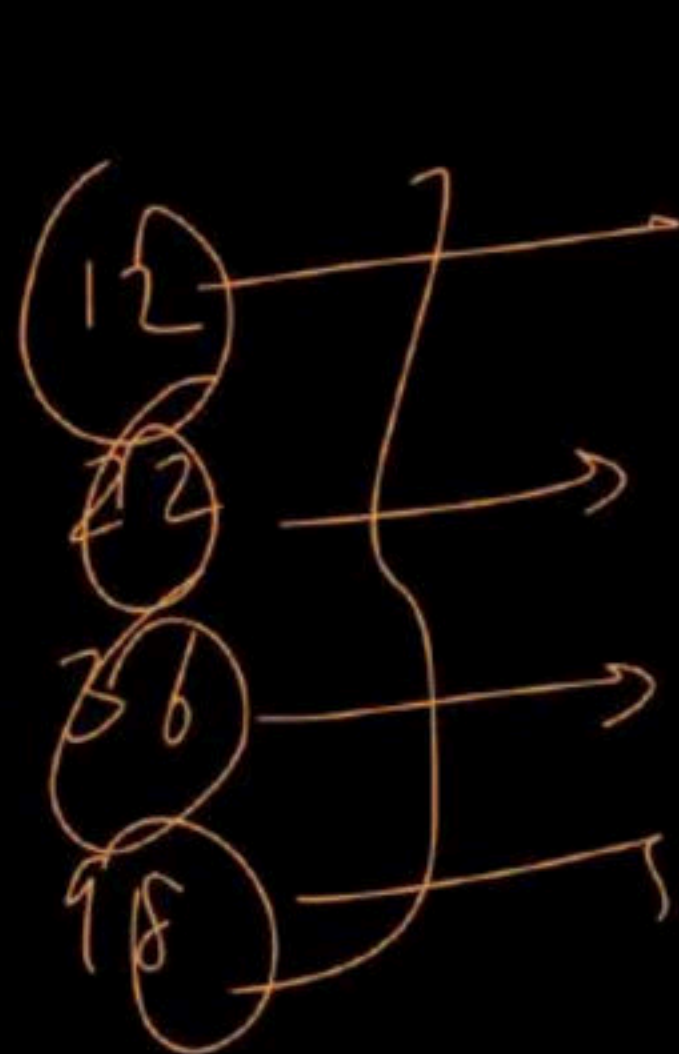
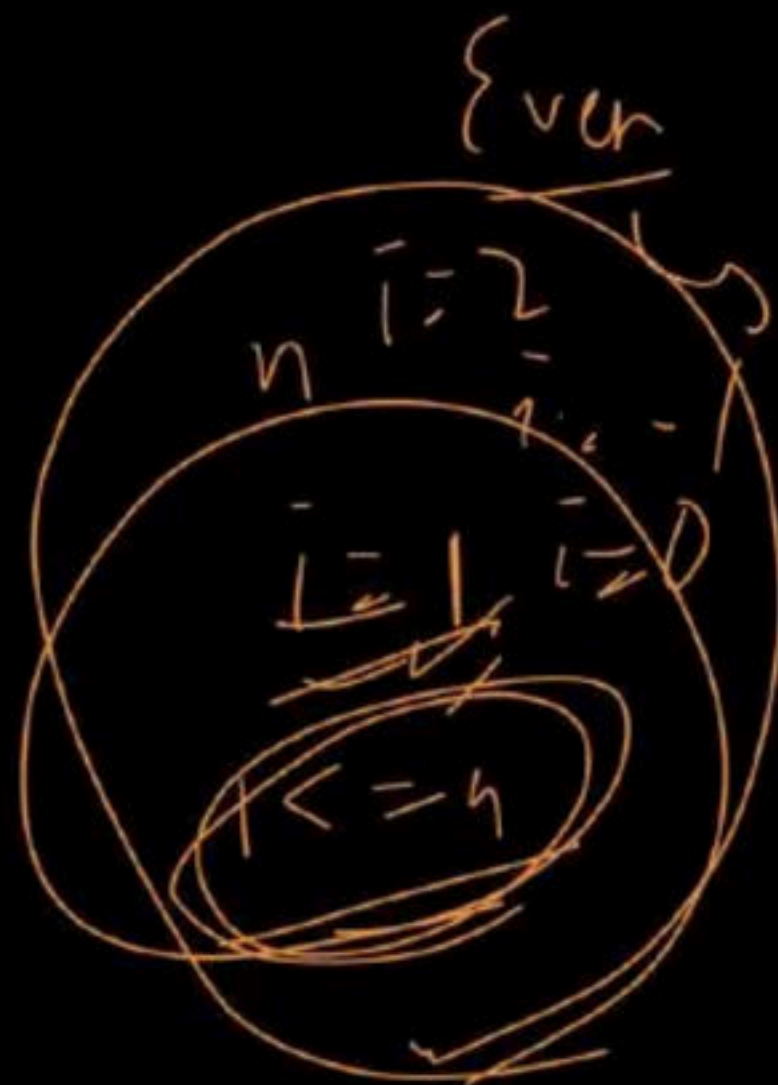
4 → if $i \leq n$

- read num
- $sum = sum + num$
- $i = i + 1$
- go to 4

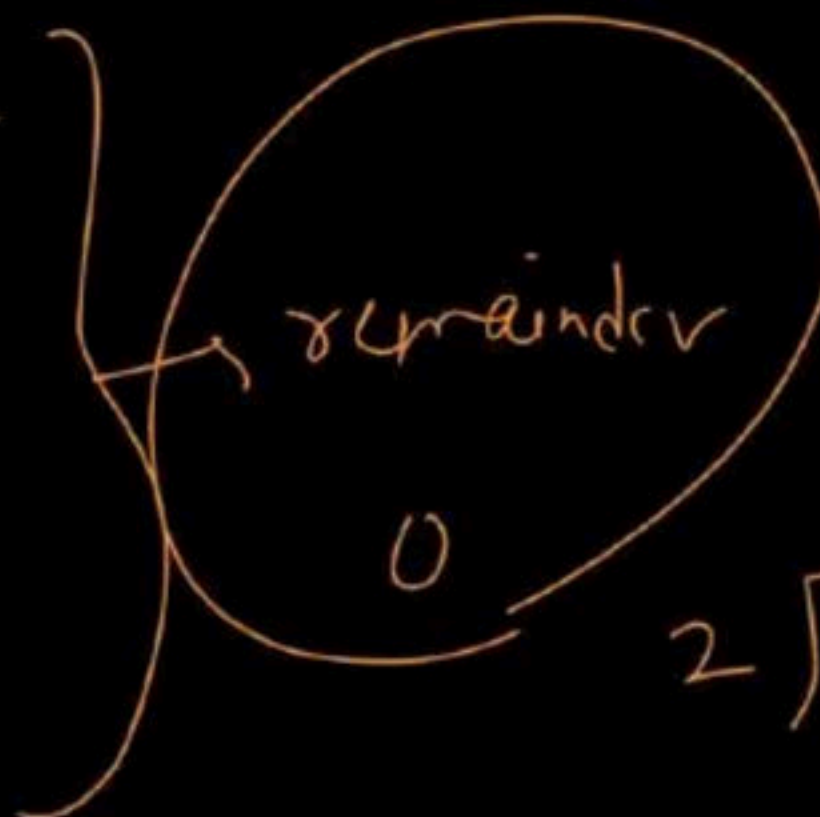


Design Flowchart - Print Even numbers from 1 to N

$$i = 0$$



2
div



$$2 \overline{) 12}$$
$$\underline{12}$$

remainder $\rightarrow 0$

$n = 10$

$i = 1$



$1 / 2$



odd

2



$2 / 2$



Even



P

3



$3 / 2$



odd

α

4



$4 / 2$



Even

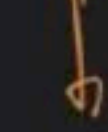


P

5



$5 / 2$



odd

α

6



$6 / 2$



Even



P

7



$7 / 2$



odd

α

8



$8 / 2$



Even

α

9



$9 / 2$



Σ

α

10



$10 / 2$



Even

α

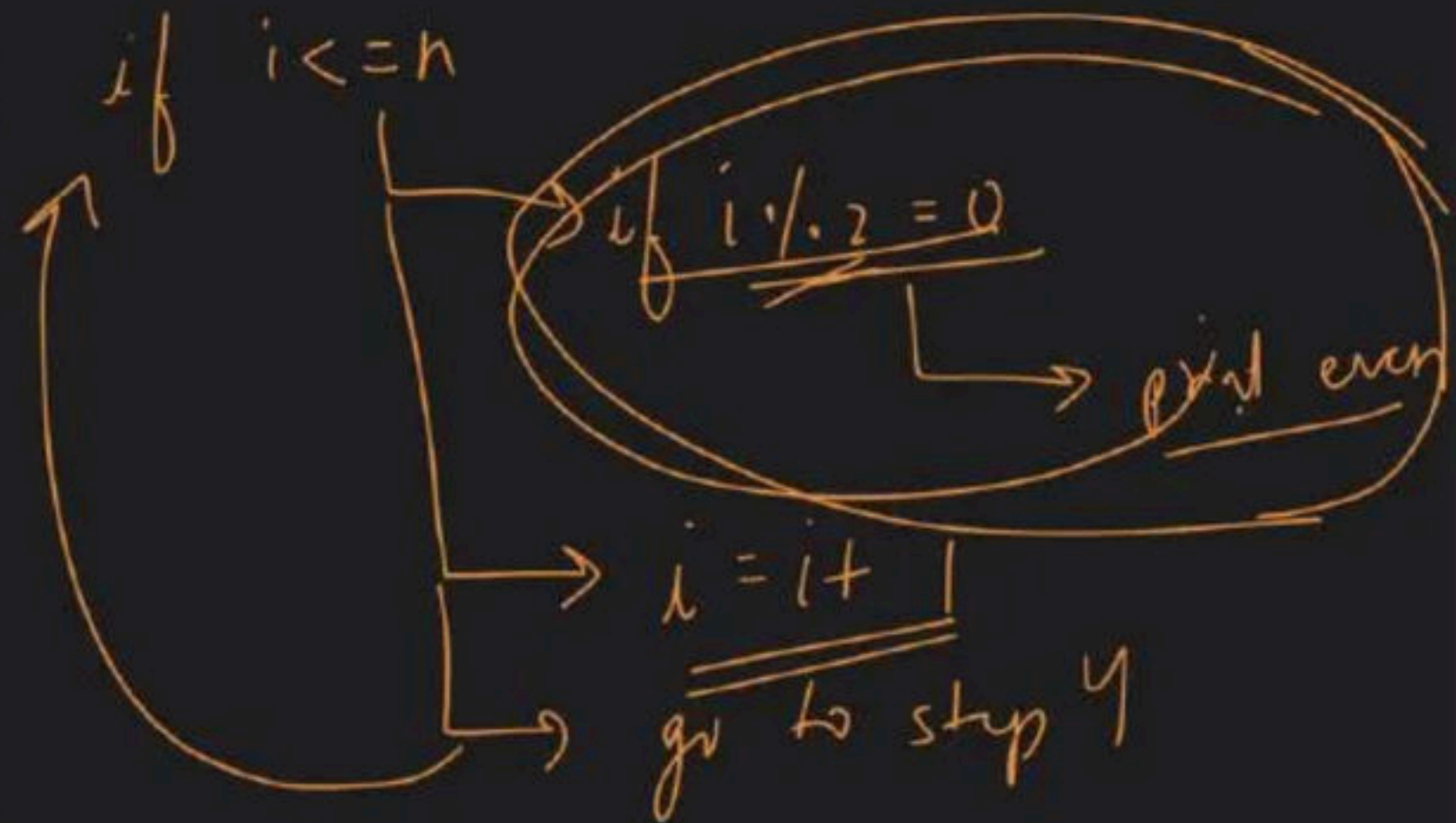
$\alpha / 2$



Σ

α

- (1) Start
- (2) read n
- (3) $i = 1$
- (4) if $i \leq n$



end

$n=7$
 $i=1$

$2 \leq 7$

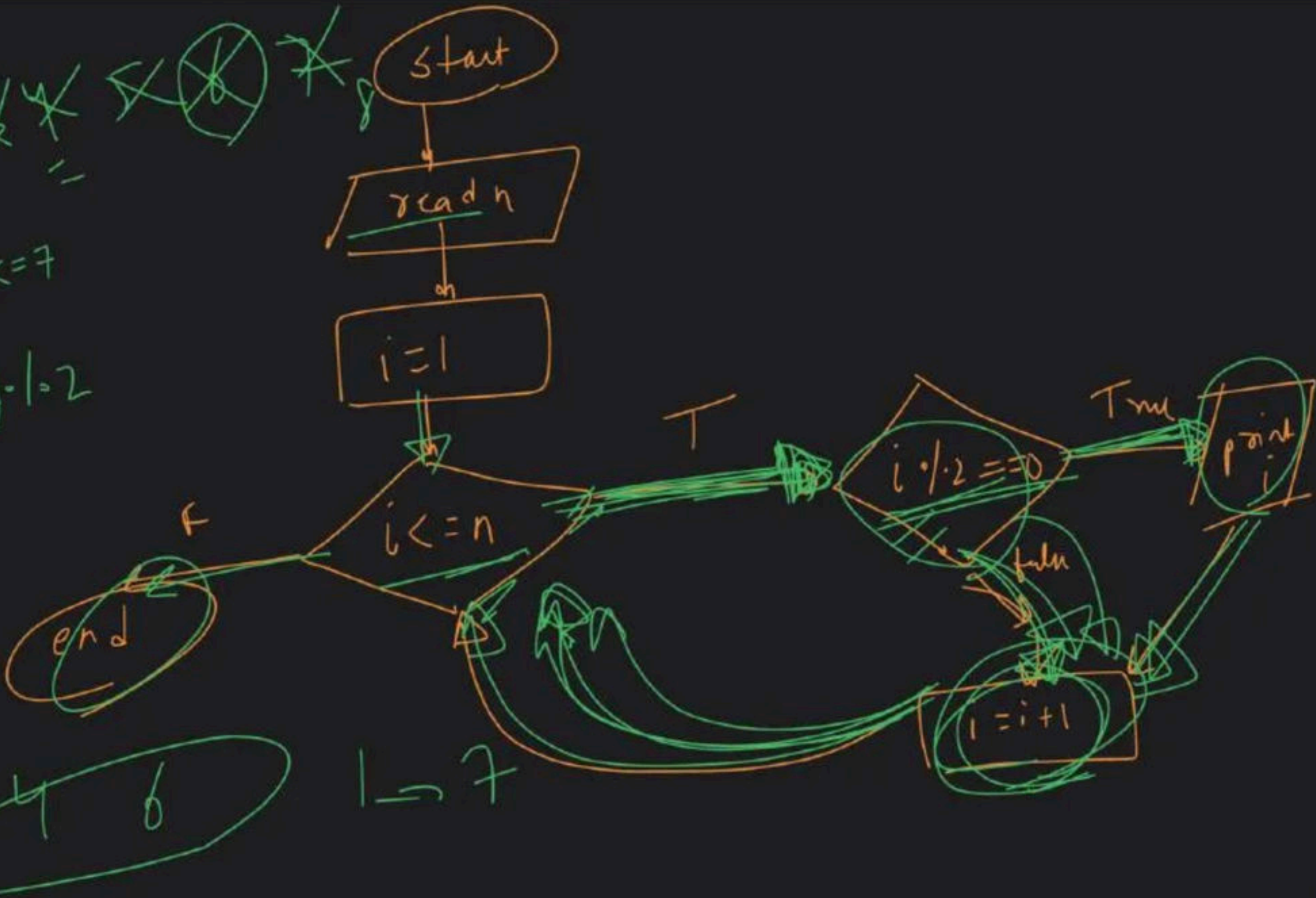
$(2-1)=2$

$1 \leq 7$

$1 \div 2 = 0$
odd

2 4 6

1 7



$$\underline{12} \div \underline{2} = \underline{6}$$

Modulo

Modulus

remainder

$$12 \div 2 = 0$$

↑
rem

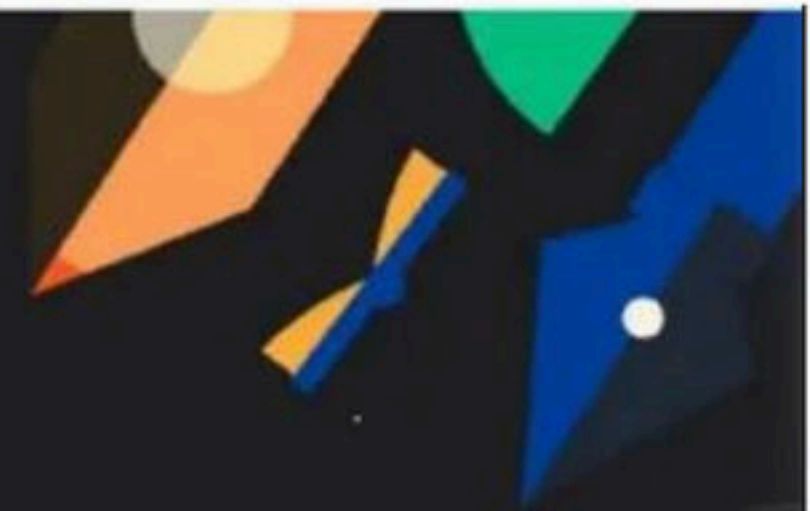
$$\text{num} \div 2$$

$$\begin{aligned} \rightarrow 0 &\rightarrow \underline{\text{Even}} \\ \rightarrow !0 &\rightarrow \underline{\text{Odd}} \end{aligned}$$

Assignment:

Design Flowchart for below:

- • Multiply 2 number after taking input
 - • Perimeter of triangle
 - • Find Simple Interest
 - • Find Compound Interest
 - • Print Counting from N to 1
 - • Find Factorial of a Number
 - • Check if a number is Prime or not
 - • Check Valid Triangle or not
 - • Print Max of 2 number
- 



L2-Basics of Programming

Special class

L2 - Basics of Programming

- Variables and Operators**
- Write your 1st Program**

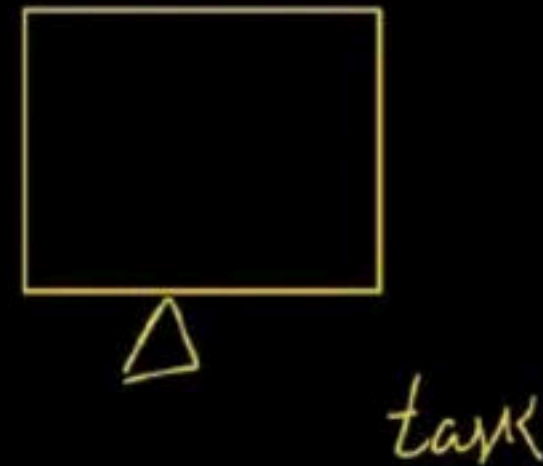
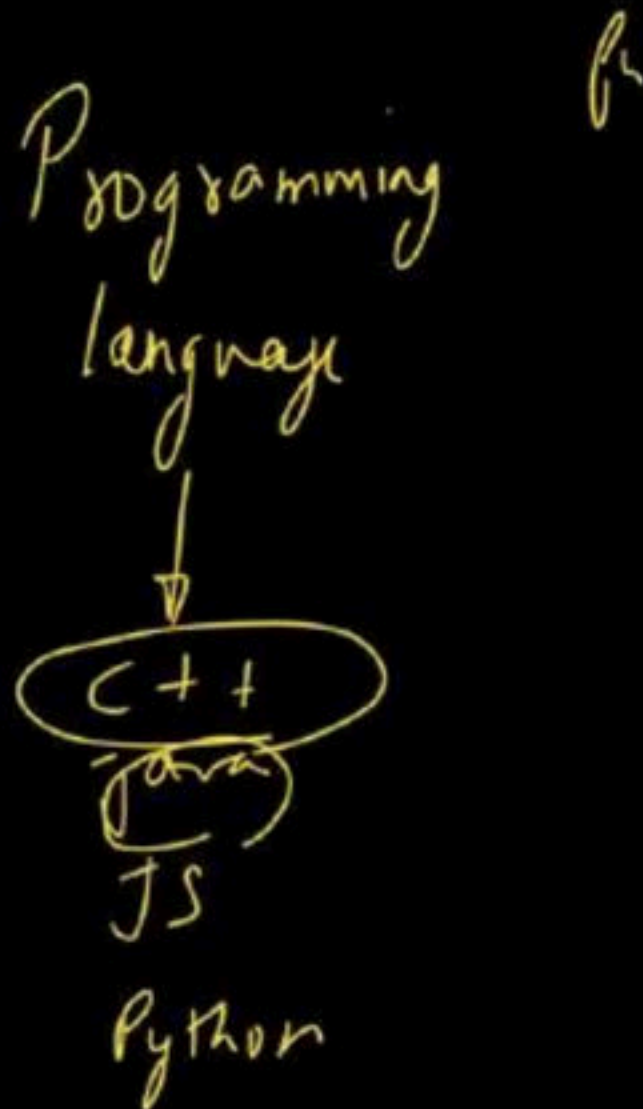
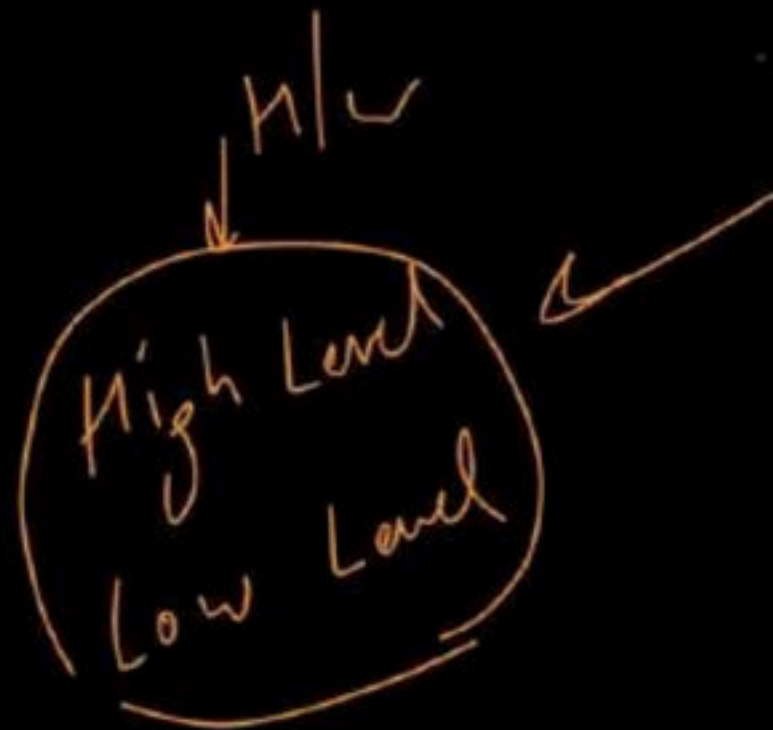
—> by Codehelp

Programming Language ?

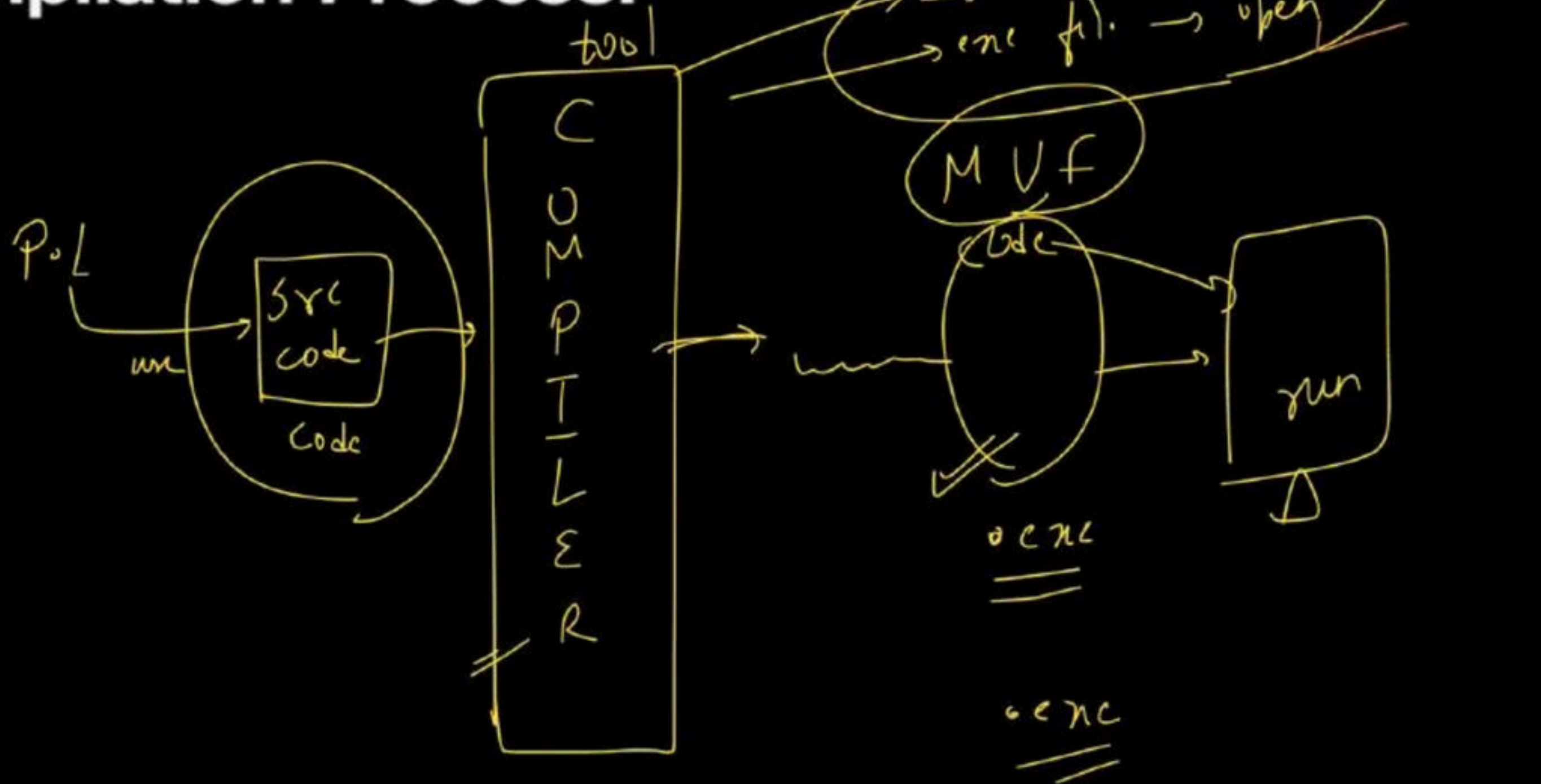
What ?

Why ?

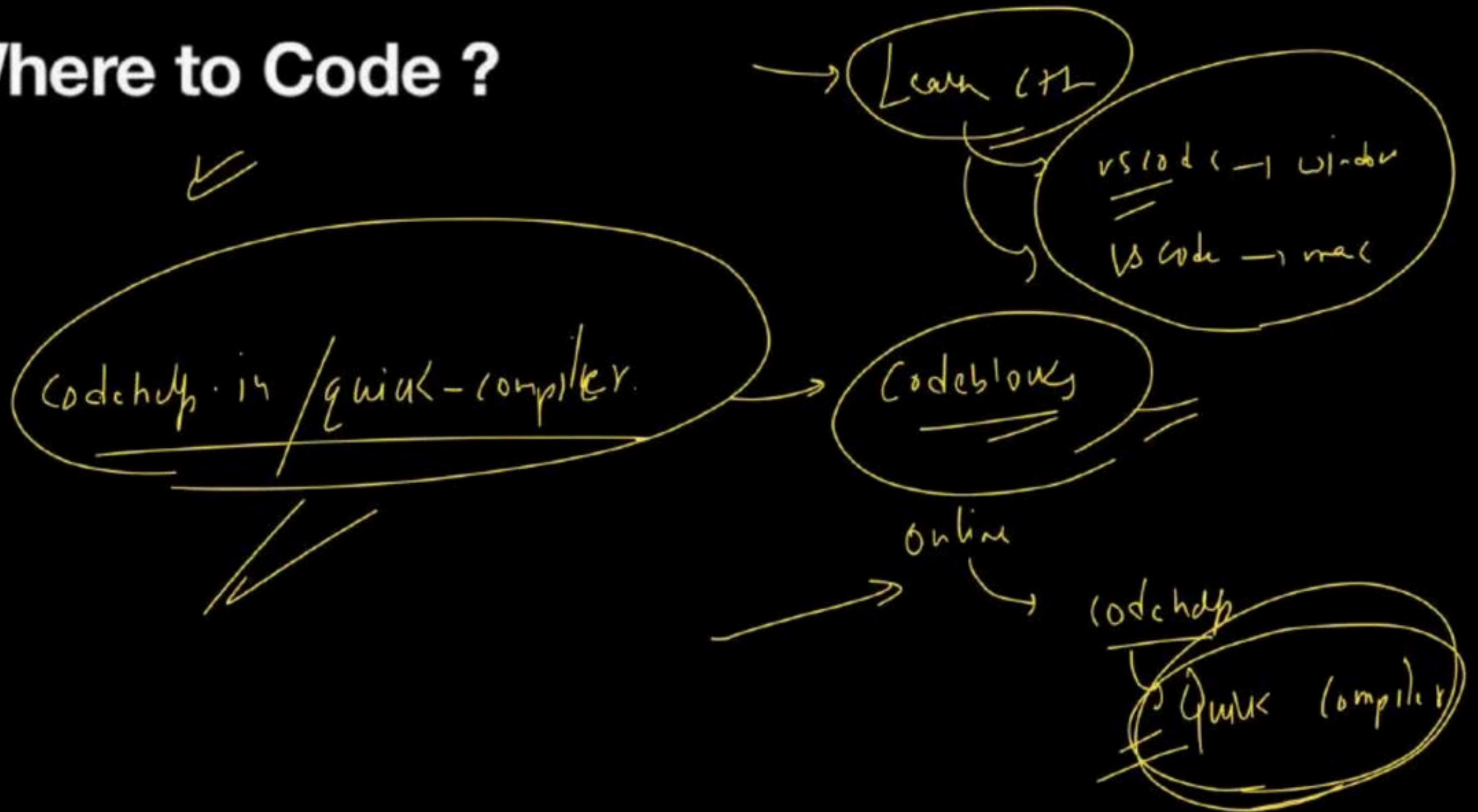
Standard



Compilation Process:



Where to Code ?

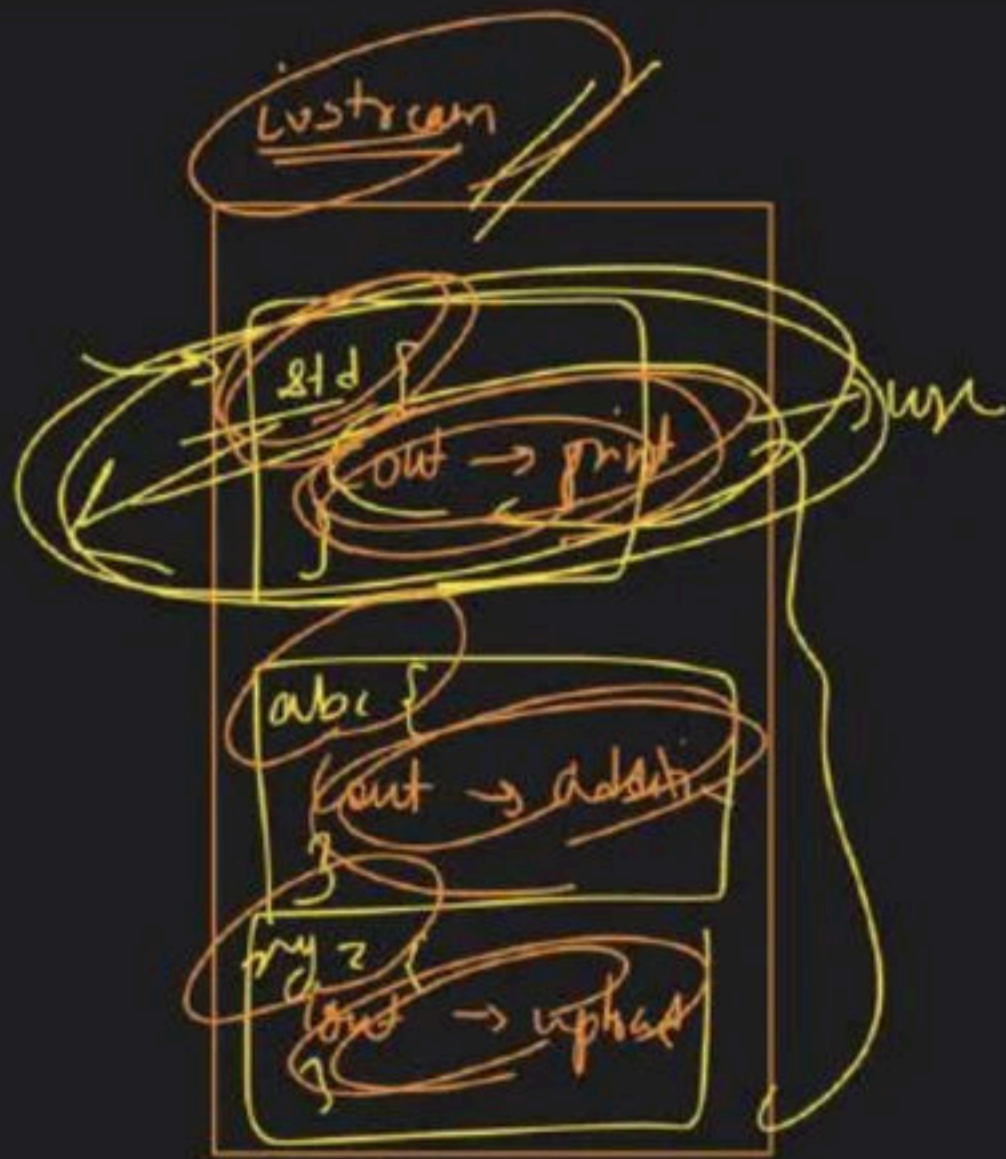


Your 1st Code:

C++
↳ Starting point
↳ int main

Starting point





```
→ #include <iostream>  
using namespace std;
```

```
int main()
```

```
cout << "Love";
```

```
return 0;  
}
```

Starting point

return type

function name

int main ()

{

~~~~~  
~~~~~  
~~~~~  
~~~~~

return 0;

}

integer →

-3, -2, -1, 0, 1, 2
3

function

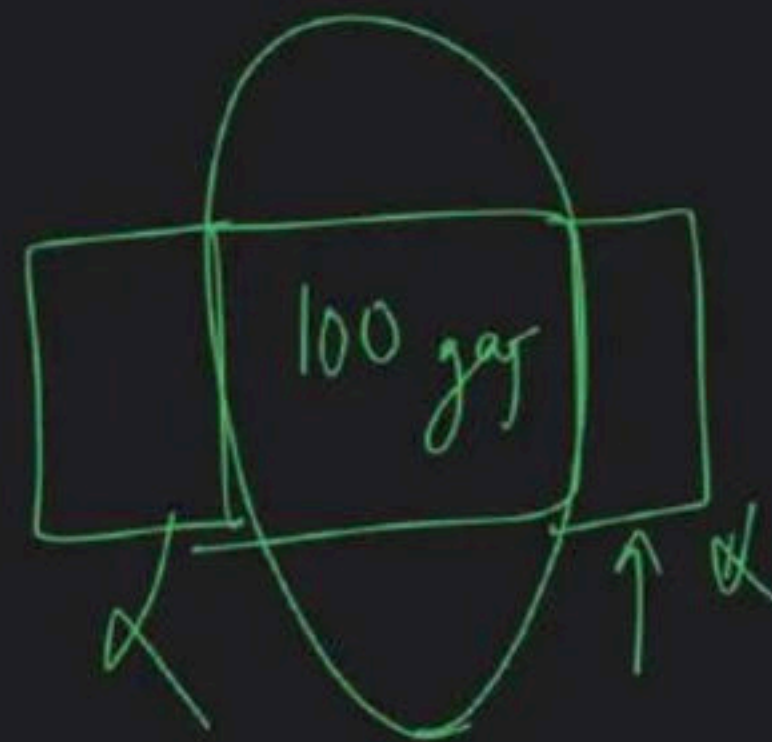
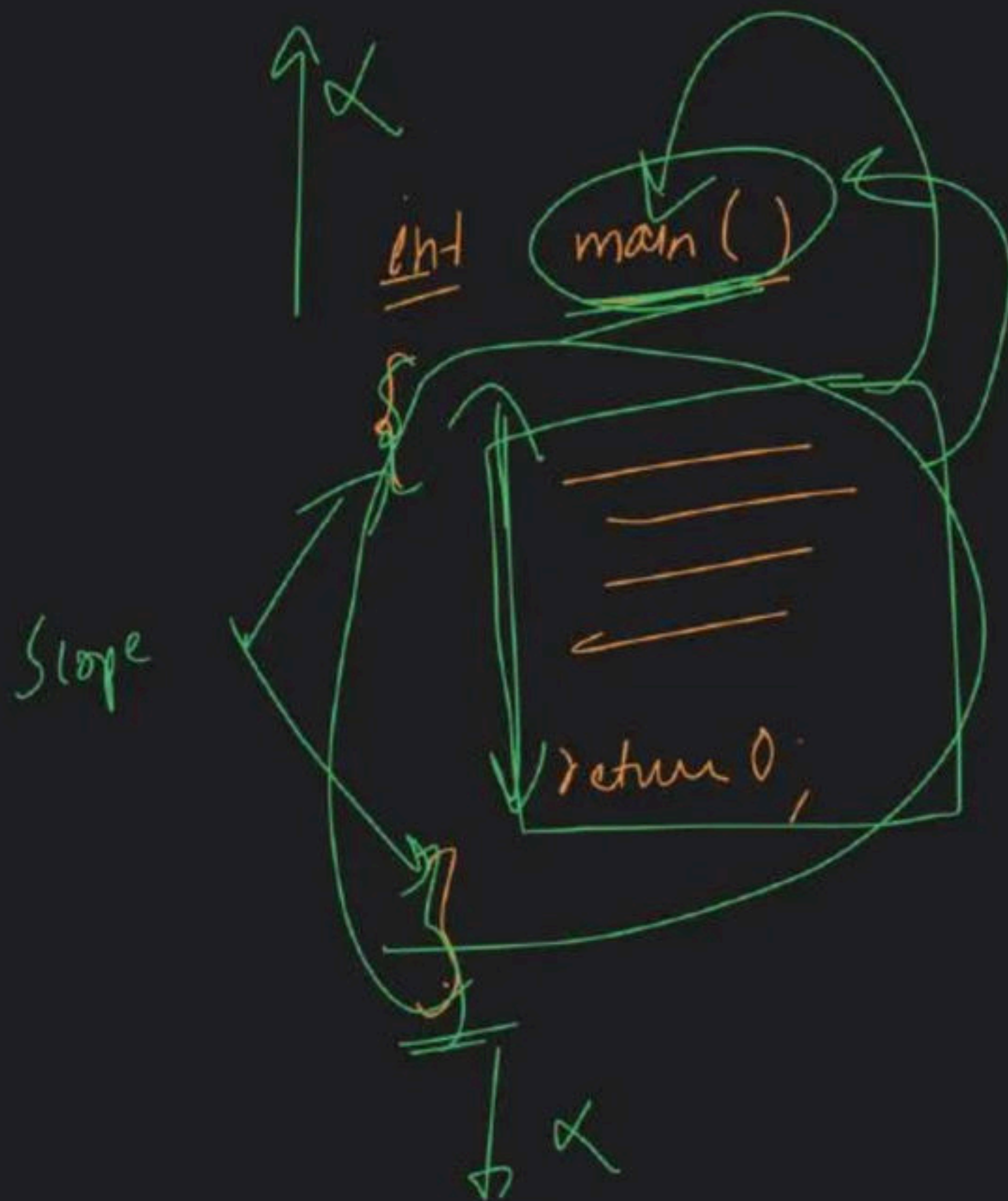
providing

may or
may not be

function

output

Call



C++ plg

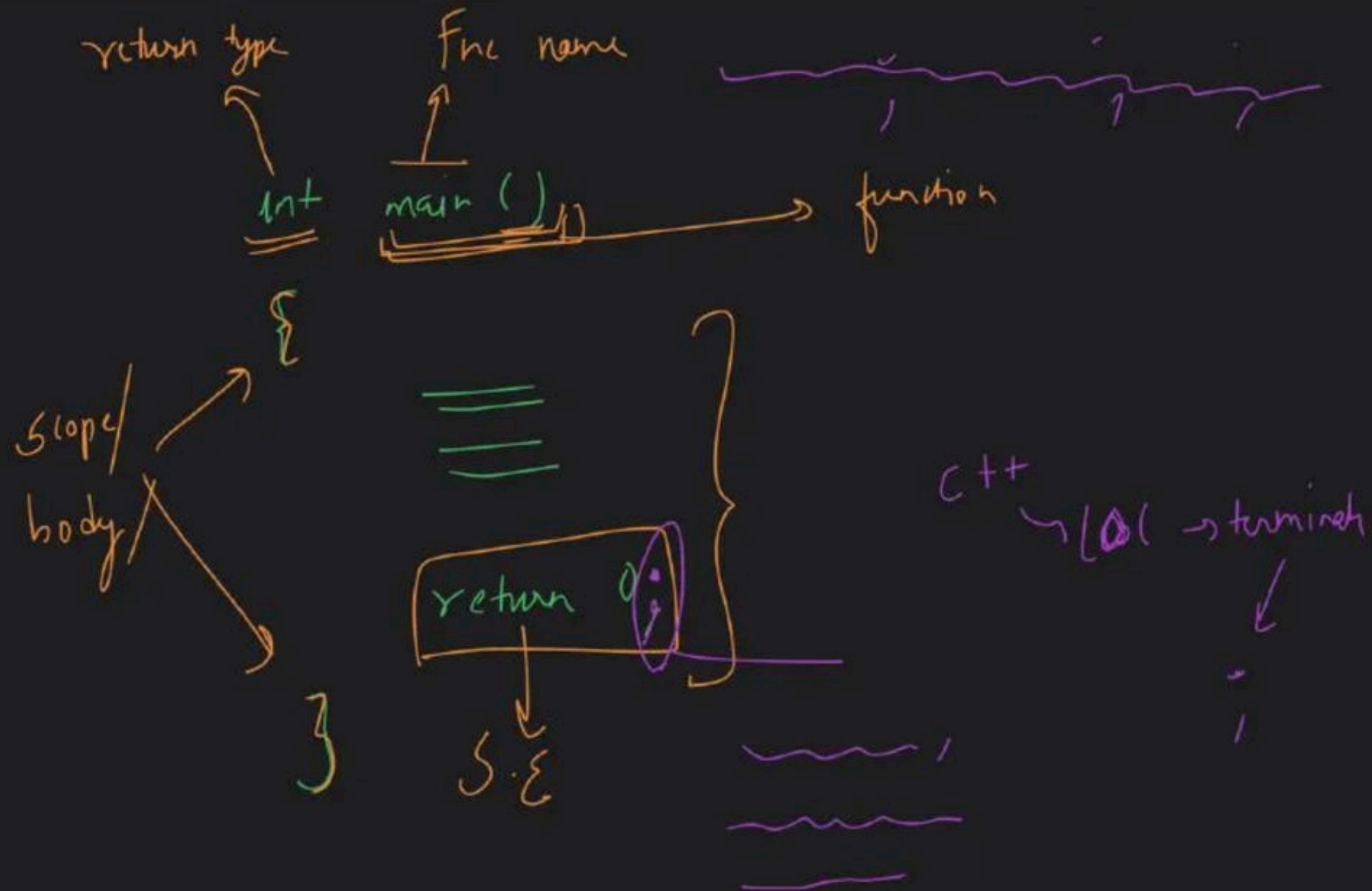
successfully execution

0-5

main()
{
 //
 //
 //
 //
 //
}

return 0





Print

file → ostream

Keyword/object/method

Love Babbar

cout

cout << "Love Babbar";

assume

prudently

definition

gdi gpt

`#include <iostream>`
`using namespace std;`

namespace

std

`int main()`
{
 `cout << "Love";`
 `return 0;`
}

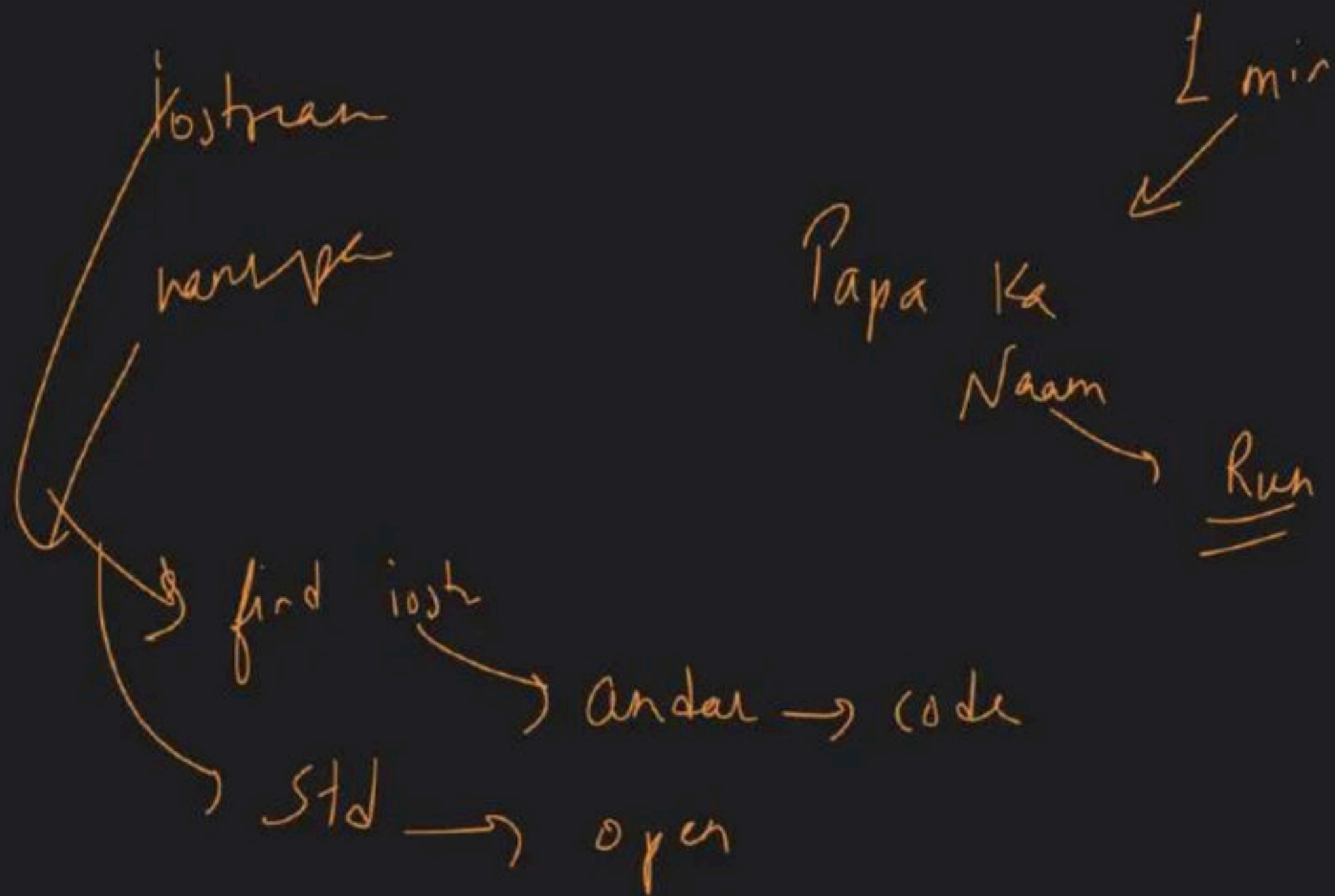
Syntax
↓
rule

std

`cout`

`cout`
100

`cout`
100



5
2
2

cout << 5
cout << 2;

endl

endl

cout << 5 << endl

cout << endl << endl << 2 << endl;

alternative

H/W → newline character

52

(52)

5
2

Input / Output in C++ :

a → char or
ab (-) string

cout << 5; → 5
cout << "5"; → 5
cout << (2+3); → 5 input/read
cout << "2+3"; → 2+3

cin

cin >>

Keyword

out but / print

cin >> var

cout

cout <<

cout << 2;

cout << "Mohit";

cout << (2+3);

cout << "2+3";

"Papa"

"lap 177"

input

cin

cin >> ramuh

bit → 0, 1

byte → 8 bit

00001010

address

cin >> ramuh

cin >> ramuh

5

ramuh

207

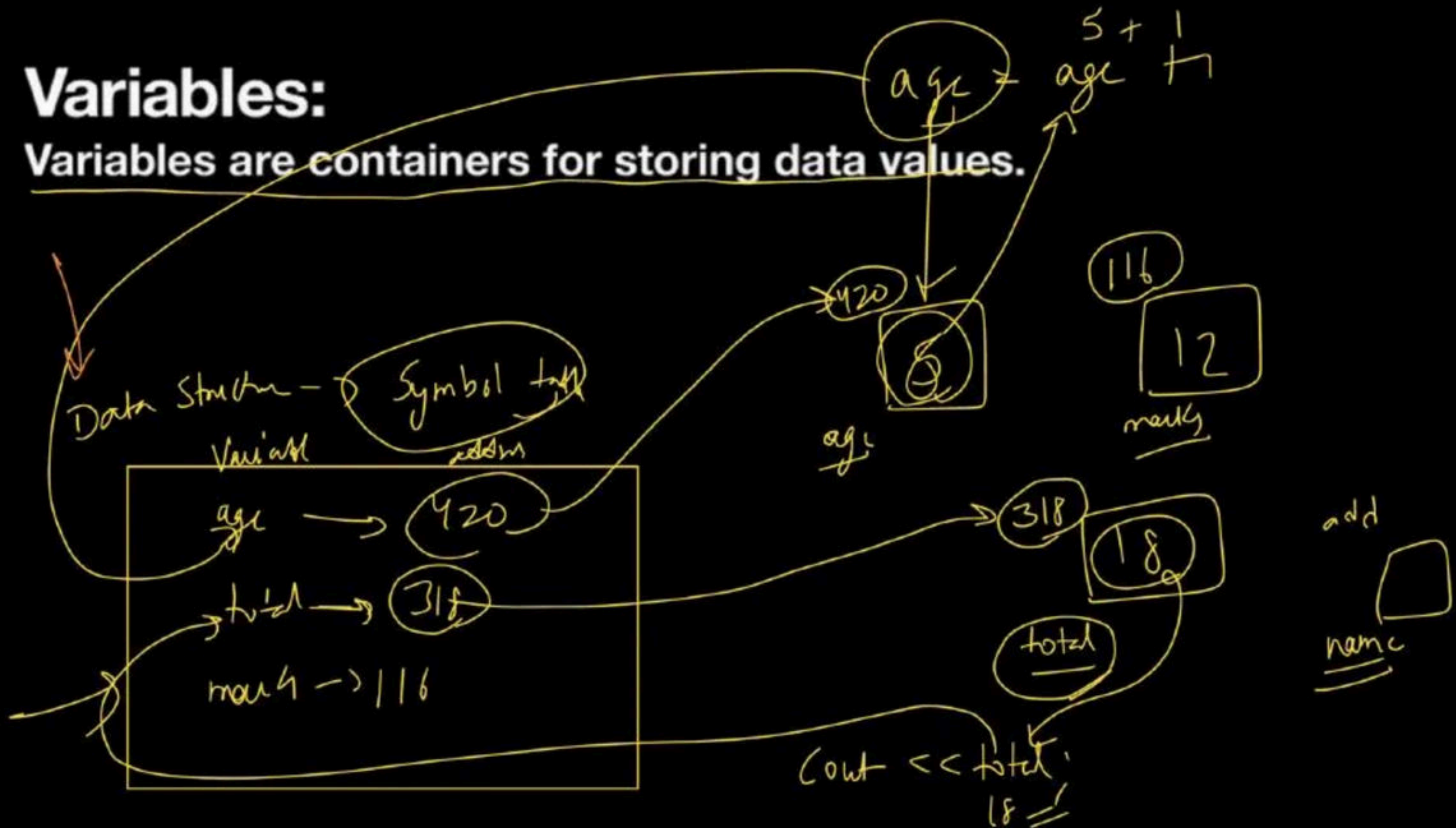
8

Variable
name

ramuh

Variables:

Variables are containers for storing data values.

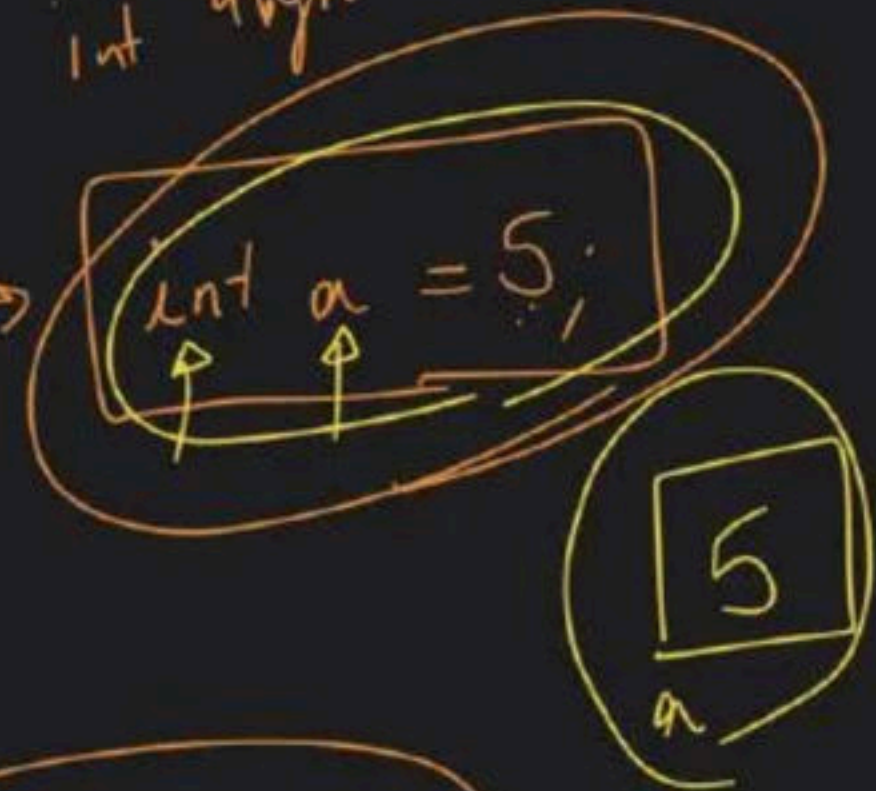
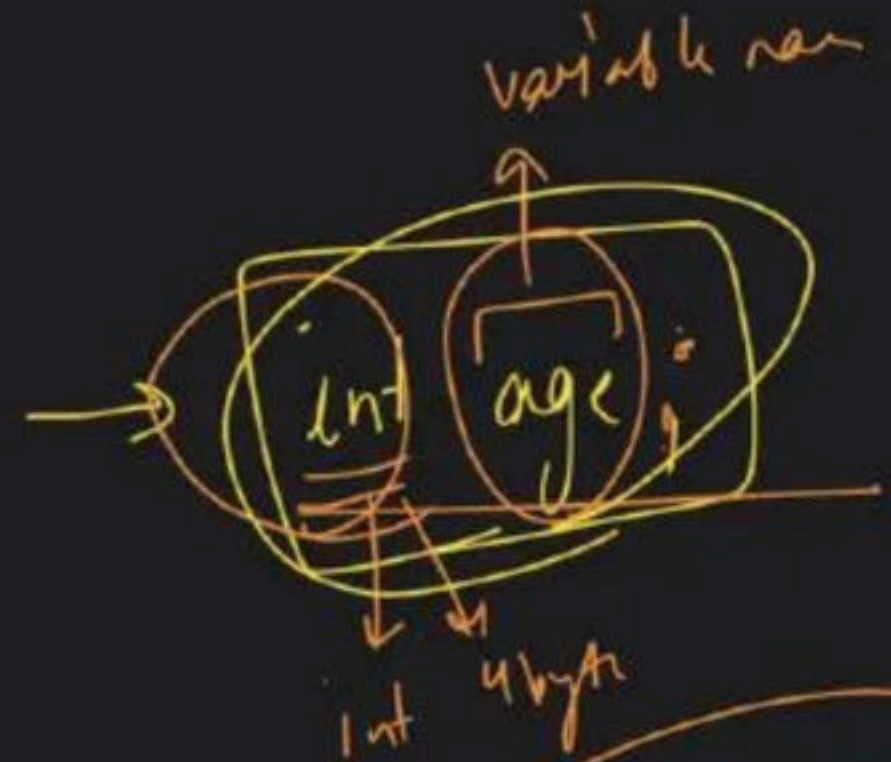


Variable

declaration

definition

initialisation



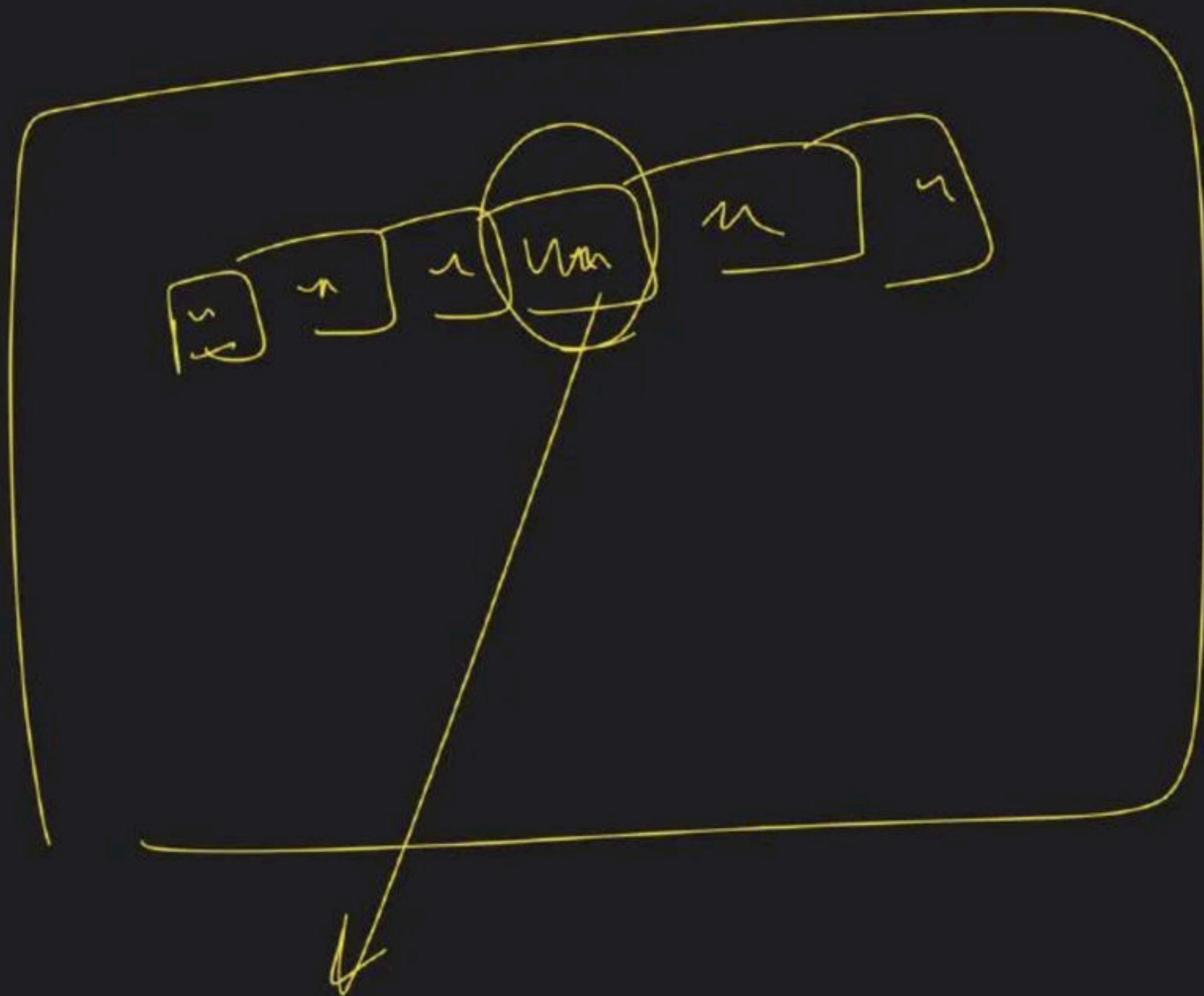
garbage Value



`float marks;`

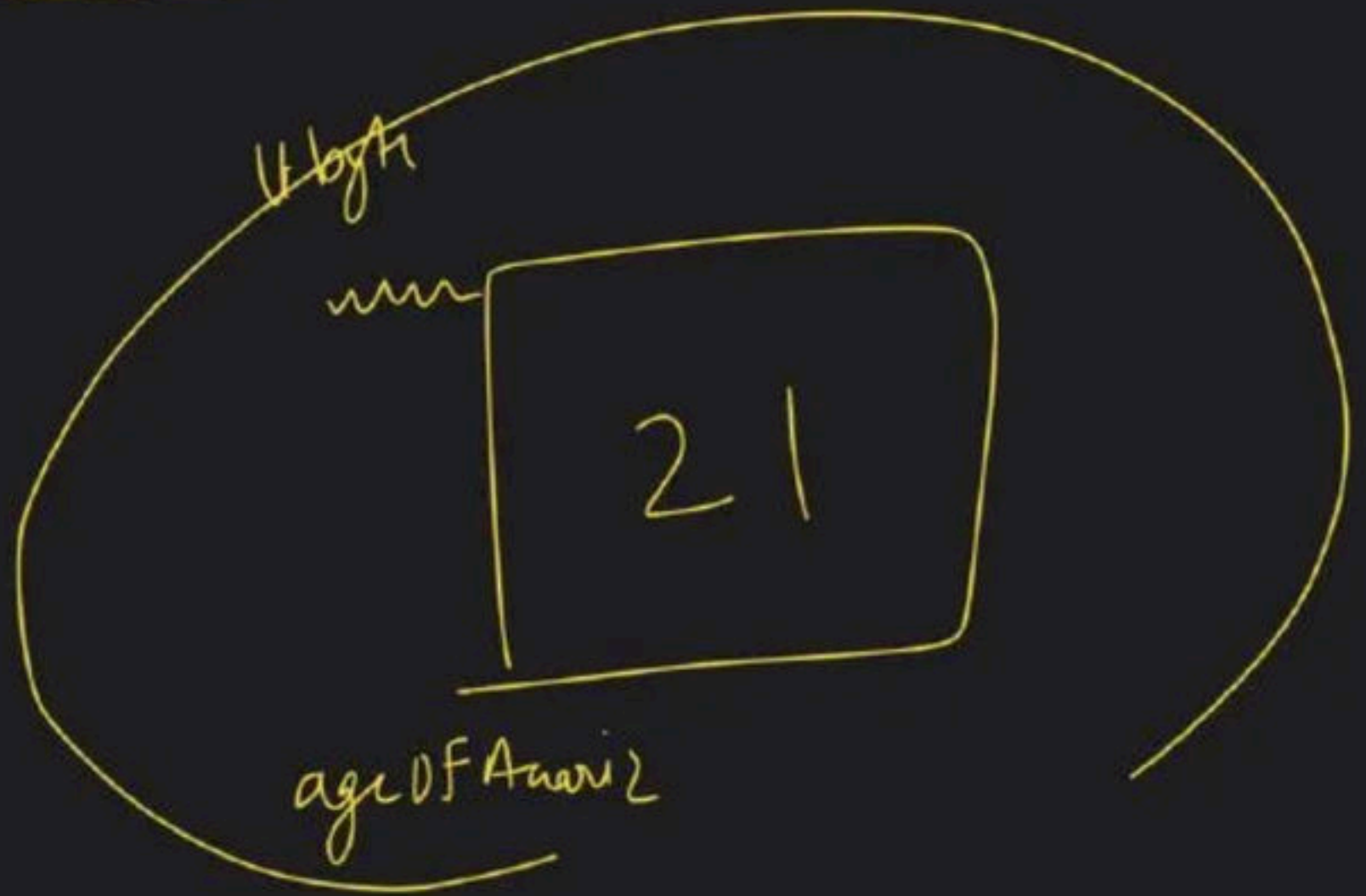
`char grade;`

`bool flag;`

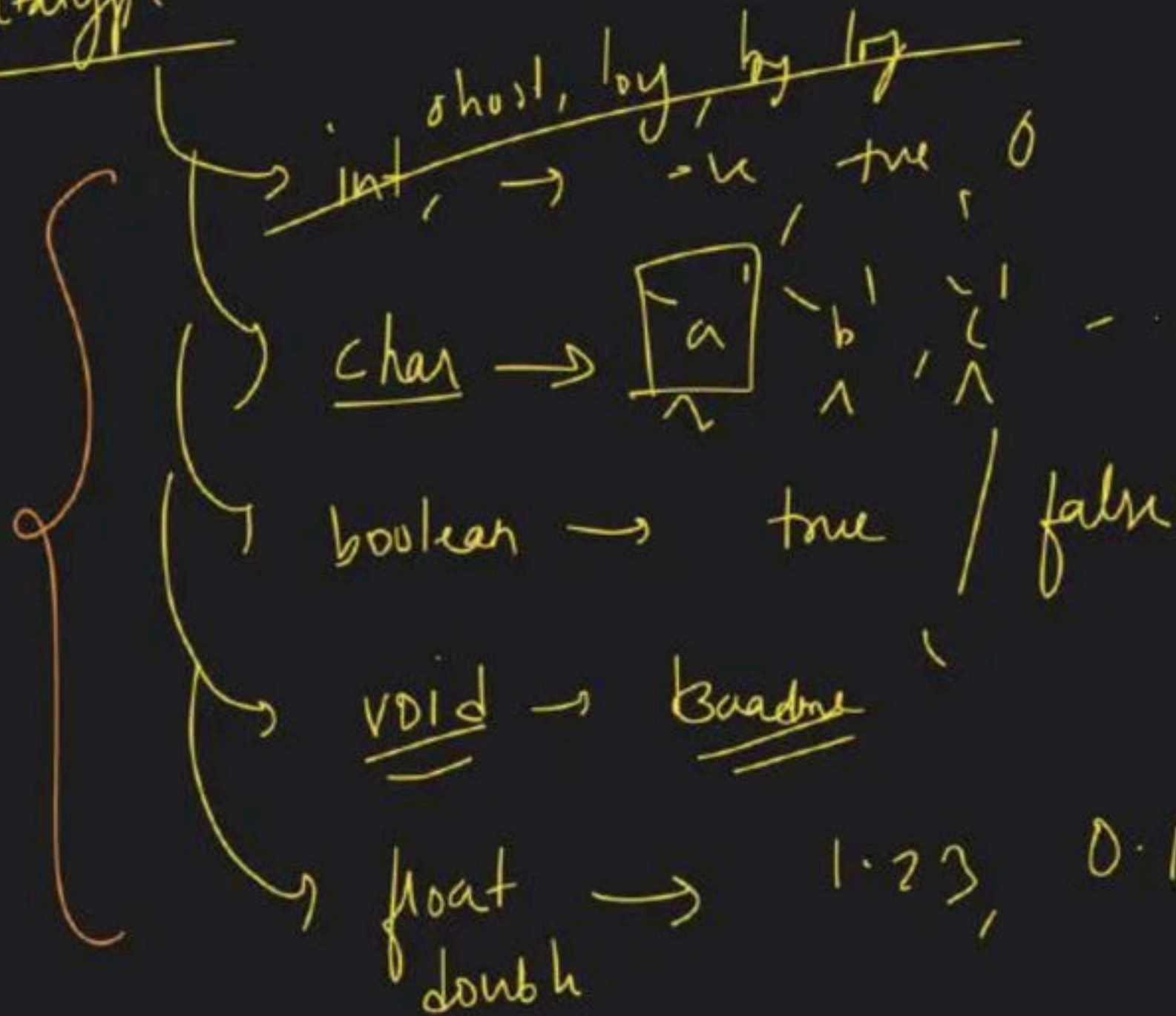


`int age OF Aariz` = 21 ;

int 4 byte
↑ ↑
a = 5
↓
datatype



datatype



11 - - - 1

1.23, 0.17, 3.142077215

bool babbar = ~~true~~;
= 1;

true $\leftrightarrow$ 1
false $\leftrightarrow$ 0

char grade = '+';

Datatypes:

The data type specifies the size and type of information the variable will store

- Diagram

C Basic Data Types	32-bit CPU		64-bit CPU	
	Size (bytes)	Range	Size (bytes)	Range
char	1	-128 to 127	1	-128 to 127
short	2	-32,768 to 32,767	2	-32,768 to 32,767
int	4	-2,147,483,648 to 2,147,483,647	4	-2,147,483,648 to 2,147,483,647
long	4	-2,147,483,648 to 2,147,483,647	8	9,223,372,036,854,775,808-9,223,372,036,854,775,807
long long	8	9,223,372,036,854,775,808-9,223,372,036,854,775,807	8	9,223,372,036,854,775,808-9,223,372,036,854,775,807
float	4	3.4E +/- 38	4	3.4E +/- 38
double	8	1.7E +/- 308	8	1.7E +/- 308

ASCII table

ASCII values

char $\rightarrow$ -128 to 127

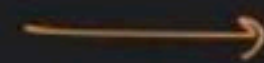
a b c d
+ - -

char grade = 65

cout << grade
↓
(a)

a → 65
b → 66
c
d
A → 67
B
C
+

`char ch = 'a'`
Signed



-128 to 127

`unsigned char = 'a'`

0 → 255

datatype

range

$n = \text{no of bits}$

char $\rightarrow$ 1 byte $\rightarrow$ 8 bits

$n = 8$

Signed $\rightarrow -2^{n-1} \rightarrow 2^{n-1} - 1$

unsigned $\rightarrow 0 \rightarrow 2^n - 1$

$0 \rightarrow 2^{8-1} \rightarrow 0 \rightarrow 255$
 $0 \rightarrow 255$

signed $= -2^{8-1} \rightarrow 2^{8-1} - 1$
 $-2^7 \rightarrow 2^7 - 1$
 $-128 \rightarrow 128 - 1$

$-128 \rightarrow 127$

Variable Naming Conventions:

Naming Conventions rules for Variables are:

- It should begin with an alphabet.
- There may be more than one alphabet, but without any spaces between them.
- Digits may be used but only after alphabet.
- No special symbol can be used except the underscore () symbol. When multiple words are needed, an underscore should separate them.
- No keywords or command can be used as a variable name.
- All statements in C++ language are case sensitive. Thus a variable A (in uppercase) is considered different from a variable declared a (in lowercase).

main

int main

cout

int cout

~~float x = 3.14;~~

Day

age 1
age 2
age 3

int a+b+c = 10

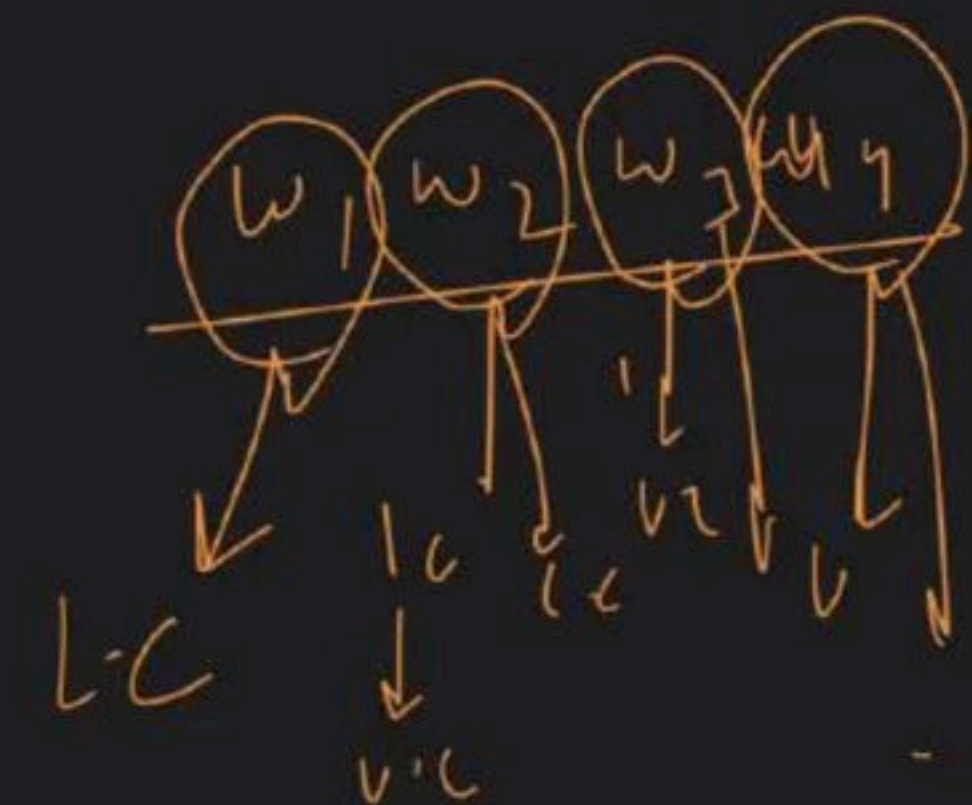
age 1 of 10

age_of_10

int age_of_car

int (love) = 5 ;

int (Love) = 5 ;



age Of Anariz

low babbar

low Babbar

harsh Kapoor

agn
↓
loween

@
↓
UL

Anus
↓
v1c

aditya Kumar

love_babbar

Assignment:

Difference in storage of +ve and -ve integers ?

sign bit leftmost

n/w
0 → +ve
1 → -ve number

int a = 5
101

int a = ~~10~~
1010

00000000

00000000

00000000

0000101

00000000

00000000

00000000

0001010

1's compliment

2's compliment

Signed vs Unsigned Integers:

Operators:

Operators are used to perform operations on variables and values.

- Arithmetic [+, -, *, /, %, ++, --]
- Relational [==, !=, >, <, >=, <=]
- Assignment [=]
- Logical [&&, ||, !]
- Bitwise [&, |, ~, ^, >>, <<]

int a = 5
int b = 5

int a = 5
int b = 5
cout << a * b;

Symbol → operation

+ → add
- → sub
* → mult
/ → divid

50/05
no

int a = 5

Q2 → Logical AND → $TT \rightarrow T$

Condition

boolean

a	b	o/p
T	T	T
T	F	F
F	T	F
F	F	F

a Δ b

Condition

o/p → T/F

→ Logical or → ~~at least~~ at least 1 is true → o/p → T

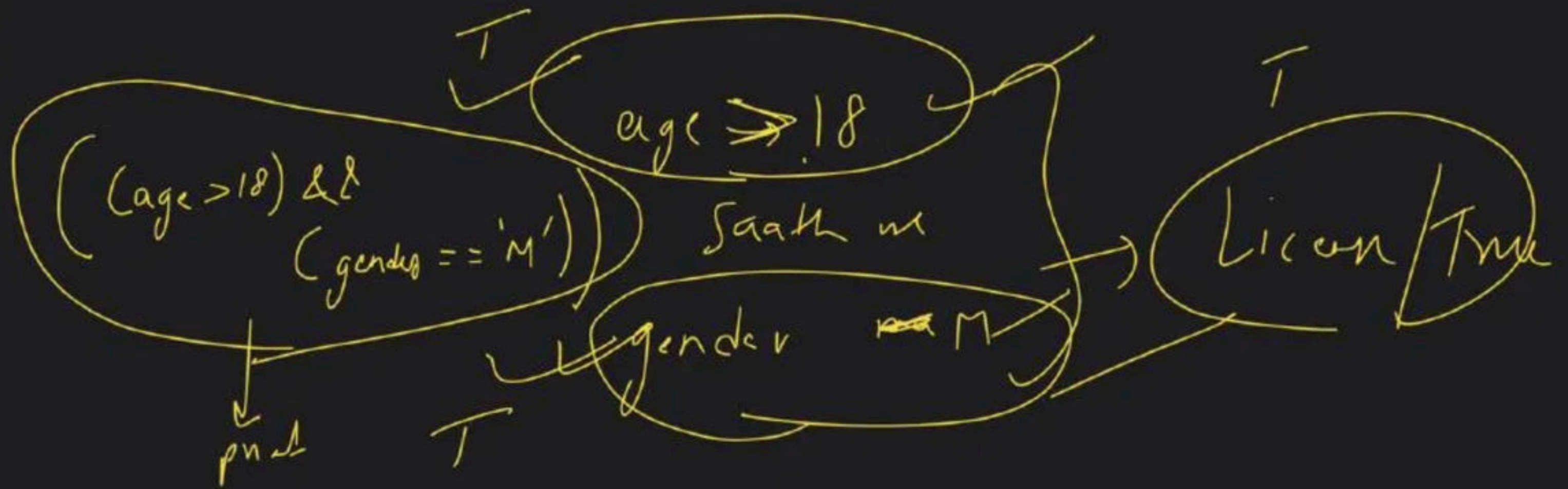
a || b

a	b	o/p
T	T	T
F	T	T
T	F	T
F	F	F

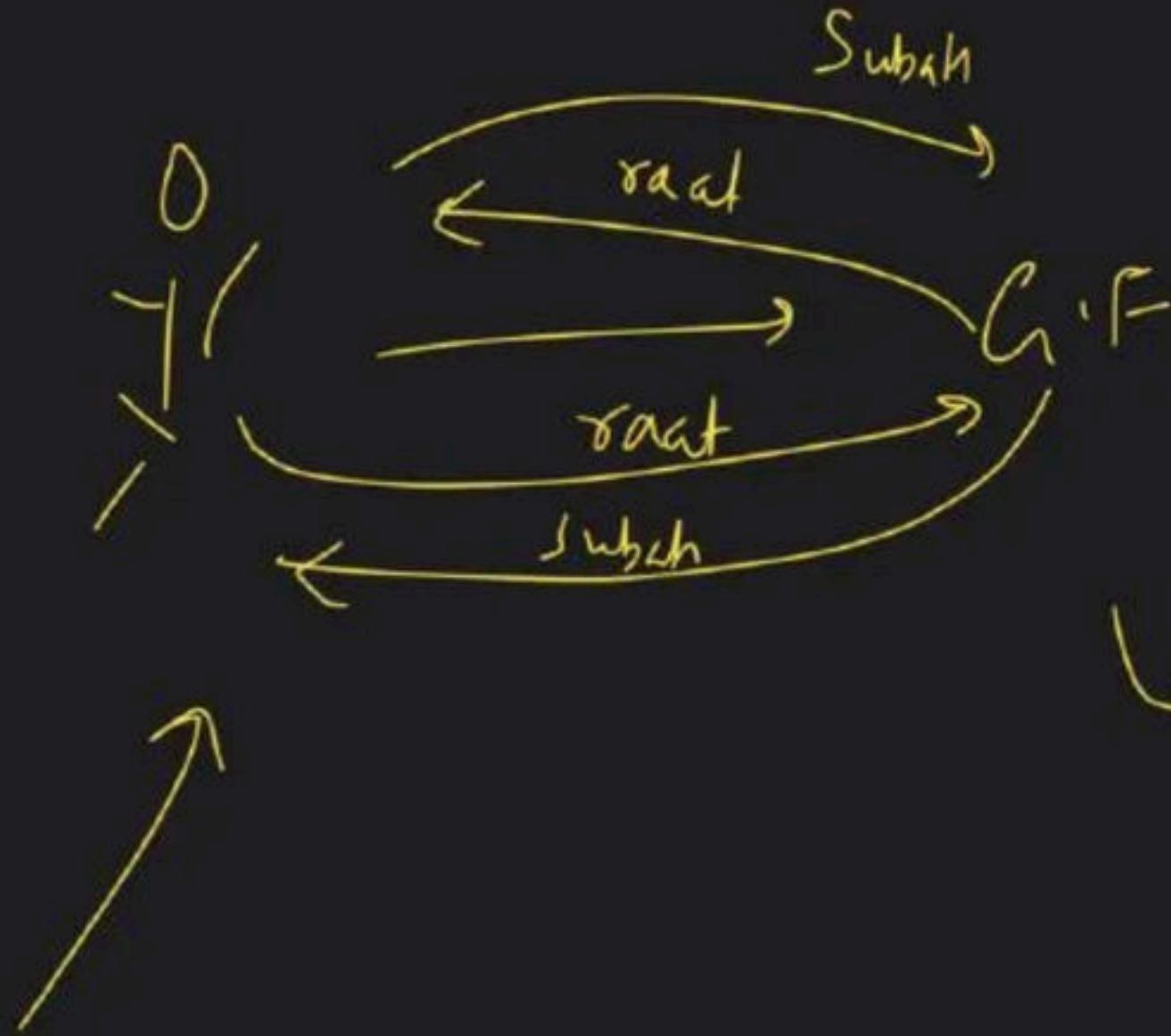
a	!a
T	F
F	T

Logical NOT

Reverse flip



[a
else → False
(age > 18) || (gender == 'M')



!! (age > 18)

↓

print

Assignment:

Complete week-"Learn C++" recorded videos.

Kel -> gym

- 32 bit vs 64 bit Architecture ✓
- Typecasting: Implicit and Explicit ✓
- Binary and Decimal Number System ✓

Name: 17

discore

ticket

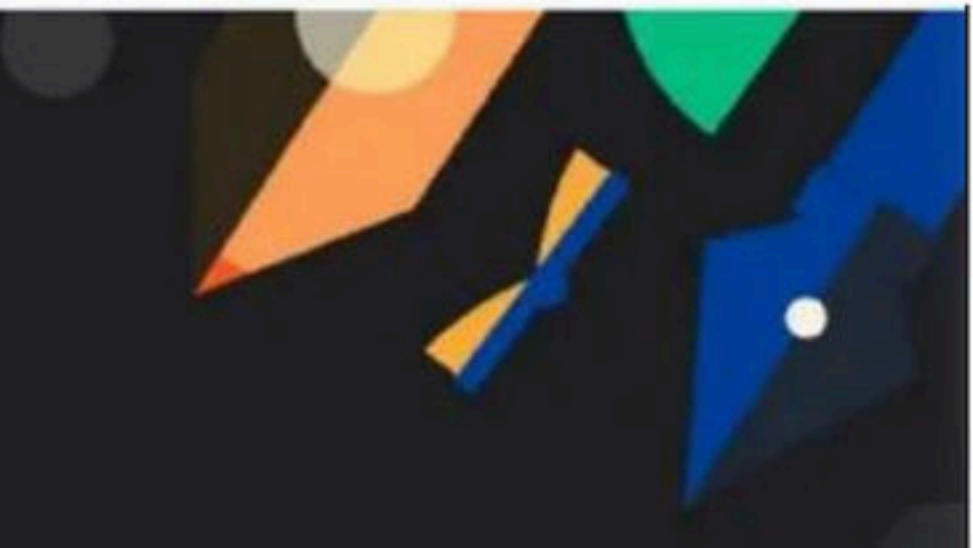
join link

discore

Learn

C++

< >



L3-Basics of Programming

Special class

L3 - Basics of Programming

Operators, Conditionals & Loops

—> by Codehelp

Operators:

1 week → No. 1/7, /m 2

Arithmetic → +, -, *, /, %

++, --

Relational → ==, !=, >, <, >=, <=

assignment (=)

logical operators (&&, ||, !)

Bitwise

Bitwise Operator

learn C++ → Operators in C++

45 min

&, |, ^, ~, >>, << → entire

Arithmetic

Binary Op

2 operands

operands

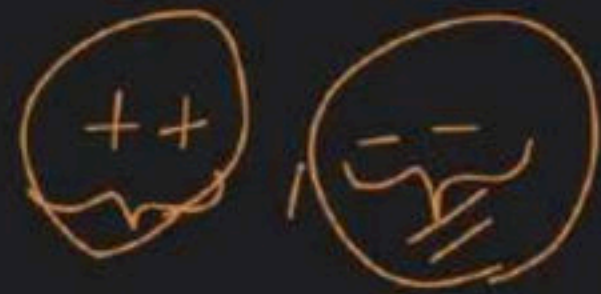


operator = +

+
-
*
/
0/0

Unary Op

1 operand



Comment
↳ Links

$++$ → increment op

pre-increment

① Pehle increment Karo

② fer use Karo

~~5~~ 6

int a = 5;

cout << (++a);

⑥

post-increment

① Pehle use Karo

② fer increment Karo

a

$++$ a

a $++$

int a = 5;

cout << (a++);

⑤

cout << a → 6

~~5~~ 6

a

-- → decrement

pre-decrement → $(--a)$

post-decrement → $(a--)$

pre-decrement → ① Pehle decrement Karo
② fir use Karo

int a = 5;

cout << (--a); → 4

→ cout << 4; → 4

4 ~~5~~

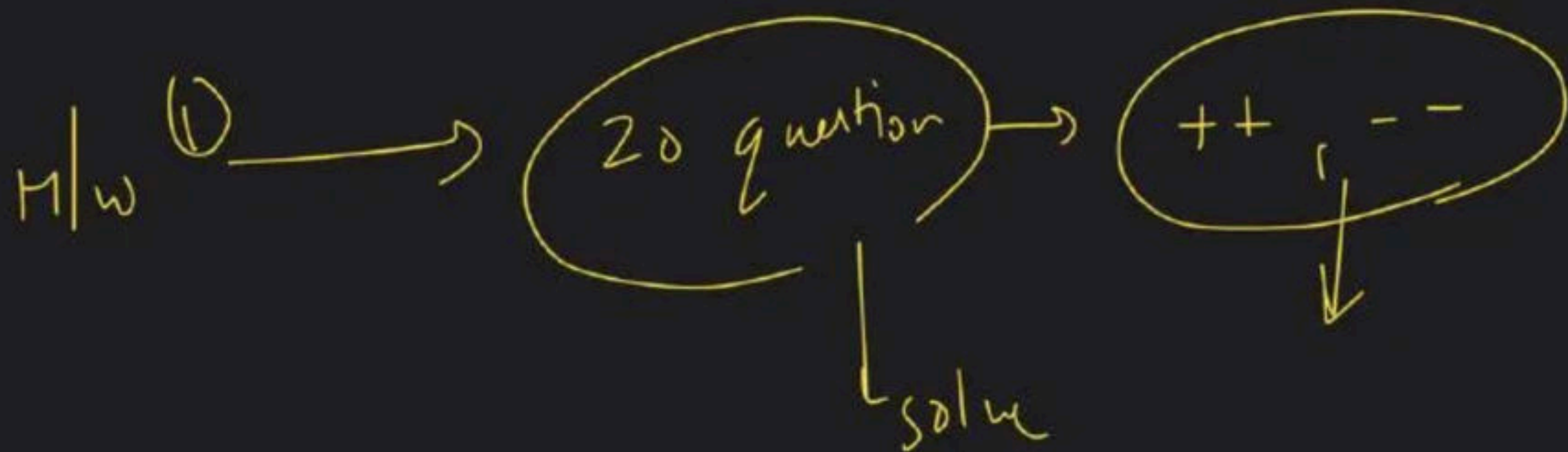
post-decrement $\rightarrow$ ① Fehl, um Karo
② for decrement Karo

int a = 5;

cout << (a--); $\rightarrow 5$

~~5~~ 4
a

~~cout << a+5;~~ $\rightarrow 9$



② Red Quiz → unlocked

Question : —

2 min

short-circuit
condition

int a = 5; —→

cout << (++a); —→ 6

cout << a; —→ 6

cout << (a++); —→ 6

cout << a; —→ 7

cout << (--a); —→ 6

cout << <; —→ 1

cout << (a--); —→ 6

cout << a; —→ 5

5
5
5

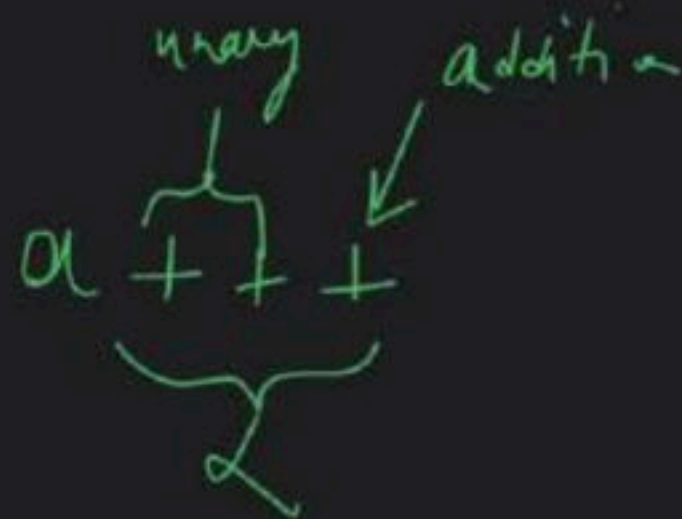
H/W

any → Lalitha

int val =

((++a) * (a++)) + (--a) * (a--)

cout << val;



$$\left\{ \begin{array}{l} a+++b \\ (a++) + b \end{array} \right.$$

$$\text{val} = \frac{(a++)(++a)}{a++ ++a}$$

$$\frac{a++ ++a}{}$$

$$\left[\left[2 + (3 \times 4) \right] + 5 \right]$$

exp →

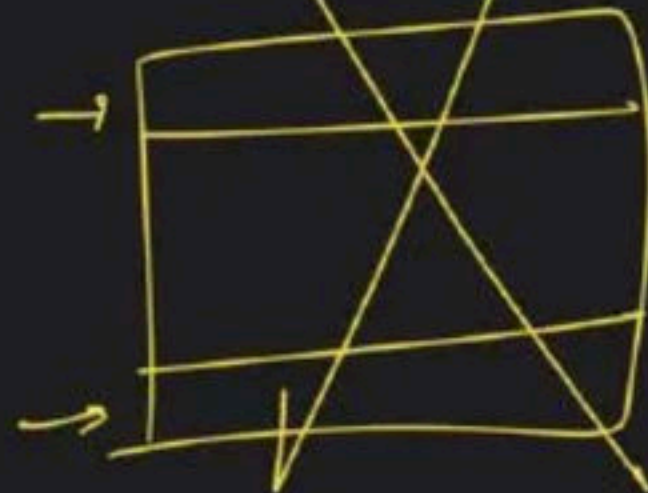
Use Brackets

~~Operator~~

~~table~~

~~Precedence~~

~~& Associativity~~



$$a = \overset{4}{\cancel{5}}, b = \cancel{4}^8$$

$$(\underline{\quad\quad\quad} a) \star (\underline{++b}) \star 5$$

$$4 \times 8 \times \underline{5}$$

$$\overset{4}{\cancel{4}} \times \overset{4}{0}$$

$$(160)$$

$$a = \overset{4}{\cancel{5}}, b = \cancel{4}^7$$

$$(\underline{\underline{\quad\quad\quad}} a) \star (\underline{\underline{b++}})$$

$$4 \times 6$$

$$(24)$$

$$a = 5$$

$$\rightarrow (- - a) \star (a - -)$$

4

+

4

18

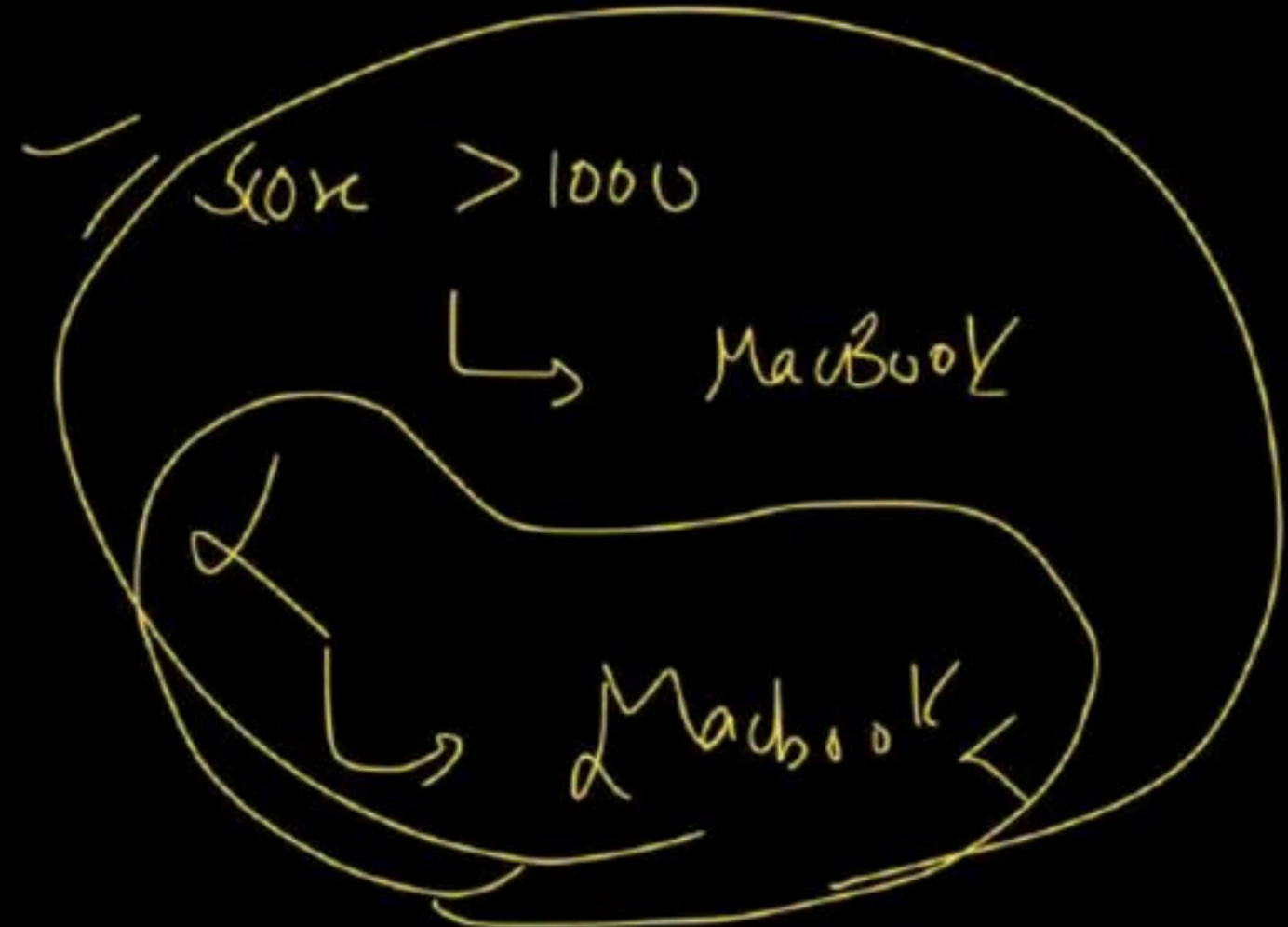
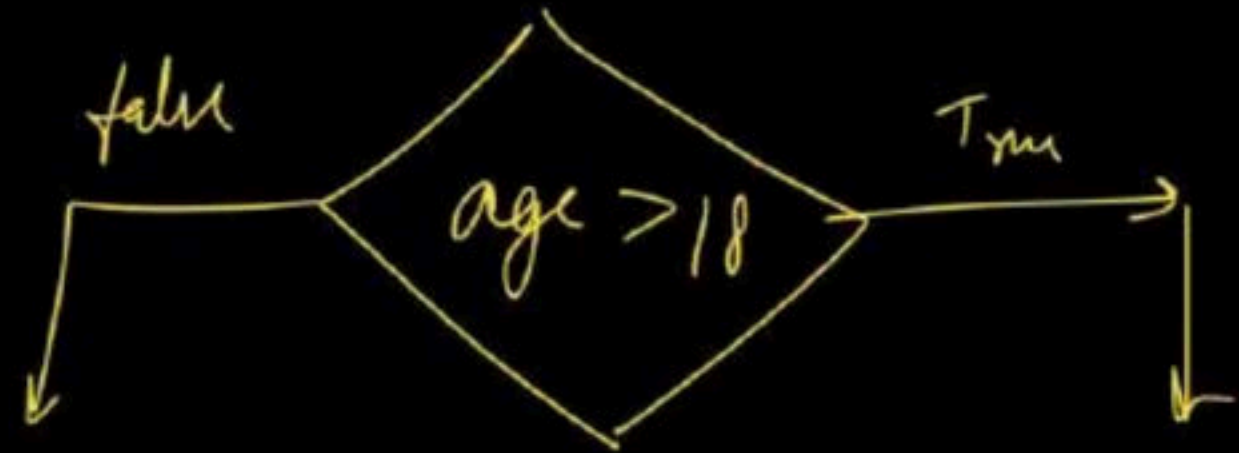
3
7
8

even
count

$$\begin{array}{r} 99999 - - - 9 \rightarrow ? \\ \hline 13 \end{array}$$

Conditionals: //

- Basic if block
- If-else block
- If-elseif-else block
- Nested if else
- Switch case
- Ternary Operator



Switch

switch (expression)

{ ~~case 0:~~

~~case 1:~~

~~case 13:~~

default:

}

value

(constant)

1, 2, 3, 4, 5, 6, 7

switch (index)

case 1:

cout << "Monday"

case 2:

cout << "Tuesday"

default:

cout << "Sunday"

}

Switch (inden)

{

case 0 : { int a = 15;

}

}

Swift

Switch (expression) → Kya Kya → Rules

Case  → Rules

Video ← Switch Case

Learn
(+1)

Ternary Operator

Syntax

```
if ( score > 1000 )  
    cout <= "MB";  
else  
    cout << "No MB";
```

ternary

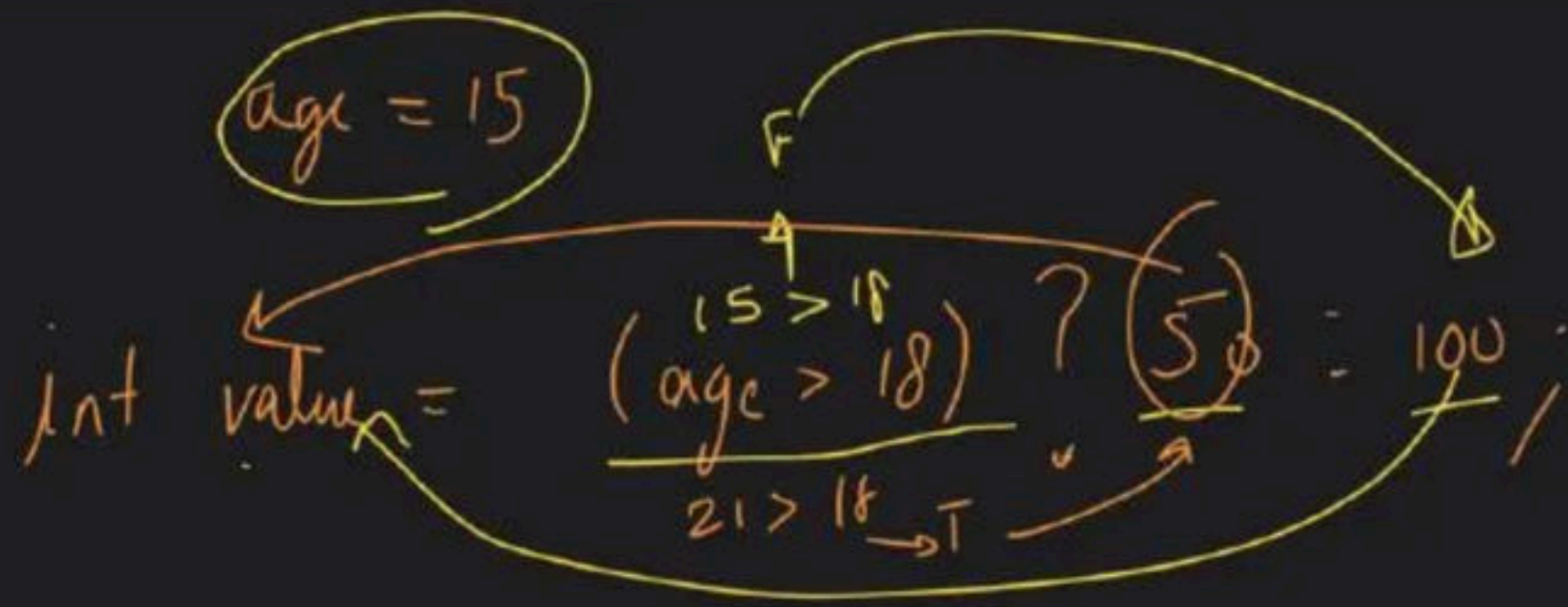
else

if Condition ? logic : logic ;

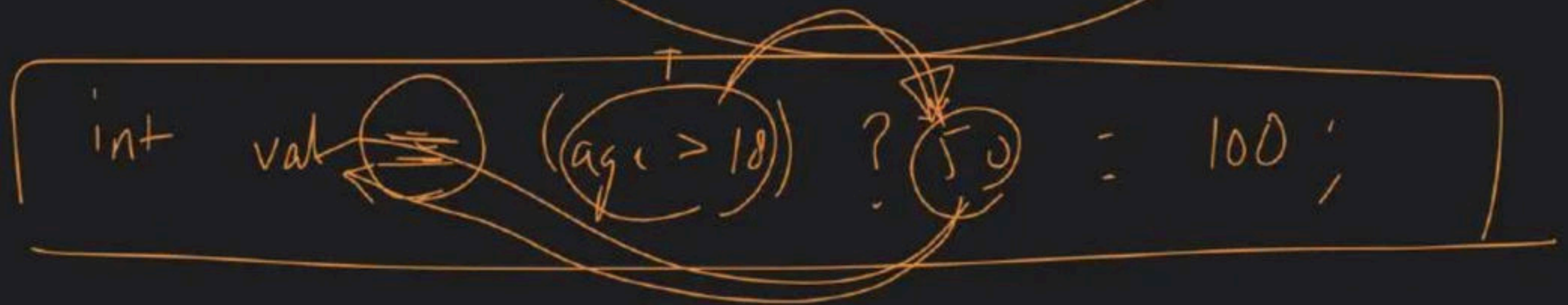
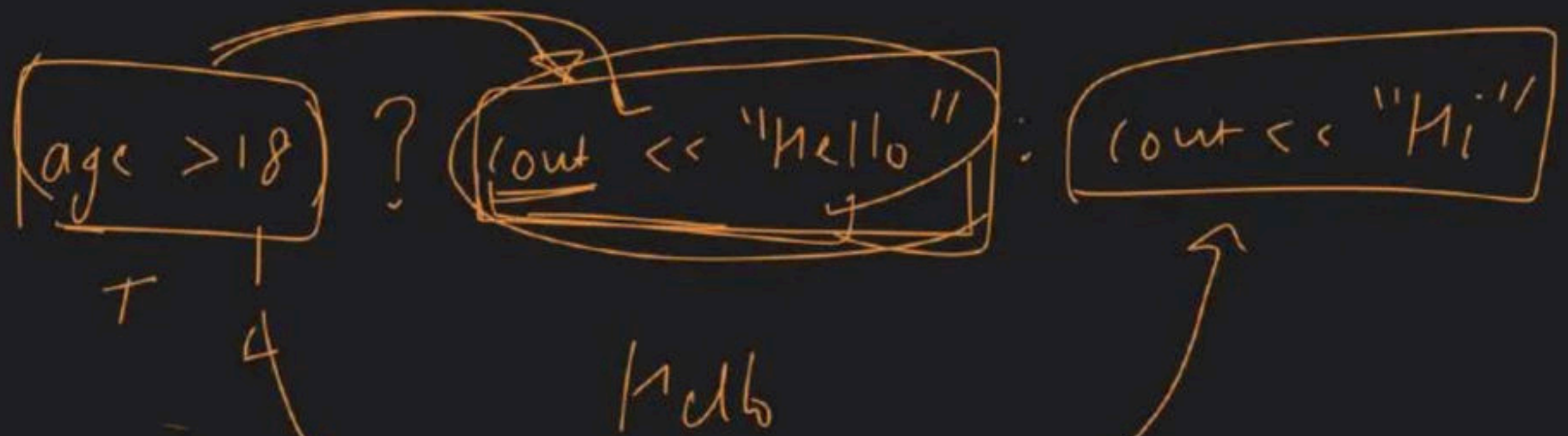
F ↑ T ↑

↓ ↓

↓ ↓



~~val~~ value 9



`cout << val; → 50`

val 50

if ()

{

if ()

{

if ()
{
}

}

}

if ()

{

if ()

{
}

do {
}

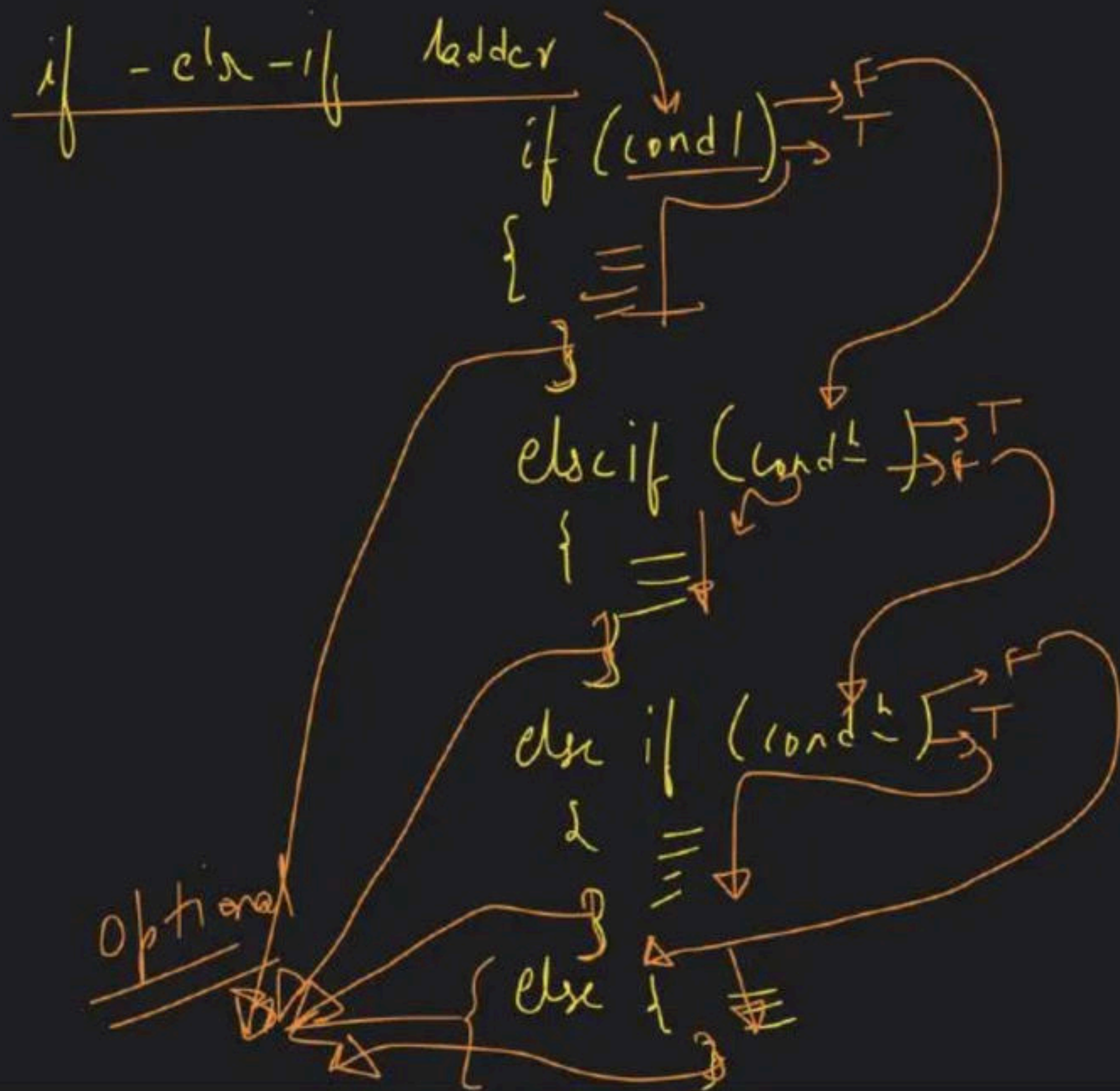
}

do

{
}

if ()

do



Restaurant → 1 Vacancy → LMEF



$\alpha \leftarrow C_1$

$\alpha \leftarrow C_2$

Bhot badi $\leftarrow C_3$

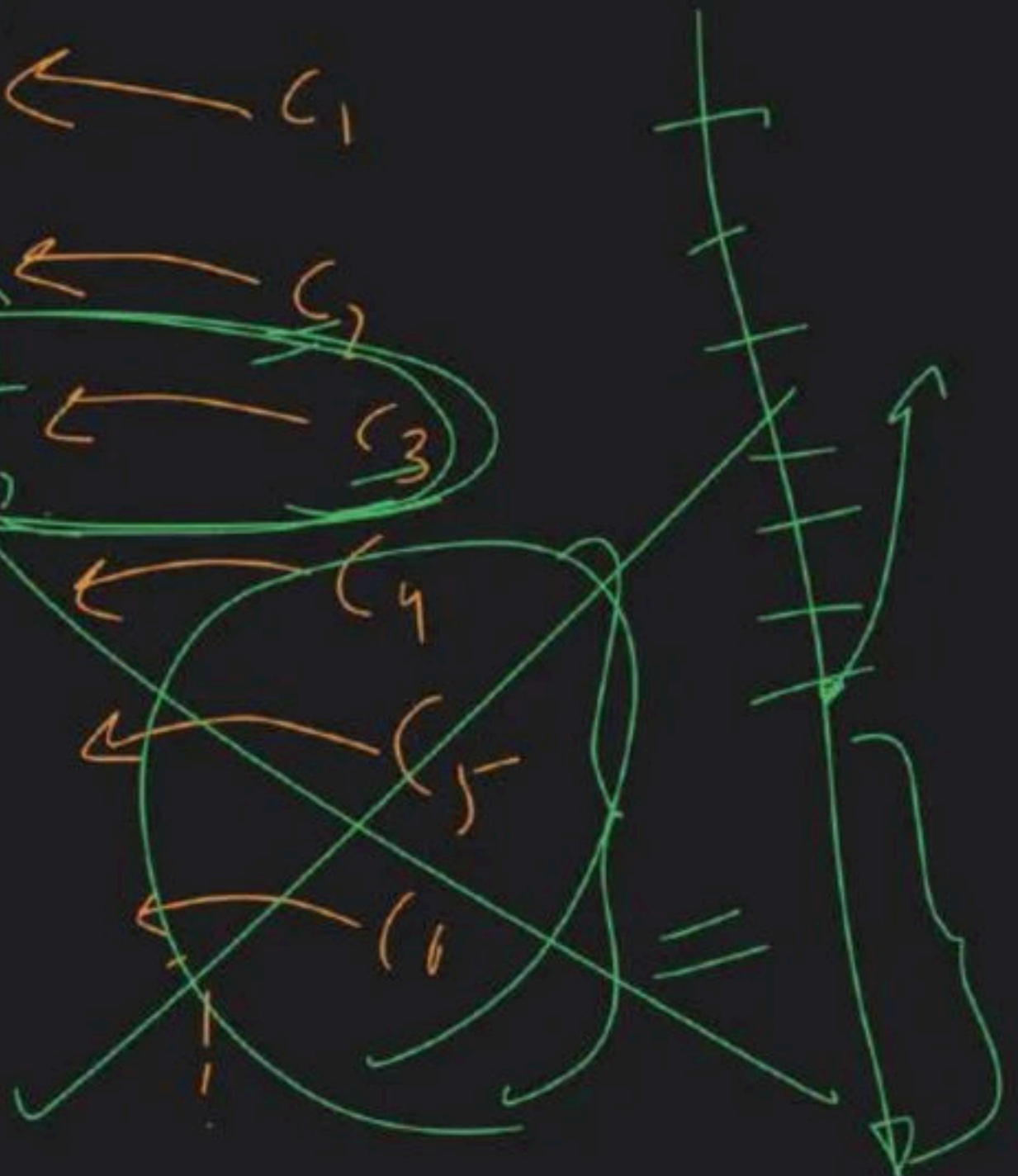
$\leftarrow C_4$

$\leftarrow C_5$

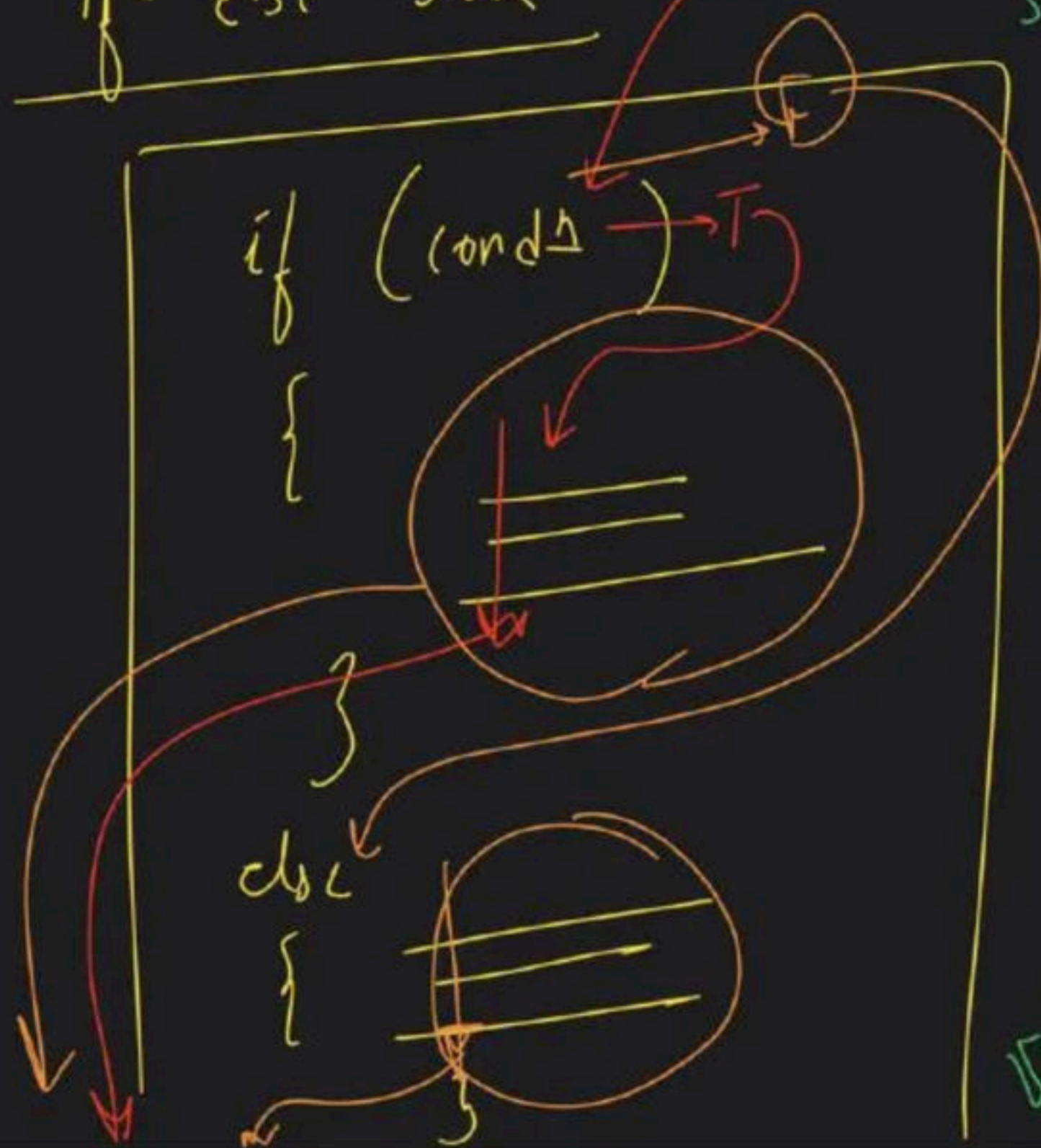
$\leftarrow C_6$

0000-110!

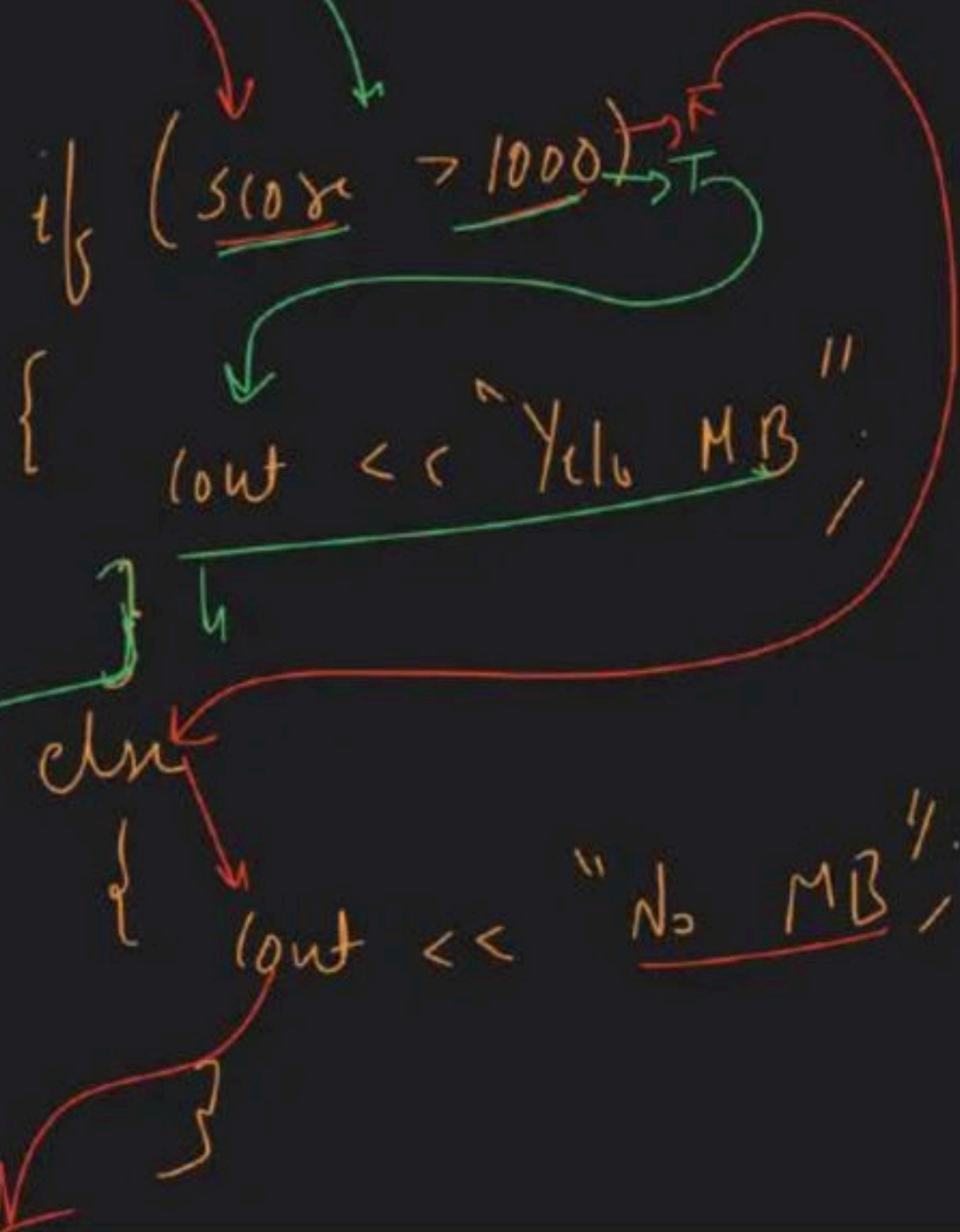
Bitwin



if-else block



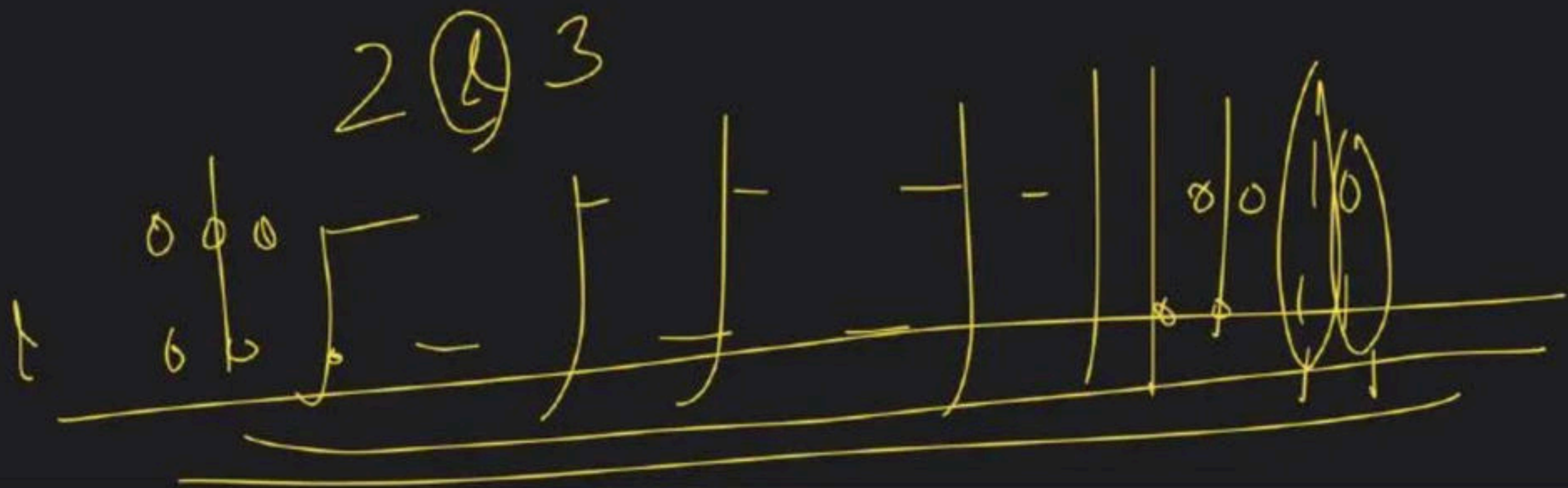
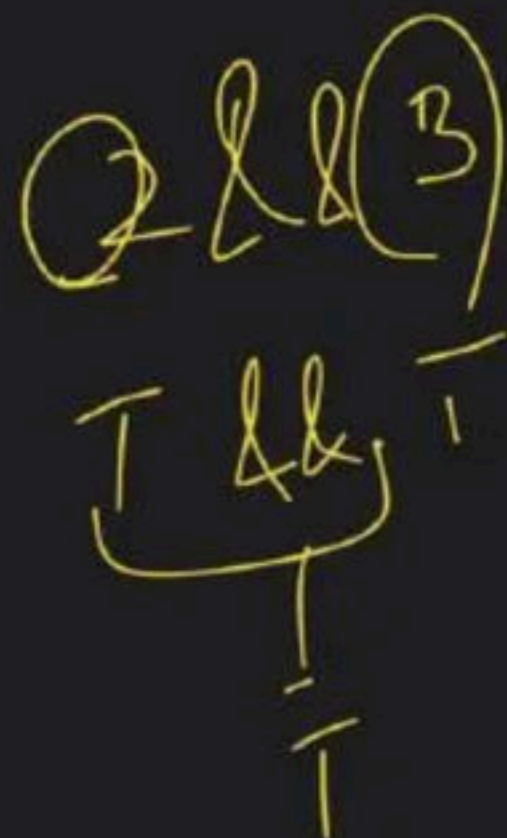
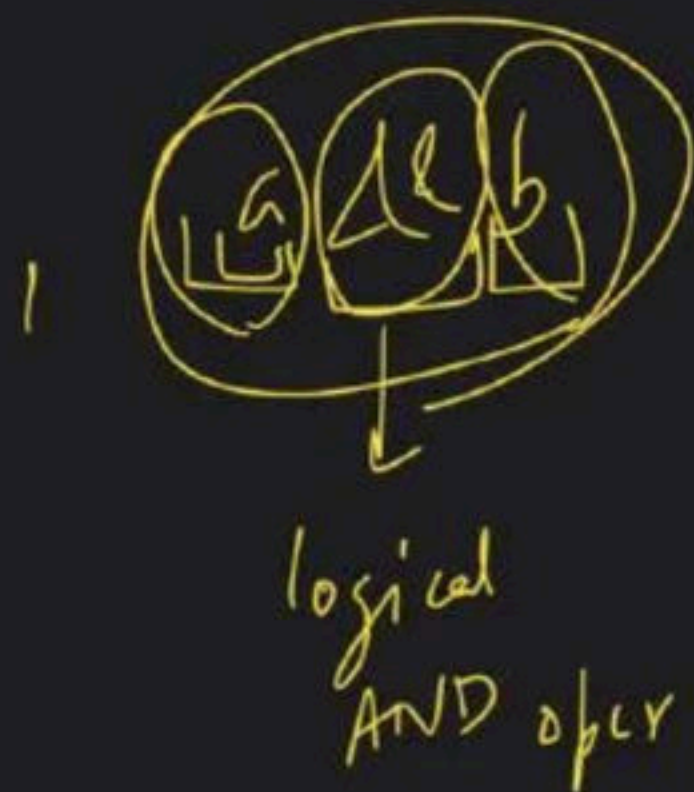
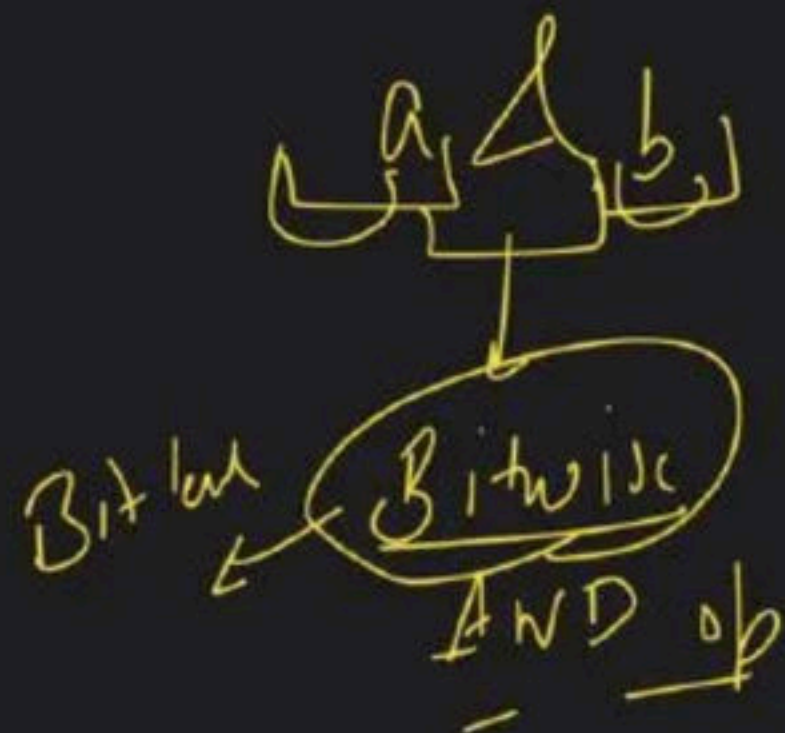
Slon = S_{po} → No MB
Slon = ~~S_{po}~~ → Ylw MB




```
if (age >= 18)
{
  cout << "Yelo lik";
}
```

```
if (score > 1000)
{
  cout << "Yelo MacBook";
}
```

```
if (age >= 18 & score > 1000)
{
  cout << "Bike";
}
```



↖ nested if

```
if (age >= 18)
```

```
{
```

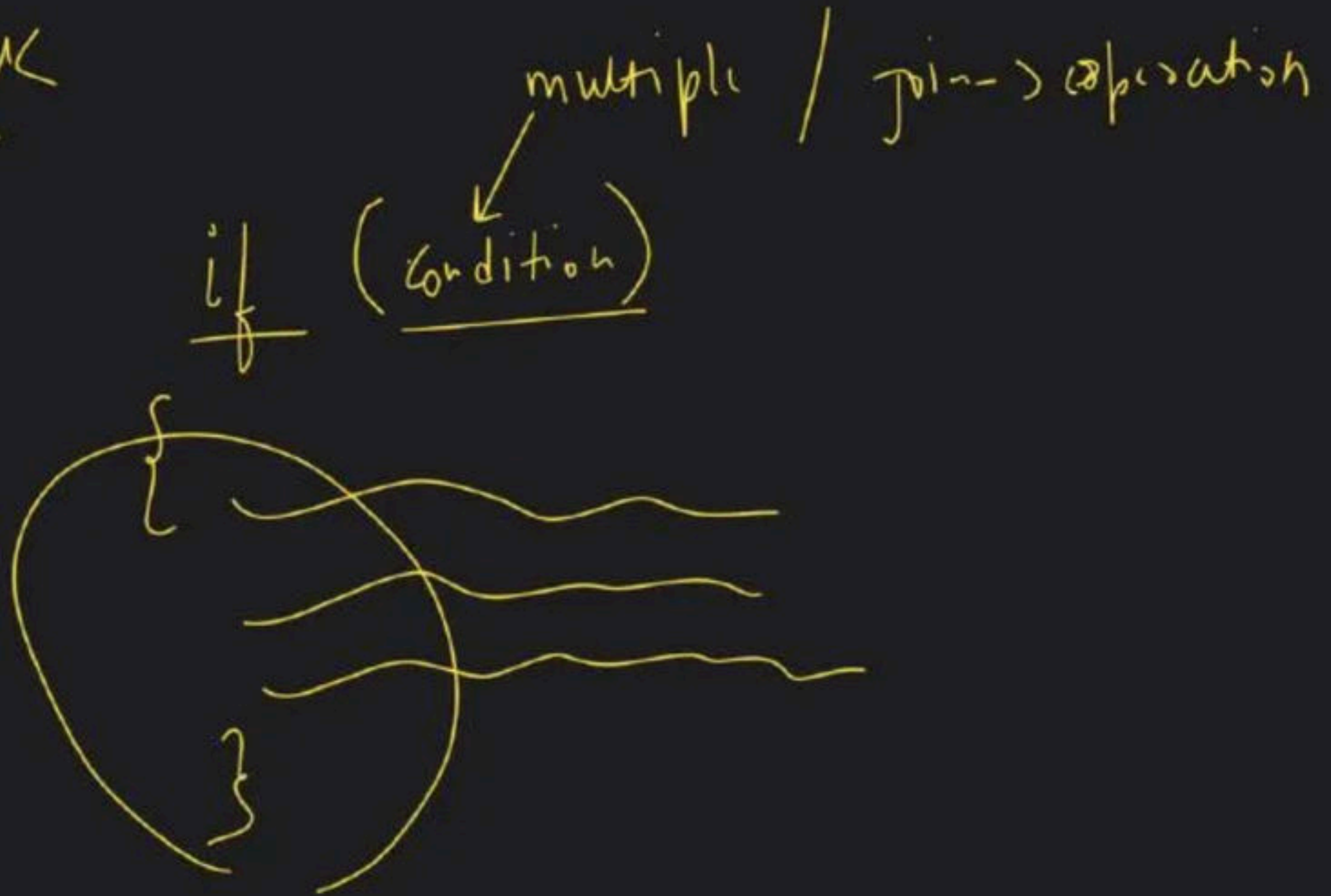
```
  if (score > 1000)
```

```
    {  
      cout << "Bike"  
    }
```

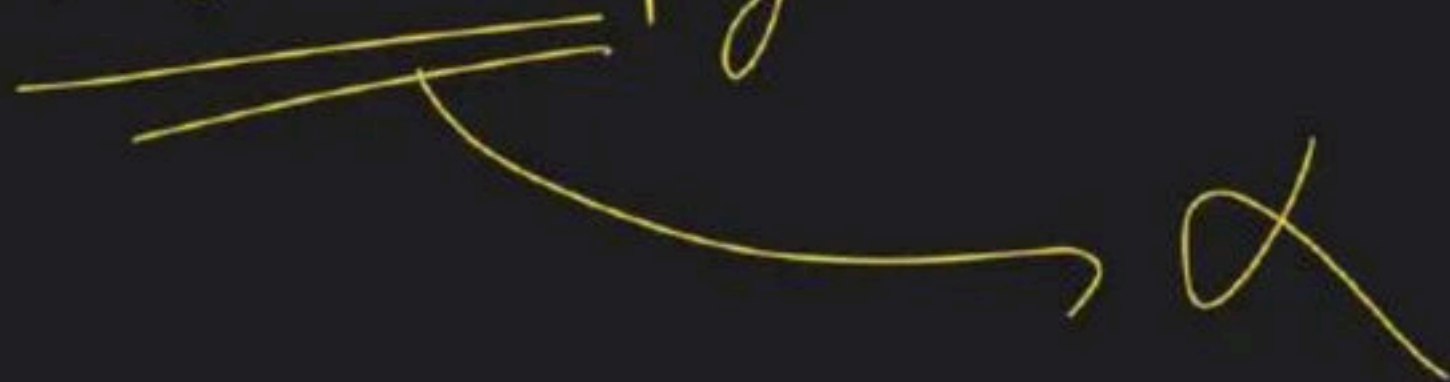
```
}
```


Basic - if Block

Syntax →

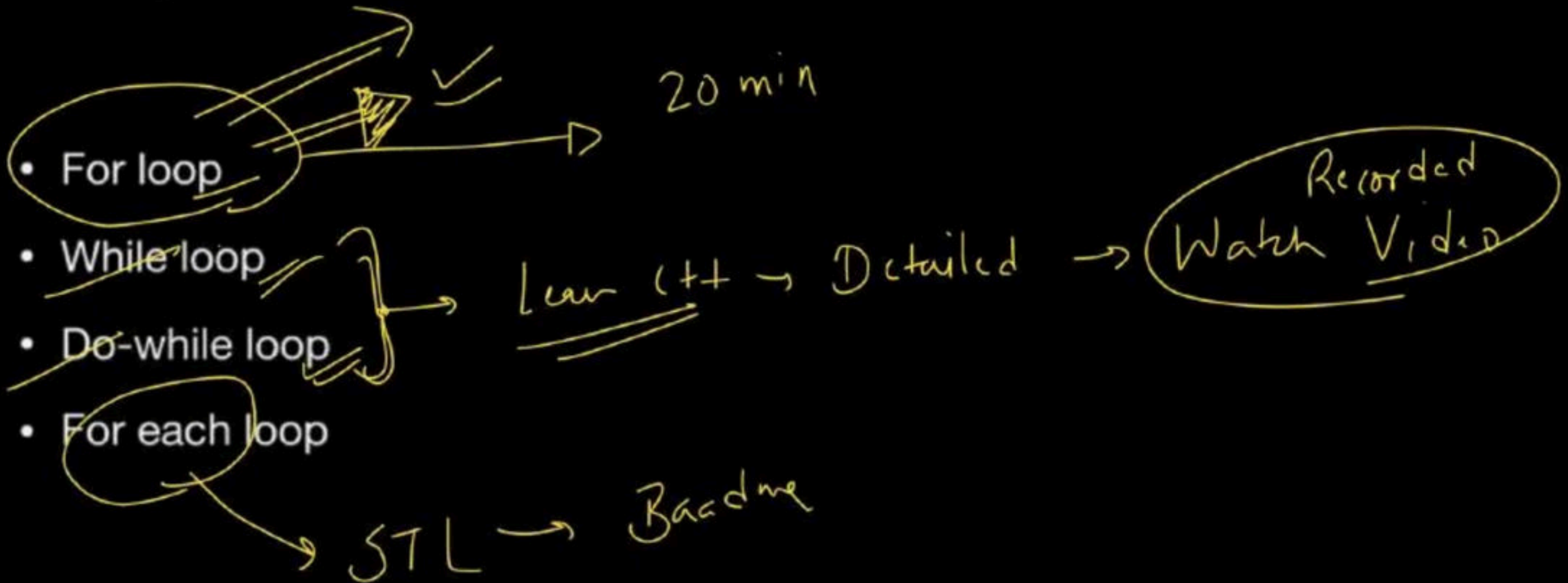


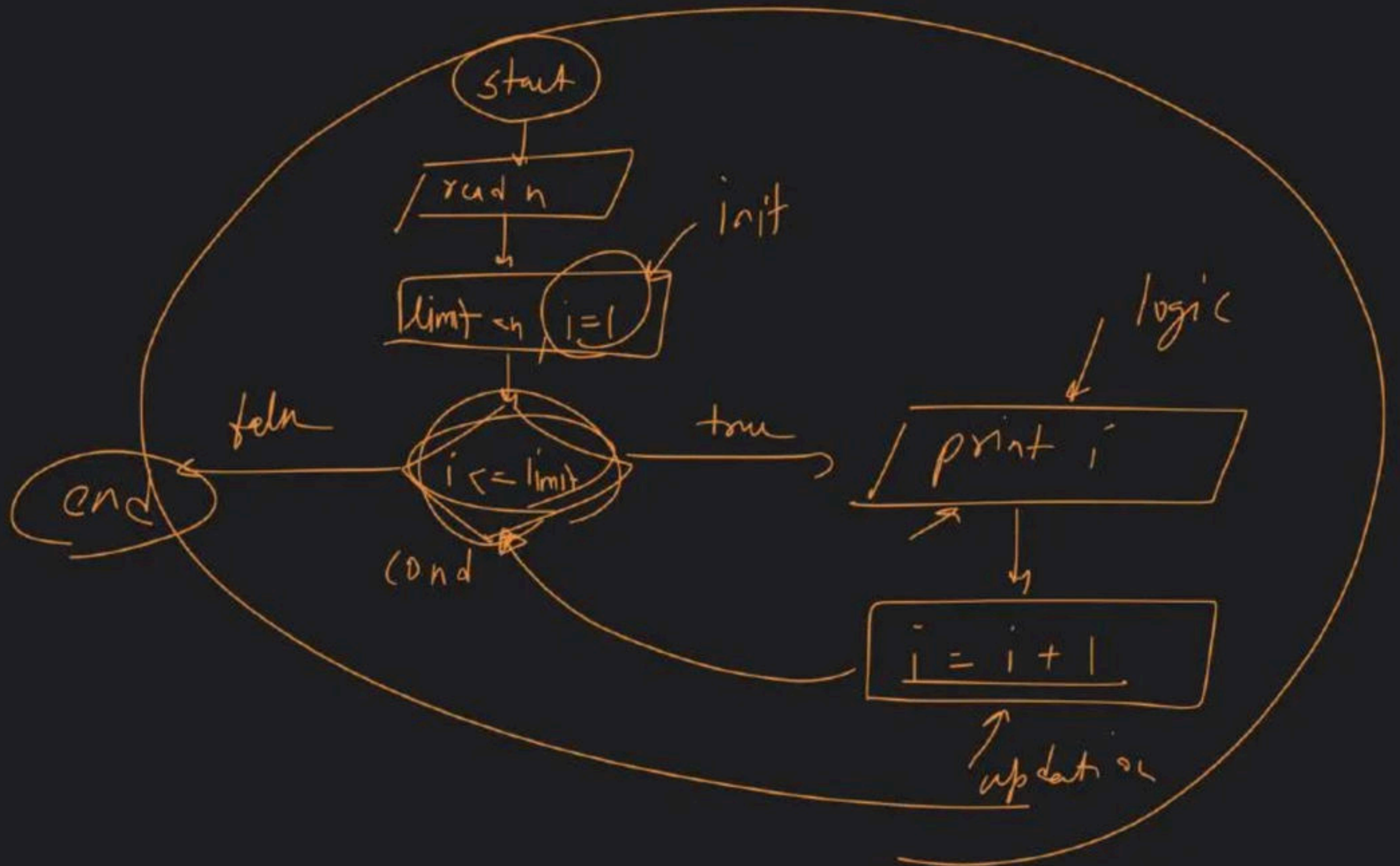
Variable Scoping



functions //

Loops:



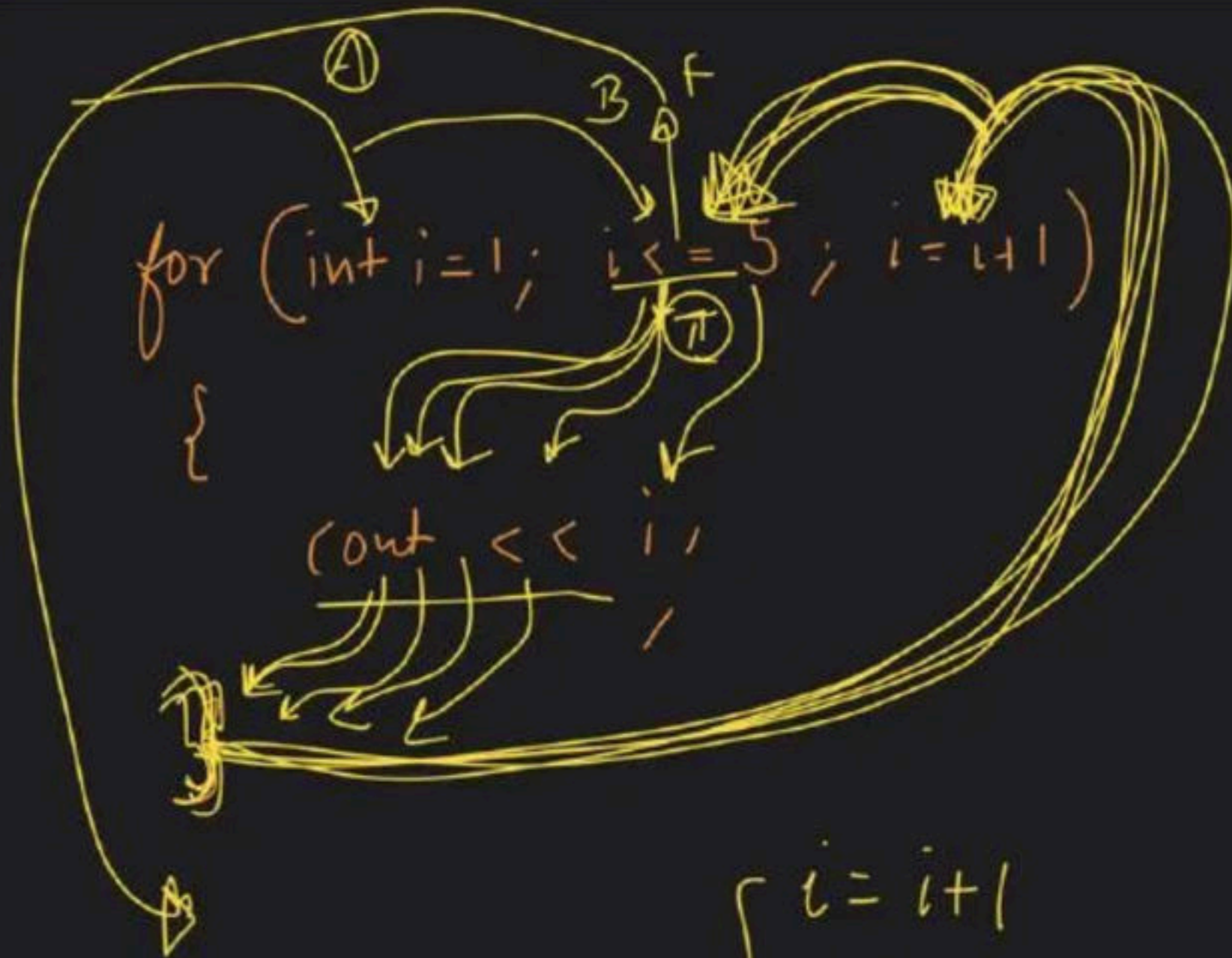


→ For Loop →
↓
Syntax

$n=5$
 $limit=5$

```
for ( int i = 1;  
      initialisation;  
      i <= limit;  
      i = i + 1;  
      update )  
{  
    _____  
    _____  
    _____  
    cout << "Hi";  
}
```

~~cout << i;~~



i = 1

1 <= 5

T

print 1

i = 1 + 1 = 2

2 <= 5

T

print 2

i = 2 + 1 = 3

3 <= 5

T

print 3

i = 3 + 1 = 4

4 <= 5

T

print 4

i = 4 + 1 = 5

5 <= 5

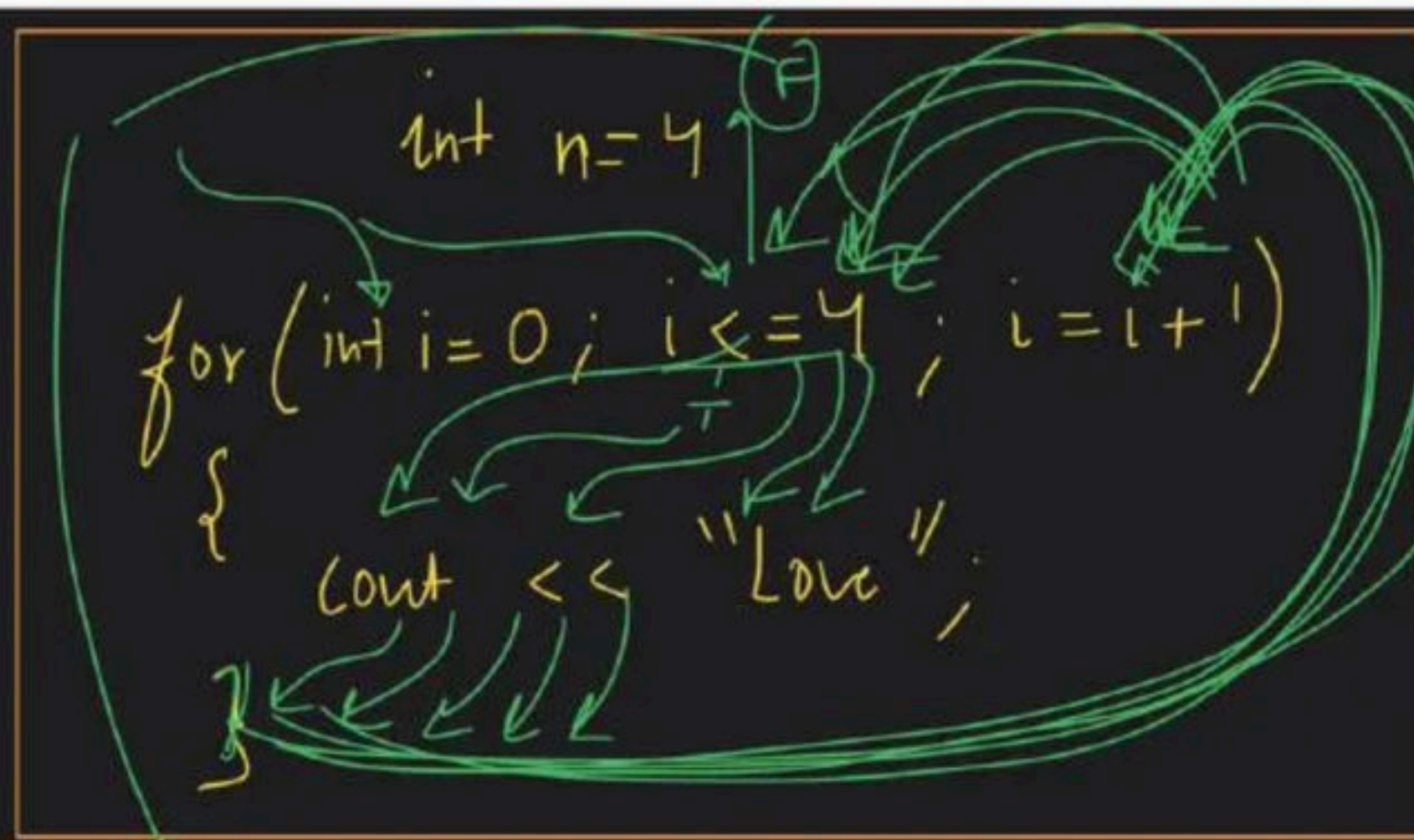
T

print 5

i = 5 + 1 = 6

6 <= 5

F



$n=4$

$i=0$

$0 \leq 4 \rightarrow T$

Love

$i=0+1=1$

$1 \leq 4 \rightarrow T$

Love

$i=1+1=2$

$2 \leq 4 \rightarrow T$

Love

$i=2+1=3$

$3 \leq 4 \rightarrow T$

Love

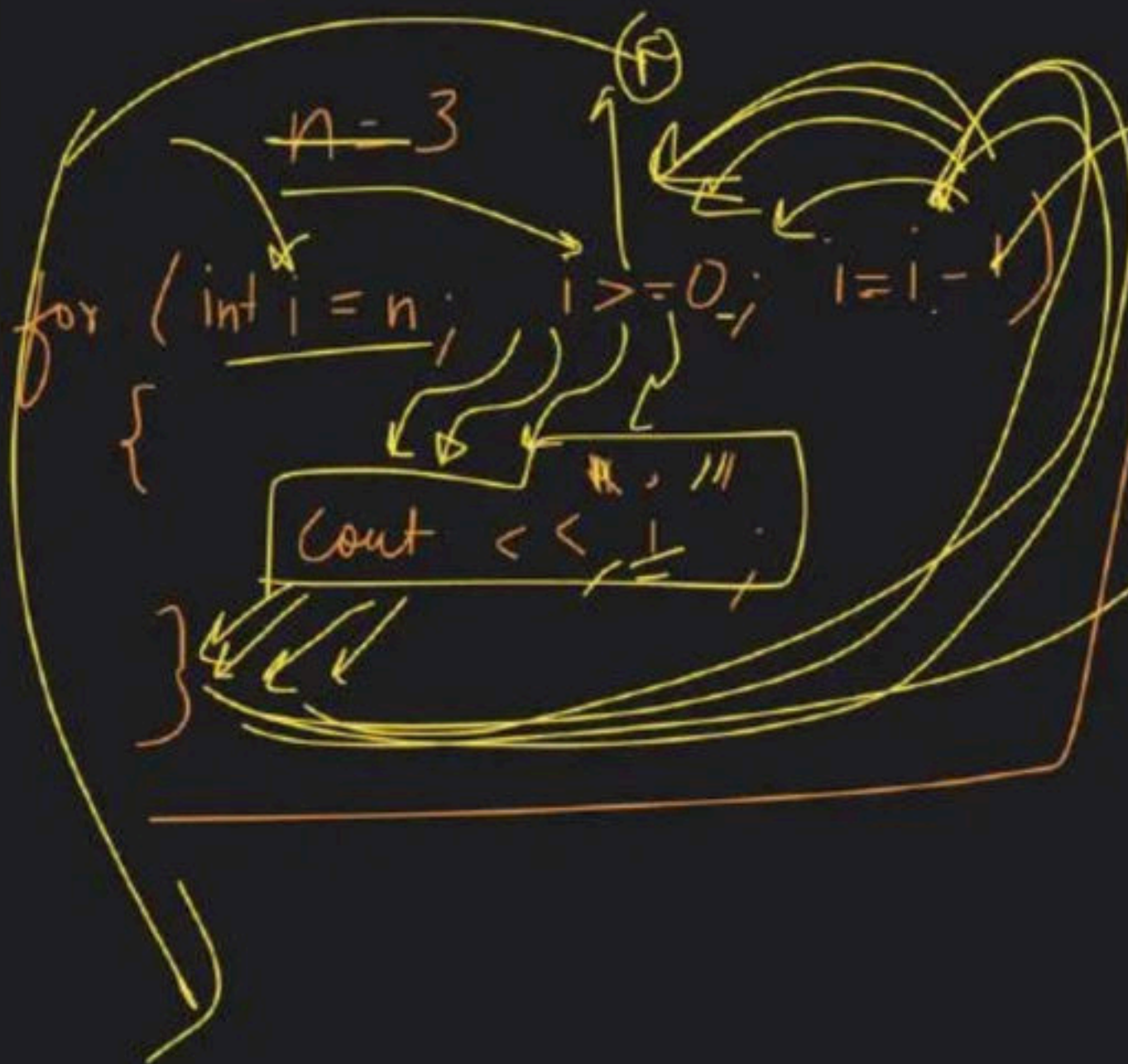
$i=3+1=4$

$4 \leq 4 \rightarrow T$

Love

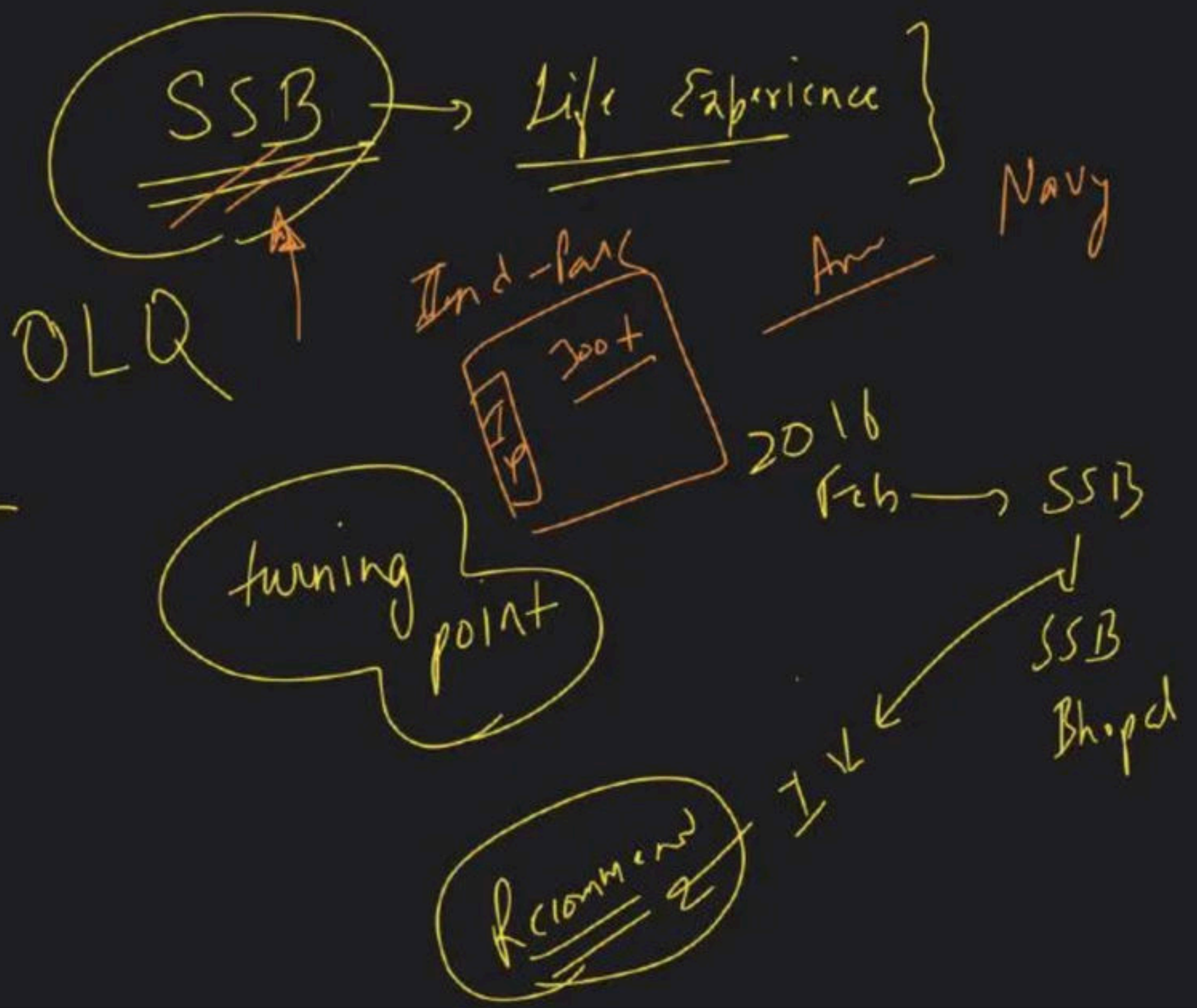
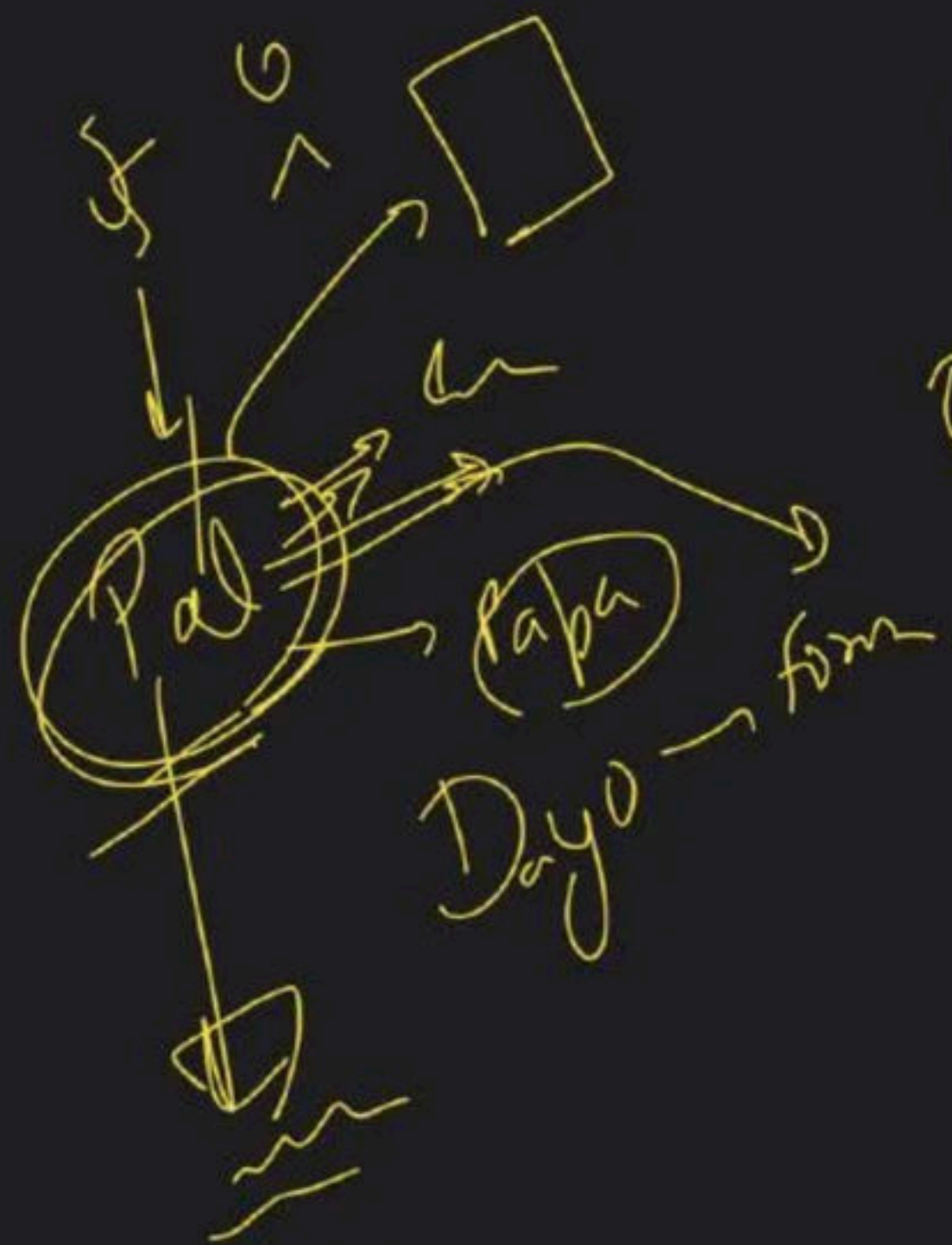
$i=4+1=5$

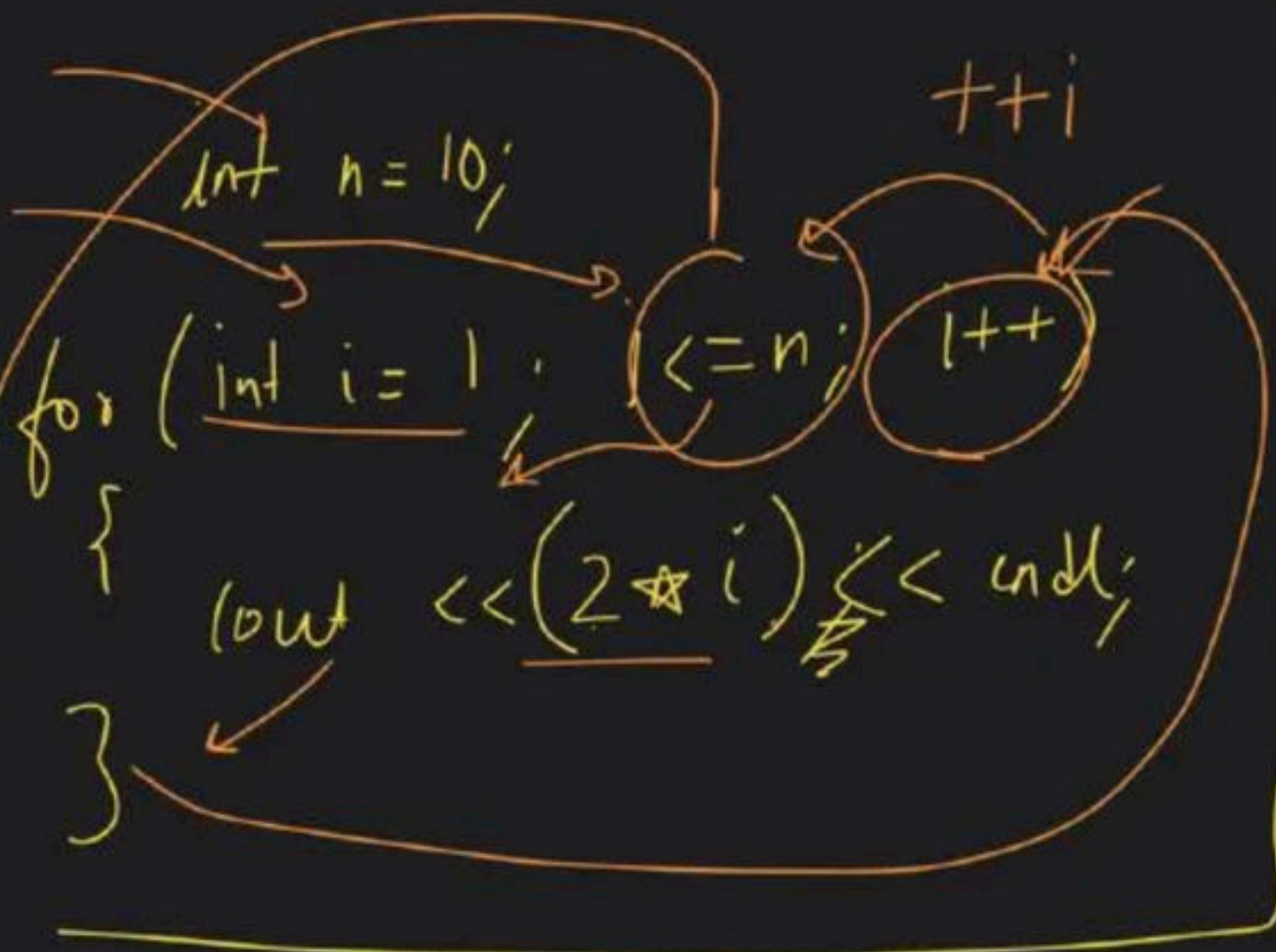
$5 \leq 4 \rightarrow F$



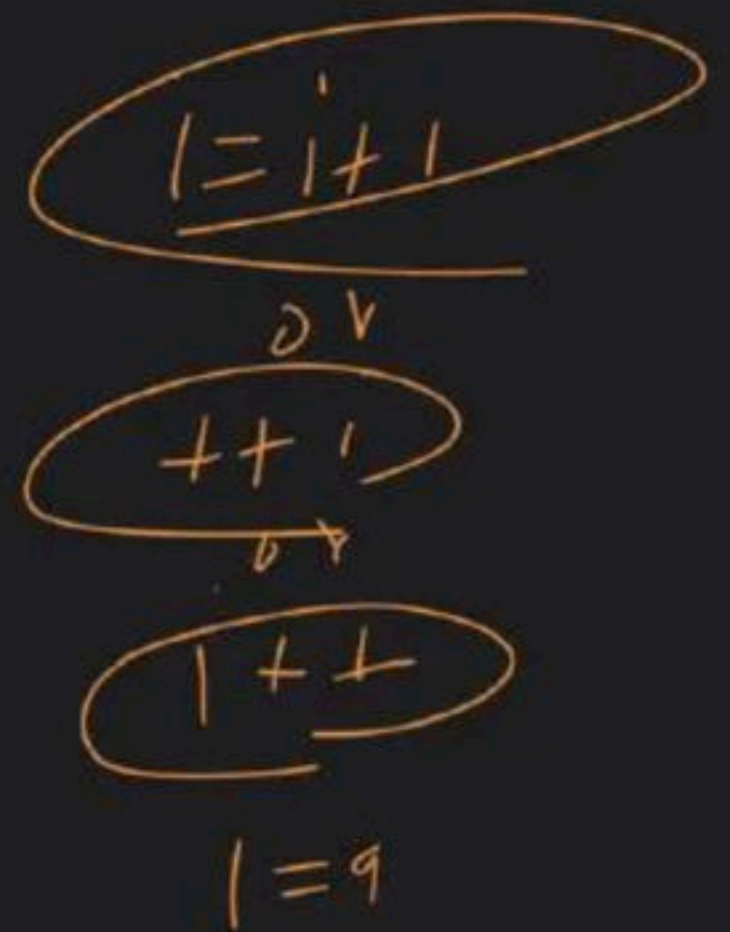
$n=3$
 $i=3$
 $3 \geq 0 \rightarrow T$
 i
 $i=3-1=2$
 $2 \geq 0 \rightarrow T$
 i
 $i=2-1=1$
 $1 \geq 0 \rightarrow T$
 i

$i=1-1=0$
 $0 \geq 0 \rightarrow T$
 i
 $i=0-1=-1$
 $-1 \geq 0 \rightarrow F$





$n = 10$
 $i = 1$
 $1 \leq 10 \rightarrow T$
 $i = 2$
 $2 \leq 10 \rightarrow T$
 $i = 3$
 $i = 4$



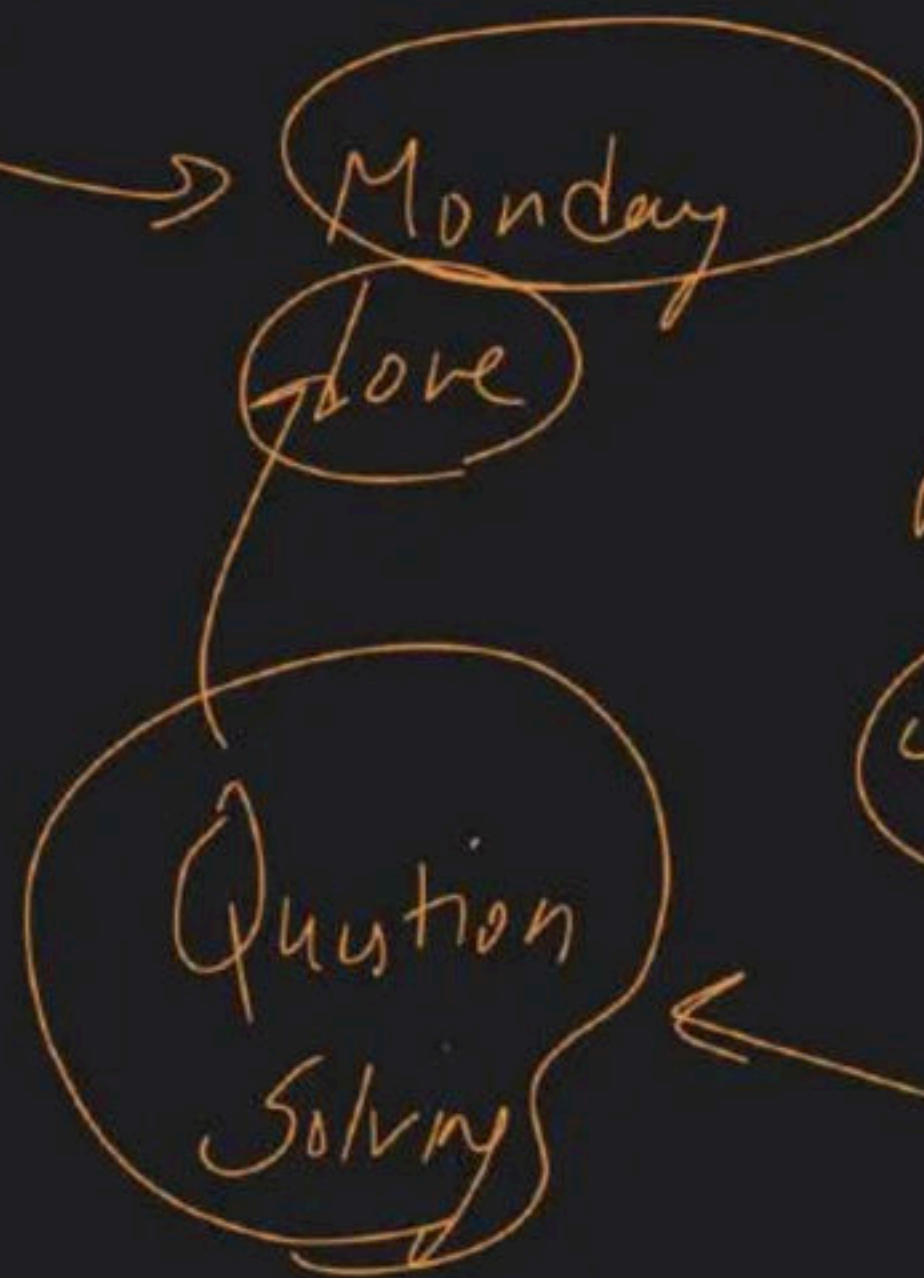
$i = 5$
 $i = 6$
 $i = 7$
 $i = 8$

$i = 10$
 $10 \leq 10 \rightarrow T$
 $i = 11$
 $11 \leq 10 \rightarrow F$

1 min

~~2~~
~~4~~
~~6~~
~~8~~
~~10~~
~~12~~
~~14~~
~~16~~
~~18~~
~~20~~

Patterns



16th Saturday → 9
↓
D-5
↓
LK

parro
Sunday → 2 pm
↓
MS → LK